SECTION PROPERTIES

REV E 01–DEC–1998

QUICK REFERENCE HYPERLINK LIST

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THE ABOVE INCORPORATES THE LAST TECHNICAL REVIEW DATE FOR THIS STANDARD.

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0 REVISION NOTICE

This revision notice provides a brief description of the changes made within this standard. This standard should be reviewed in detail to determine the total extent of the revision. Areas that contain technical changes (changes in requirements) are noted in the margin of this standard with a revision bar.

Revision Synopsis: Revision E which corrects formatting errors in Revision D. This revision also corrects plane surface and solid views as noted below.

Revision Description:

a. Corrected the illustration for the nose rib in FIGURE 3–1, page 39.

b. Corrected the illustrations for the spherical sector and the spherical segment in FIGURE 3–3, page 64.

1 INTRODUCTION

“Section Properties” can be defined as those mathematical attributes of the cross–sectional geometry of a structural element or assembly which are required in order that a relationship between an applied load and its resulting stresses and deflections or an estimate of the element’s or assembly’s stiffness can be established. Loads applied to elements or assemblies are normally resolved into convenient components which will then fall into one or more of the following four categories:

- axial load
- transverse shear
- longitudinal bending
- torsion (or twisting moment)

FIGURE 1–1 is an index to section properties set up to provide the user a listing of geometric shapes such as:

- plane surfaces
- thin rods formed into various shapes
- solids
- thin shells

The circled number for each shape in FIGURE 1–1 denotes the location of its section properties in FIGURE 3–1, FIGURE 3–2, FIGURE 3–3. or FIGURE 3–4.

This BDM contains the section properties which are required for the analysis of:

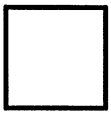
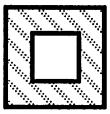
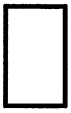
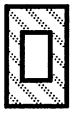
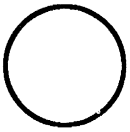
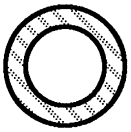
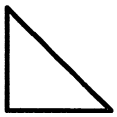
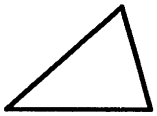
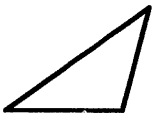
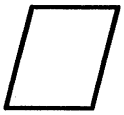
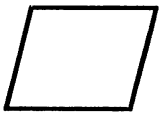
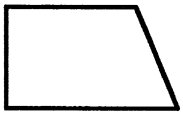
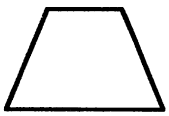
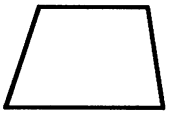
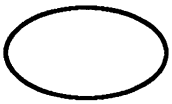
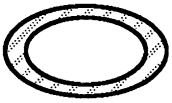
Axial Load	–	Cross Sectional Areas (A)
Transverse Shear	–	First Moment of Area, (Q) together with:
Longitudinal Bending	–	Second Moment of Area (I)

This latter category includes straight beams only; the properties required for the analysis of curved beams and also for that of torsion have, for reasons of convenience, been placed in the BDM’s wherein the appropriate methods are given.

This BDM also covers some aspects of the estimation of the weights of structural and nonstructural elements which require certain additional properties of entire elements, typically surface areas and volumes, and formulas for the estimation of these are given as are those required for the determination of the locations of the centers of gravity and the rotational inertias of both solid bodies and thin shells.

1 INTRODUCTION (CONTINUED)

FIGURE 1–1 SECTION PROPERTIES INDEX (CONTINUED ON NEXT PAGE)

Plane Surfaces (FIGURE 3–1)			
SQUARE  1	HOLLOW SQUARE  2	RECTANGLE  3	HOLLOW RECTANGLE  4
CIRCLE  5	HOLLOW CIRCLE  6	SEMI–CIRCLE 7	HOLLOW SEMI–CIRCLE 8
RIGHT TRIANGLE  9	EQUILATERAL TRIANGLE 10	ISOSCELES TRIANGLE 11	OBLIQUE TRIANGLE  12
OBTUSE ANGLED TRIANGLE  13	RHOMBUS  14	PARALLELOGRAM  15	RIGHT ANGLE TRAPEZOID  16
ISOSCELES TRAPEZOID  17	OBLIQUE TRAPEZOID  18	ELLIPSE  19	HOLLOW ELLIPSE  20

1 INTRODUCTION (CONTINUED)

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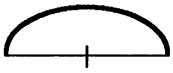
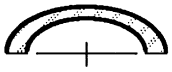
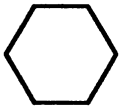
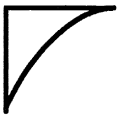
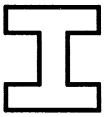
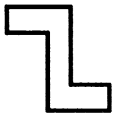
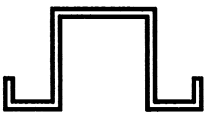
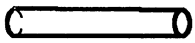
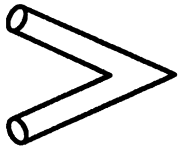
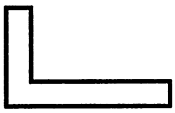
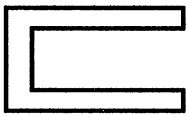
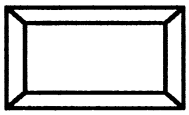
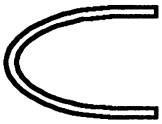
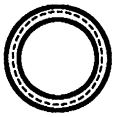
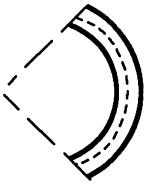
Plane Surfaces (FIGURE 3–1)			
SEMI–ELLIPSE  21	HOLLOW SEMI–ELLIPSE  22	PARABOLIC SEGMENT 23	PARABOLIC HALF–SEGMENT 24
NOSE RIB 25	REGULAR HEXAGON  26	REGULAR OCTAGON 27	CIRCULAR SECTOR 28
HOLLOW CIRCULAR SECTOR 29	CIRCULAR SEGMENT 30	CIRCULAR COMPLEMENT (90° ARC) 31	ELLIPTIC COMPLEMENT 32
COMPLEMENT OF HALF PARABOLA  33	PARABOLIC FILLET IN RIGHT ANGLE 34	SYMMETRICAL I SECTION  35	SYMMETRICAL Z SECTION  36
SYMMETRICAL T SECTION 37		SYMMETRICAL HAT SECTION  38	

SECTION PROPERTIES

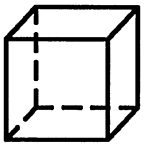
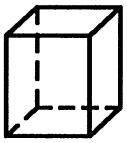
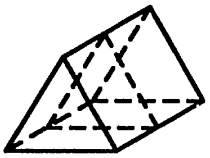
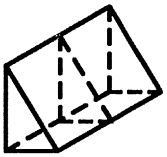
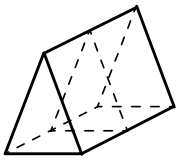
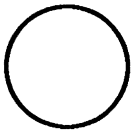
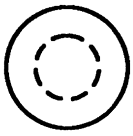
1 INTRODUCTION (CONTINUED)

FIGURE 1–1 SECTION PROPERTIES INDEX (CONTINUED)

Thin Rods Formed Into Various Shapes (FIGURE 3–2)			
STRAIGHT ROD  1	INCLINED ROD NOT THROUGH C. G. AXIS 2	BROKEN ROD  3	“L” ROD  4
“U” ROD  5	RECTANGULAR ROD  6	ELLIPTIC ROD 7	PARABOLIC ROD  8
CIRCULAR ROD  9	SEMI–CIRCULAR ROD 10	SEGMENT OF A CIRCULAR ROD  11	

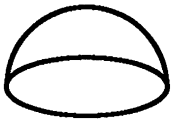
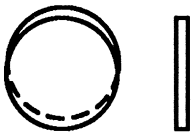
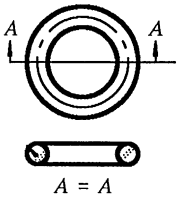
1 INTRODUCTION (CONTINUED)

FIGURE 1–1 SECTION PROPERTIES INDEX (CONTINUED)

Solids (FIGURE 3–3)		
CUBE  ①	RECTANGULAR PRISM  ②	RIGHT RECTANGULAR PYRAMID ③
EQUILATERAL TRIANGULAR PRISM  ④	RIGHT ANGLED WEDGE  ⑤	ISOSCELES WEDGE  ⑥
RIGHT CIRCULAR CYLINDER ⑦	HOLLOW RIGHT CIRCULAR CYLINDER ⑧	RIGHT CIRCULAR CONE ⑨
FRUSTUM OF A CONE ⑩	SPHERE  ⑪	HOLLOW SPHERE  ⑫

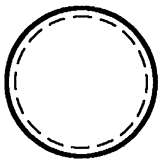
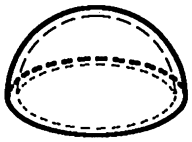
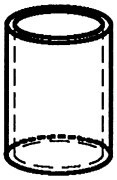
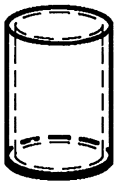
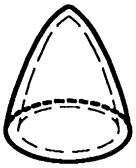
1 INTRODUCTION (CONTINUED)

FIGURE 1–1 SECTION PROPERTIES INDEX (CONTINUED)

Solids (FIGURE 3–3)		
HEMISPHERE  13	ELLIPTICAL CYLINDER 14	ELLIPSOID 15
PARABOLOID OF REVOLUTION 16	ELLIPTIC PARABOLOID 17	THIN CIRCULAR LAMINA  18
TORUS  19	SPHERICAL SECTOR 20	SPHERICAL SEGMENT 21

1 INTRODUCTION (CONTINUED)

FIGURE 1–1 SECTION PROPERTIES INDEX (CONTINUED)

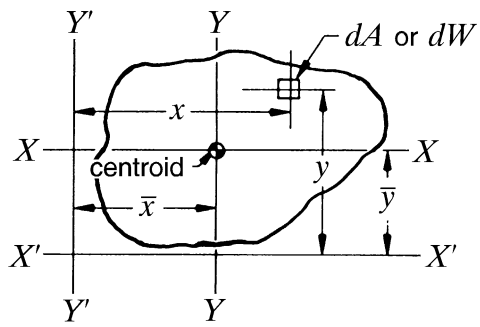
Thin Shells (FIGURE 3–4)	
SPHERICAL SHELL  ①	HEMISPHERICAL SHELL  ②
LATERAL CYLINDRICAL SHELL  ③	TOTAL CYLINDRICAL SHELL  ④
CIRCULAR CONICAL SHELL ⑤	FRUSTUM OF CIRCULAR CONICAL SHELL ⑥
OGIVE OF REVOLUTION SHELL  ⑦	PARABOLIC SHELL ⑧

2 BASIC EQUATIONS

2.1 Area, Moments of Inertia, and Torsion Constants

See FIGURE 2–1 for equations and FIGURE 2–2 for application of these equations.

FIGURE 2–1 AREA, MOMENTS OF INERTIA, AND TORSION CONSTANTS (CONTINUED ON NEXT PAGE)



Area	$A = \sum dA$	**Product of Inertia**	$I_{xy} = \sum (x - \bar{x})(y - \bar{y})dA$
Weight	$W = \sum dW$	**Polar Moment of Inertia** (where x_o and y_o are distances between the shear center and the centroidal axis, see FIGURE 5–1)	About centroidal axis: $I_p = I_{xx} + I_{yy}$ About shear center: $I_{scp} = I_{xx} + I_{yy} + A(x_o^2 + y_o^2)$
Centroid	$\bar{y} = \frac{\sum ydA}{A}$ $\bar{x} = \frac{\sum xdA}{A}$		
Weight Moment of Inertia (For unit thickness)	$I_{wxx} = \sum (y - \bar{y})^2 dW + \sum I_{wox}$ $I_{wyy} = \sum (x - \bar{x})^2 dW + \sum I_{woy}$ $I_{wzz} = WI_{zz}/A$	**Radius of Gyration**	About any axis zz: $\rho_{zz} = \sqrt{\frac{I_{zz}}{A}}$
Area First Moment Q is the sum of moments of elements on one side of selected plane.	$Q_{xxi} = \lvert \sum (y - \bar{y})dA \rvert$ $Q_{yyj} = \lvert \sum (x - \bar{x})dA \rvert$		About centroidal axis: $\rho_{xx} = \sqrt{\frac{I_{xx}}{A}}$ $\rho_{yy} = \sqrt{\frac{I_{yy}}{A}}$
Area Moment of Inertia (Where I_o is the moment of inertia of dA or dW about its own centroidal axis)	About centroidal axis: $I_{xx} = \sum y^2 dA + \sum I_{ox} - \bar{y}^2 \sum dA$ $I_{yy} = \sum x^2 dA + \sum I_{oy} - \bar{x}^2 \sum dA$ About any axis parallel to the centroidal axis: $I_{x'x'} = I_{xx} + A\bar{y}^2$ $I_{y'y'} = I_{yy} + A\bar{x}^2$		Polar radius of gyration about shear center: $\rho_{scp} = \sqrt{\frac{I_{scp}}{A}}$
		Weight Radius of Gyration	About centroidal axis: $\rho_{xx} = \sqrt{\frac{I_{wxx}}{W}}$ $\rho_{yy} = \sqrt{\frac{I_{wyy}}{W}}$

SECTION PROPERTIES

2.1 Area, Moments of Inertia, and Torsion Constants (Continued)

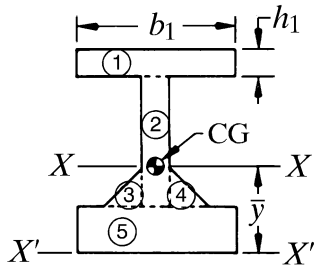
FIGURE 2–1 AREA, MOMENTS OF INERTIA, AND TORSION CONSTANTS (CONCLUDED)

Torsion Constant	For open thin walled sections: $J = \frac{1}{3}\sum bt^3$ For closed thin walled sections: $J = \frac{4A^2}{\sum \frac{b}{t}}$ where $A =$ enclosed area $b =$ middle line length of flange or web $t =$ thickness For all round sections: $J = I_p$

2.1 Area, Moments of Inertia, and Torsion Constants (Continued)

FIGURE 2–2 SECTION PROPERTIES – EXAMPLE PROBLEM

Given:



Find: Find the section properties about a horizontal axis through the centroid of the areas shown in adjacent figure.

Solution: Divide the area into parts for which the section properties can be found in FIGURE 3–1. Tabulate results as shown below:

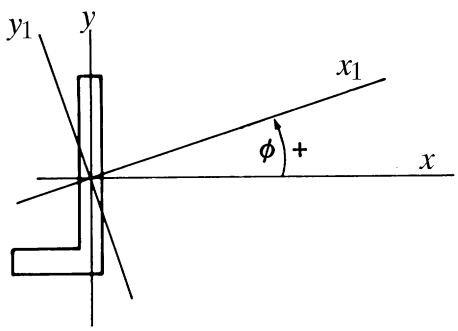
Part	W	H	y	dA	ydA	y^2dA	I_{o_x}
1	b_1	h_1	y_1	b_1h_1	A_1y_1	$A_1y_1^2$	$b_1h_2^3/12$
2	b_2	h_2	y_2	b_2h_2	A_2y_2	$A_2y_2^2$	$b_2h_2^3/12$
3	b_3	h_3	y_3	$b_3h_3/2$	A_3y_3	$A_3y_3^2$	$b_3h_3^3/36$
4	b_4	h_4	y_4	$b_4h_4/2$	A_4y_4	$A_4y_4^2$	$b_4h_4^3/36$
5	b_5	h_5	y_5	b_5h_5	A_5y_5	$A_5y_5^2$	$b_5h_5^3/12$
			$\sum$	dA	ydA	y^2dA	I_{o_x}

H = Height
W = Width

For answers, use applicable formulas shown in FIGURE 2–1.

2.2 Moments and Products of Inertia About Inclined Axes

FIGURE 2–3 MOMENTS AND PRODUCTS OF INERTIA ABOUT INCLINED AXES



$$I_{x_1} = I_{xx}\cos^2\phi - I_{xy}\sin 2\phi + I_{yy}\sin^2\phi$$

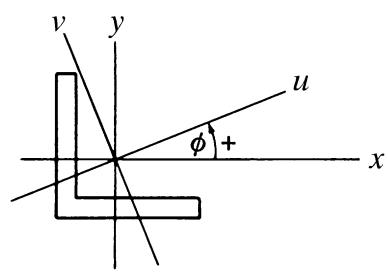
$$I_{y_1} = I_{xx}\sin^2\phi + I_{xy}\sin 2\phi + I_{yy}\cos^2\phi$$

$$I_{x_1y_1} = I_{xy}\cos 2\phi + \frac{I_{xx} - I_{yy}}{2}\sin 2\phi$$

where x_1 and y_1 are the inclined axes, and ϕ is the angle between the x axis and the x_1 axis.

2.3 Principal Axes

FIGURE 2–4 PRINCIPAL AXES



The principal axes are the axes about which I_{xx} and I_{yy} are a minimum and maximum, and $I_{xy} = 0$. When I_{xx}, I_{yy} and I_{xy} are known, the angle ϕ between the x axis and a principal axis and $I_{\max}$ and $I_{\min}$ can be found as follows :

$$\phi = \frac{1}{2}\tan^{-1}\left(\frac{2I_{xy}}{I_{yy} - I_{xx}}\right)$$

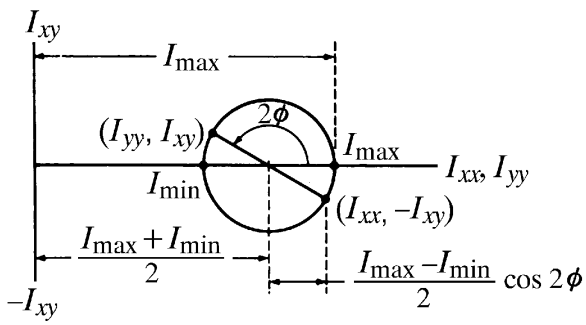
$$I_{\max} = \frac{I_{xx} + I_{yy}}{2} + \sqrt{{I_{xy}}^2 + \left(\frac{I_{xx} - I_{yy}}{2}\right)^2}$$

$$I_{\min} = \frac{I_{xx} + I_{yy}}{2} - \sqrt{{I_{xy}}^2 + \left(\frac{I_{xx} - I_{yy}}{2}\right)^2}$$

For example, in the illustration, v and u are the principal axes, and $I_{uv} = 0$.

2.4 Mohr's Circle for Moment of Inertia

FIGURE 2–5 MOHR'S CIRCLE FOR MOMENT OF INERTIA



The above relationships can be obtained graphically as follows:

Plot (I_{yy}, I_{xy}) and $(I_{xx}, -I_{xy})$ and draw a circle through these points with the center on the I_{xx}, I_{yy} axis. I_{max}, I_{min}, and 2ϕ can be read from the graph, as shown.

3 SECTION PROPERTIES TABLES

NOTE 1:

The following expressions do not necessarily use similar terms between each geometric shape for the purposes of using superposition.

NOTE 2:

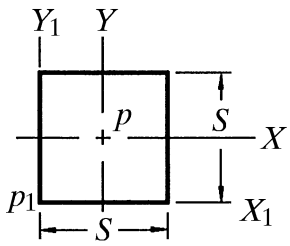
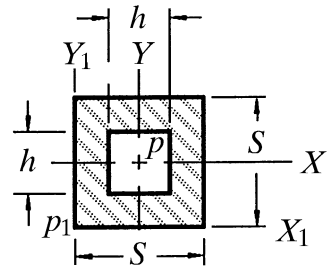
The following expressions showing coefficients expressed in their approximate form $\left(\text{i.e. } 0.894 \text{ vs. } 2/\sqrt{5}\right)$ may be suspect for error in a level of accuracy, due to round–off, that may be unacceptable to some users.

NOTE 3:

The formulas given in FIGURE 3–2 for properties of "Thin Rods Formed into Various Shapes" are approximations.

3.1 Plane Surfaces

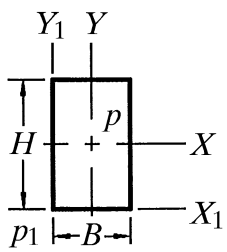
FIGURE 3–1 PLANE SURFACES (CONTINUED ON NEXT PAGE)

	SQUARE	HOLLOW SQUARE
	 ①	 ②
Area	S^2	$S^2 - h^2$
Centroid	At point p	At point p
Weight Moment of Inertia	$I_x = I_y = \frac{WS^2}{12}$ $I_{x_1} = I_{y_1} = \frac{WS^2}{3}$ $I_p = I_x + I_y = \frac{WS^2}{6}$ $I_{p_1} = I_{x_1} + I_{y_1} = \frac{2WS^2}{3}$	$I_x = I_y = \frac{W(S^2 + h^2)}{12}$ $I_{x_1} = I_{y_1} = \frac{W(4S^2 + h^2)}{12}$ $I_p = I_x + I_y = \frac{W(S^2 + h^2)}{6}$ $I_{p_1} = \frac{W(4S^2 + h^2)}{6}$
Area Moment of Inertia	$I_x = I_y = \frac{S^4}{12}$ $I_{x_1} = I_{y_1} = \frac{S^4}{3}$ $I_p = I_x + I_y = \frac{S^4}{6}$ $I_{p_1} = I_{x_1} + I_{y_1} = \frac{2S^4}{3}$	$I_x = I_y = \frac{S^4 - h^4}{12}$ $I_{x_1} = I_{y_1} = \frac{4S^4 - 3S^2h^2 - h^4}{6}$ $I_p = \frac{S^4 - h^4}{6}$ $I_{p_1} = \frac{4S^4 - 3S^2h^2 - h^4}{6}$
Radius of Gyration	$\rho_x = \rho_y = \frac{S}{2\sqrt{3}}$ $\rho_{x_1} = \rho_{y_1} = \frac{S}{\sqrt{3}}$ $\rho_p = \frac{S}{\sqrt{6}}$ $\rho_{p_1} = \frac{S\sqrt{2}}{\sqrt{3}}$	$\rho_x = \rho_y = \frac{1}{2}\sqrt{\frac{S^2 + h^2}{3}}$ $\rho_{x_1} = \rho_{y_1} = \frac{1}{2}\sqrt{\frac{4S^2 + h^2}{3}}$ $\rho_p = \sqrt{\frac{S^2 + h^2}{6}}$ $\rho_{p_1} = \sqrt{\frac{4S^2 + h^2}{6}}$

3.1 Plane Surfaces (Continued)

FIGURE 3–1 PLANE SURFACES (CONTINUED)

RECTANGLE



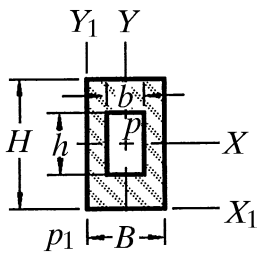
3

Area	$A = BH$	
Centroid	At point p	
Weight Moment of Inertia	$I_x = \frac{WH^2}{12}$ $I_y = \frac{WB^2}{12}$ $I_p = \frac{W}{12}(H^2 + B^2)$	$I_{x_1} = \frac{WH^2}{3}$ $I_{y_1} = \frac{WB^2}{3}$ $I_{p_1} = \frac{W}{3}(H^2 + B^2)$
Area Moment of Inertia	$I_x = \frac{BH^3}{12}$ $I_y = \frac{HB^3}{12}$ $I_p = \frac{BH}{12}(H^2 + B^2)$	$I_{x_1} = \frac{BH^3}{3}$ $I_{y_1} = \frac{HB^3}{3}$ $I_{p_1} = \frac{BH}{3}(H^2 + B^2)$
Radius of Gyration	$\rho_x = \frac{H}{2\sqrt{3}}$ $\rho_y = \frac{B}{2\sqrt{3}}$ $\rho_p = \frac{1}{2}\sqrt{\frac{H^2 + B^2}{3}}$ $\rho_{p_1} = \sqrt{\frac{H^2 + B^2}{3}}$	$\rho_{x_1} = \frac{H}{\sqrt{3}}$ $\rho_{y_1} = \frac{B}{\sqrt{3}}$

3.1 Plane Surfaces (Continued)

FIGURE 3–1 PLANE SURFACES (CONTINUED)

HOLLOW RECTANGLE

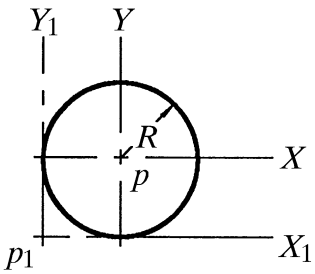
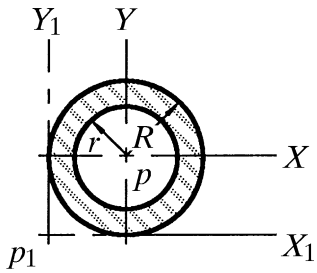


4

Area	$A = BH - bh$	
Centroid	– – –	
Weight Moment of Inertia	$I_x = \frac{W}{12}\left(\frac{BH^3 - bh^3}{BH - bh}\right)$ $I_y = \frac{W}{12}\left(\frac{HB^3 - hb^3}{BH - bh}\right)$ $I_p = I_x + I_y$	$I_{x_1} = I_x + \frac{WH^2}{4}$ $I_{y_1} = I_y + \frac{WB^2}{4}$ $I_{p_1} = I_{x_1} + I_{y_1}$
Area Moment of Inertia	$I_x = \frac{BH^3 - bh^3}{12}$ $I_y = \frac{HB^3 - hb^3}{12}$ $I_p = I_x + I_y$	$I_{x_1} = \frac{BH^3}{3} - \frac{bh(3H^2 + h^2)}{12}$ $I_{y_1} = \frac{HB^3}{3} - \frac{bh(3B^2 + b^2)}{12}$ $I_{p_1} = I_{x_1} + I_{y_1}$
Radius of Gyration	$\rho_x = \sqrt{\frac{BH^3 - bh^3}{12(BH - bh)}}$ $\rho_y = \sqrt{\frac{HB^3 - hb^3}{12(BH - bh)}}$ $\rho_p = \sqrt{\frac{I_p}{BH - bh}}$	$\rho_{x_1} = \sqrt{\frac{I_{x_1}}{BH - bh}}$ $\rho_{y_1} = \sqrt{\frac{I_{y_1}}{BH - bh}}$ $\rho_{p_1} = \sqrt{\frac{I_{p_1}}{BH - bh}}$

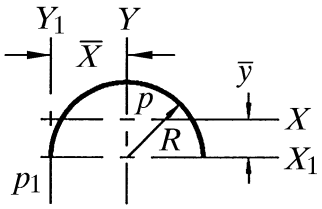
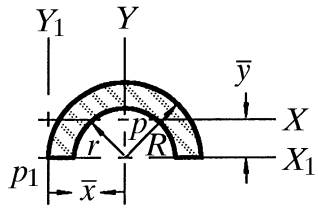
3.1 Plane Surfaces (Continued)

FIGURE 3–1 PLANE SURFACES (CONTINUED)

	CIRCLE	HOLLOW CIRCLE
	 5	 6
Area	$A = \pi R^2 \quad (D = 2R)$	$A = \pi(R^2 - r^2)$
Centroid	– – –	– – –
Weight Moment of Inertia	$I_x = I_y = \frac{WD^2}{16} = \frac{WR^2}{4}$ $I_{x_1} = I_{y_1} = 1.25WR^2$ $I_p = \frac{WD^2}{8} = \frac{WR^2}{2}$	$I_x = I_y = \frac{W(R^2 + r^2)}{4}$ $I_p = \frac{W(R^2 + r^2)}{2}$ $I_{x_1} = I_{y_1} = \frac{W(5R^2 + r^2)}{4}$
Area Moment of Inertia	$I_x = I_y = \frac{\pi D^4}{64}$ $I_{x_1} = I_{y_1} = \frac{5\pi D^4}{64}$ $I_p = \frac{\pi D^4}{32}$ $I_{p_1} = \frac{5\pi D^4}{32}$	$I_x = I_y = \frac{\pi(R^4 - r^4)}{4}$ $I_p = \frac{\pi(R^4 - r^4)}{2}$ $I_{x_1} = I_{y_1} = \frac{\pi(5R^4 - 4R^2r^2 - r^4)}{4}$ $I_{p_1} = \frac{\pi(5R^4 - 4R^2r^2 - r^4)}{2}$
Radius of Gyration	$\rho_x = \rho_y = \frac{D}{4}$ $\rho_{x_1} = \rho_{y_1} = \frac{D\sqrt{5}}{4}$ $\rho_p = \frac{D}{2\sqrt{2}}$ $\rho_{p_1} = D\sqrt{\frac{5}{8}}$	$\rho_x = \rho_y = \frac{\sqrt{R^2 + r^2}}{2}$ $\rho_p = \sqrt{\frac{R^2 + r^2}{2}}$ $\rho_{x_1} = \rho_{y_1} = \frac{1}{2}\sqrt{5R^2 + r^2}$ $\rho_{p_1} = \sqrt{\frac{I_{p_1}}{A}}$

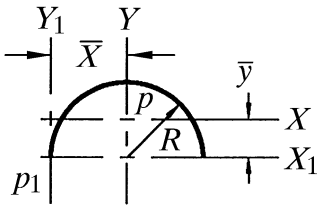
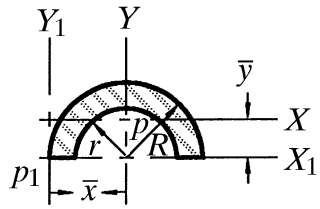
3.1 Plane Surfaces (Continued)

FIGURE 3–1 PLANE SURFACES (CONTINUED)

	SEMI–CIRCLE	HOLLOW SEMI–CIRCLE
	 7	 8
Area	$A = \frac{\pi R^2}{2} \quad (D = 2R)$	$A = \frac{\pi(R^2 - r^2)}{2}$
Centroid	$\bar{y} = \frac{4R}{3\pi} \quad \bar{x} = \frac{R}{2}$	$\bar{y} = \frac{4(R^3 - r^3)}{3\pi(R^2 - r^2)}$
Weight Moment of Inertia	$I_x = \left(\frac{1}{4} - \frac{16}{9\pi^2}\right)WR^2$ $I_{x_1} = \frac{WD^2}{16}$ $I_y = \frac{WD^2}{16}$ $I_{y_1} = \frac{5WD^2}{16}$ $I_p = \frac{2WD^2}{25}$ $I_{p_1} = \frac{3WD^2}{8}$	$I_x = \frac{W(R^2 + r^2)}{4} - \bar{y}^2 W \quad I_{x_1} = \frac{W(R^2 + r^2)}{4}$ $I_y = \frac{W(R^2 + r^2)}{4} \quad I_{y_1} = \frac{W(5R^2 + r^2)}{4}$ $I_p = I_x + I_y \quad I_{p_1} = \frac{W(3R^2 + r^2)}{2}$
Area Moment of Inertia	$I_x = \left(\frac{\pi}{8} - \frac{8}{9\pi}\right)R^4$ $I_{x_1} = \frac{\pi R^4}{8}$ $I_y = \frac{\pi R^4}{8}$ $I_{y_1} = \frac{5\pi R^4}{8}$ $I_p = \left(\frac{\pi}{4} - \frac{8}{9\pi}\right)R^4$ $I_{p_1} = \frac{3\pi R^4}{4}$	$I_x = \frac{\pi}{8}(R^4 - r^4) - \frac{\pi(R^2 - r^2)}{2}\bar{y}^2$ $I_{x_1} = \frac{\pi}{8}(R^4 - r^4)$ $I_y = \frac{\pi}{8}(R^4 - r^4)$ $I_{y_1} = \frac{\pi(R^2 - r^2)(5R^2 + r^2)}{8}$ $I_p = I_x + I_y \quad I_{p_1} = I_{x_1} + I_{y_1}$

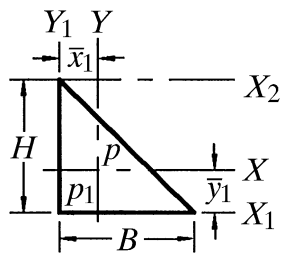
3.1 Plane Surfaces (Continued)

FIGURE 3–1 PLANE SURFACES (CONTINUED)

SEMI–CIRCLE (CONTINUED)		HOLLOW SEMI–CIRCLE (CONTINUED)
 7		 8
Radius of Gyration	$\rho_x = R\sqrt{\frac{1}{4} - \frac{16}{9\pi^2}}$ $\rho_{x_1} = \frac{R}{2}$ $\rho_y = \frac{R}{2}$ $\rho_{y_1} = \frac{R\sqrt{5}}{2}$ $\rho_p = 0.566R$ $\rho_{p_1} = 1.225R$	$\rho_x = \sqrt{\frac{2I_x}{\pi(R^2 - r^2)}}$ $\rho_{x_1} = \sqrt{\frac{R^2 + r^2}{2}}$ $\rho_y = \sqrt{\frac{R^2 + r^2}{2}}$ $\rho_{y_1} = \sqrt{\frac{2I_{y_1}}{\pi(R^2 - r^2)}}$ $\rho_p = \sqrt{\frac{2I_p}{\pi(R^2 - r^2)}}$ $\rho_{p_1} = \sqrt{\frac{2I_{p_1}}{\pi(R^2 - r^2)}}$

3.1 Plane Surfaces (Continued)

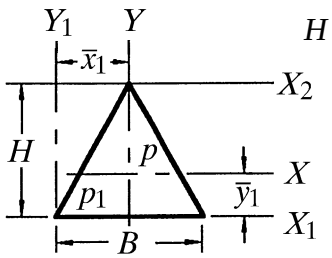
FIGURE 3–1 PLANE SURFACES (CONTINUED)

RIGHT TRIANGLE  9		
Area	$A = \frac{BH}{2}$	
Centroid	$\bar{x}_1 = \frac{B}{3}$	$\bar{y}_1 = \frac{H}{3}$
Weight Moment of Inertia	$I_x = \frac{WH^2}{18}$ $I_{x_2} = \frac{WH^2}{2}$ $I_y = \frac{WB^2}{18}$ $I_p = \frac{W(H^2 + B^2)}{18}$	$I_{x_1} = \frac{WH^2}{6}$ $I_{y_1} = \frac{WB^2}{6}$ $I_{p_1} = \frac{W(H^2 + B^2)}{6}$
Area Moment of Inertia	$I_x = \frac{BH^3}{36}$ $I_{x_2} = \frac{BH^3}{4}$ $I_y = \frac{B^3H}{36}$ $I_p = \frac{BH(B^2 + H^2)}{36}$	$I_{x_1} = \frac{BH^3}{12}$ $I_{y_1} = \frac{B^3H}{12}$ $I_{p_1} = \frac{BH(B^2 + H^2)}{12}$
Radius of Gyration	$\rho_x = \frac{H}{\sqrt{18}}$ $\rho_{x_2} = \frac{H}{\sqrt{2}}$ $\rho_y = \frac{B}{\sqrt{18}}$ $\rho_p = \sqrt{\frac{B^2 + H^2}{18}}$	$\rho_{x_1} = \frac{H}{\sqrt{6}}$ $\rho_{y_1} = \frac{B}{\sqrt{6}}$ $\rho_{p_1} = \sqrt{\frac{B^2 + H^2}{6}}$

3.1 Plane Surfaces (Continued)

FIGURE 3–1 PLANE SURFACES (CONTINUED)

EQUILATERAL TRIANGLE



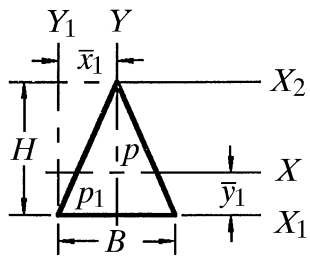
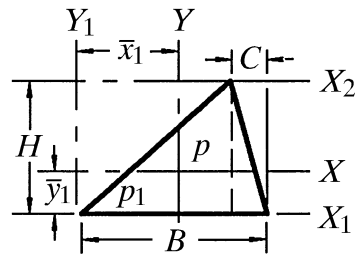
$$H = B\sqrt{\frac{3}{4}}$$

(10)

Area	$A = \frac{BH}{2}$	
Centroid	$\bar{x}_1 = \frac{B}{2}$	$\bar{y}_1 = \frac{H}{3}$
Weight Moment of Inertia	$I_x = \frac{WH^2}{18}$	$I_{x_1} = \frac{WH^2}{6}$
	$I_{x_2} = \frac{WH^2}{2}$	
	$I_y = \frac{WB^2}{24}$	$I_{y_1} = \frac{7WB^2}{24}$
	$I_p = \frac{WB^2}{12}$	$I_{p_1} = \frac{5WB^2}{12}$
Area Moment of Inertia	$I_x = \frac{BH^3}{36}$	$I_{x_1} = \frac{BH^3}{12}$
	$I_{x_2} = \frac{BH^3}{4}$	
	$I_y = \frac{HB^3}{48}$	$I_{y_1} = \frac{7B^3H}{48}$
	$I_p = \frac{B^3H}{24}$	$I_{p_1} = \frac{5B^3H}{24}$
Radius of Gyration	$\rho_x = 0.236H$	$\rho_{x_1} = \frac{H}{\sqrt{6}}$
	$\rho_{x_2} = 0.707H$	
	$\rho_y = 0.204B$	$\rho_{y_1} = 0.540B$
	$\rho_p = \frac{B}{2\sqrt{3}}$	$\rho_{p_1} = 0.646B$

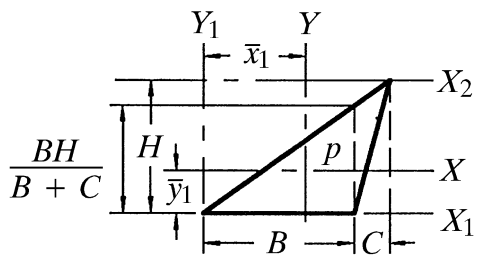
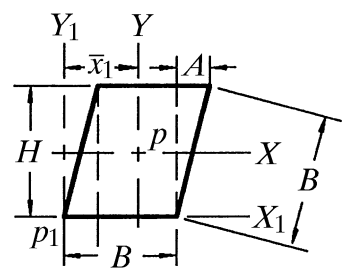
3.1 Plane Surfaces (Continued)

FIGURE 3–1 PLANE SURFACES (CONTINUED)

	ISOSCELES TRIANGLE	OBLIQUE TRIANGLE
	 11	 12
Area	$\frac{BH}{2}$	$\frac{BH}{2}$
Centroid	$\bar{x}_1 = \frac{B}{2}$ $\bar{y}_1 = \frac{H}{3}$	$\bar{x}_1 = \frac{2B - C}{3}$ $\bar{y}_1 = \frac{H}{3}$
Weight Moment of Inertia	$I_x = \frac{WH^2}{18}$ $I_{x_1} = \frac{WH^2}{6}$ $I_{x_2} = \frac{WH^2}{2}$ $I_y = \frac{WB^2}{24}$ $I_{y_1} = \frac{7WB^2}{24}$ $I_p = \frac{W(4H^2 + 3B^2)}{72}$ $I_{p_1} = \frac{W(7B^2 + 4H^2)}{24}$	$I_x = \frac{WH^2}{18}$ $I_{x_1} = \frac{WH^2}{6}$ $I_{x_2} = \frac{WH^2}{2}$ $I_y = \frac{W}{18}(B^2 + C^2 - BC)$ $I_p = \frac{W}{18}(H^2 + B^2 + C^2 - BC)$
Area Moment of Inertia	$I_x = \frac{BH^3}{36}$ $I_{x_1} = \frac{BH^3}{12}$ $I_{x_2} = \frac{BH^3}{4}$ $I_y = \frac{B^3H}{48}$ $I_{y_1} = \frac{7B^3H}{48}$ $I_p = \frac{4BH^3 + 3B^3H}{144}$ $I_{p_1} = \frac{4BH^3 + 7B^3H}{48}$	$I_x = \frac{BH^3}{36}$ $I_{x_1} = \frac{BH^3}{12}$ $I_{x_2} = \frac{BH^3}{4}$ $I_y = \frac{BH}{36}(B^2 + C^2 - BC)$ $I_{y_1} = \frac{BH}{36}\left[B^2 + C^2 - BC + 2(2B - C)^2\right]$ $I_p = \frac{BH}{36}(H^2 + B^2 + C^2 - BC)$
Radius of Gyration	$\rho_x = 0.236H$ $\rho_{x_1} = \frac{H}{\sqrt{6}}$ $\rho_{x_2} = 0.707H$ $\rho_y = 0.204B$ $\rho_{y_1} = 0.540B$ $\rho_p = 0.118\sqrt{4H^2 + 3B^2}$ $\rho_{p_1} = 0.204\sqrt{4H^2 + 7B^2}$	$\rho_x = 0.236H$ $\rho_{x_1} = \frac{H}{\sqrt{6}}$ $\rho_{x_2} = 0.707H$ $\rho_y = 0.236\sqrt{B^2 + C^2 - BC}$ $\rho_p = 0.236\sqrt{H^2 + B^2 + C^2 - BC}$

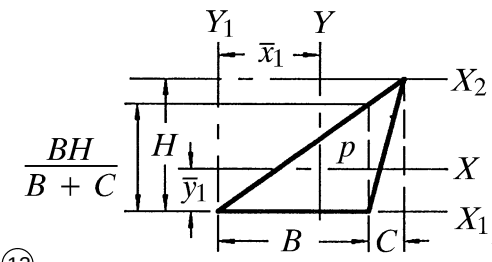
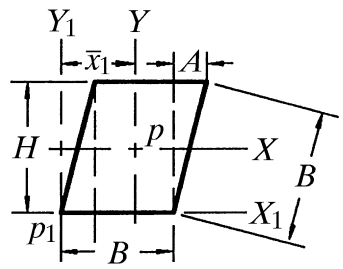
3.1 Plane Surfaces (Continued)

FIGURE 3–1 PLANE SURFACES (CONTINUED)

OBTUSE ANGLED TRIANGLE	RHOMBUS
 13	 14
Area	$\frac{BH}{2}$	BH
Centroid	$\bar{x}_1 = \frac{2B + C}{3}$ $\bar{y}_1 = \frac{H}{3}$	$\bar{x} = \frac{A + B}{2}$ $\bar{y} = \frac{H}{2}$
Weight Moment of Inertia	$I_x = \frac{WH^2}{18}$ $I_{x_1} = \frac{WH^2}{6}$ $I_{x_2} = \frac{WH^2}{2}$ $I_y = \frac{W}{18}(B^2 + BC + C^2)$ $I_{y_1} = \frac{W(3B^2 + 3BC + C^2)}{6}$ $I_p = \frac{W(H^2 + B^2 + BC + C^2)}{18}$	$I_x = \frac{WH^2}{12}$ $I_{x_1} = \frac{WH^2}{3}$ $I_y = \frac{W(A^2 + B^2)}{12}$ $I_{y_1} = \frac{W(2A^2 + 2B^2 + 3AB)}{6}$ $I_p = \frac{WB^2}{6}$ $I_{p_1} = \frac{WB(3A + 4B)}{6}$
Area Moment of Inertia	$I_x = \frac{BH^3}{36}$ $I_{x_1} = \frac{BH^3}{12}$ $I_{x_2} = \frac{BH^3}{4}$ $I_y = \frac{BH}{36}(B^2 + BC + C^2)$ $I_{y_1} = \frac{BH}{12}(3B^2 + 3BC + C^2)$ $I_p = \frac{BH}{36}(H^2 + B^2 + BC + C^2)$	$I_x = \frac{BH^3}{12}$ $I_{x_1} = \frac{BH^3}{3}$ $I_y = \frac{BH(A^2 + B^2)}{12}$ $I_{y_1} = \frac{BH(2A^2 + 2B^2 + 3AB)}{6}$ $I_p = \frac{B^3H}{6}$ $I_p = \frac{B^2H(3A + 4B)}{6}$

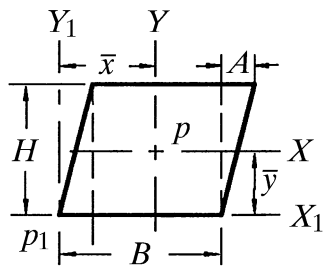
3.1 Plane Surfaces (Continued)

FIGURE 3–1 PLANE SURFACES (CONTINUED)

OBTUSE ANGLED TRIANGLE (CONTINUED)	RHOMBUS (CONTINUED)
 13	 14
Radius of Gyration	$\rho_x = 0.236H$ $\rho_{x_1} = \frac{H}{\sqrt{6}}$ $\rho_{x_2} = 0.707H$ $\rho_y = 0.236\sqrt{B^2 + BC + C^2}$ $\rho_{y_1} = \sqrt{\frac{3B^2 + 3BC + C^2}{6}}$ $\rho_p = 0.236\sqrt{H^2 + B^2 + BC + C^2}$
	$\rho_x = \frac{H}{2\sqrt{3}}$ $\rho_{x_1} = \frac{H}{\sqrt{3}}$ $\rho_y = \frac{1}{2}\sqrt{\frac{(A^2 + B^2)}{3}}$ $\rho_{y_1} = \sqrt{\frac{2A^2 + 2B^2 + 3AB}{6}}$ $\rho_p = \frac{B}{\sqrt{6}}$ $\rho_{p_1} = \sqrt{\frac{B(3A + 4B)}{6}}$

3.1 Plane Surfaces (Continued)

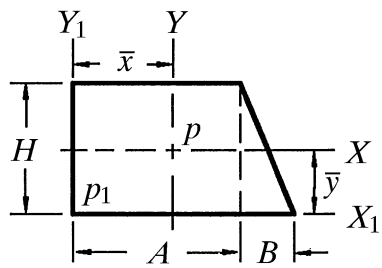
FIGURE 3–1 PLANE SURFACES (CONTINUED)

PARALLELOGRAM  (15)		
Area	BH	
Centroid	$\bar{x} = \frac{A+B}{2}$	$\bar{y} = \frac{H}{2}$
Weight Moment of Inertia	$I_x = \frac{WH^2}{12}$ $I_y = \frac{W}{12}(A^2 + B^2)$ $I_p = \frac{W}{12}(A^2 + B^2 + H^2)$ $I_{p_1} = \frac{W(2A^2 + 2B^2 + 3AB + 2H^2)}{6}$	$I_{x_1} = \frac{WH^2}{3}$ $I_{y_1} = \frac{W}{6}(2A^2 + 2B^2 + 3AB)$
Area Moment of Inertia	$I_x = \frac{BH^3}{12}$ $I_y = \frac{BH(A^2 + B^2)}{12}$ $I_p = \frac{BH}{12}(A^2 + B^2 + H^2)$	$I_{x_1} = \frac{BH^3}{3}$ $I_{y_1} = \frac{BH}{6}(2A^2 + 2B^2 + 3AB)$ $I_{p_1} = \frac{BH}{6}(2A^2 + 2B^2 + 3AB + 2H^2)$
Radius of Gyration	$\rho_x = \frac{H}{2\sqrt{3}}$ $\rho_y = \frac{1}{2}\sqrt{\frac{A^2 + B^2}{3}}$ $\rho_p = \frac{1}{2}\sqrt{\frac{A^2 + B^2 + H^2}{3}}$	$\rho_{x_1} = \frac{H}{\sqrt{3}}$ $\rho_{y_1} = \sqrt{\frac{2A^2 + 2B^2 + 3AB}{6}}$ $\rho_{p_1} = \sqrt{\frac{2A^2 + 2B^2 + 3AB + 2H^2}{6}}$

3.1 Plane Surfaces (Continued)

FIGURE 3–1 PLANE SURFACES (CONTINUED)

RIGHT ANGLE TRAPEZOID

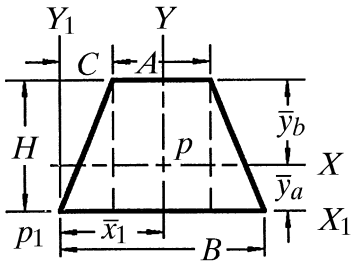


(16)

Area	$\frac{H}{2}(2A + B)$	
Centroid	$\bar{x} = \frac{3A^2 + 3AB + B^2}{3(2A + B)}$	$\bar{y} = \frac{H(3A + B)}{3(2A + B)}$
Weight Moment of Inertia	$I_x = \frac{WH^2(6A^2 + 6AB + B^2)}{18(2A + B)^2}$ $I_y = I_{y_1} - W\bar{x}^2$ $I_p = I_x + I_y$	$I_{x_1} = \frac{WH^2(4A + B)}{6(2A + B)}$ $I_{y_1} = W\frac{[2A^2(2A + 3B) + B^2(4A + B)]}{6(2A + B)}$ $I_{p_1} = I_{x_1} + I_{y_1}$
Area Moment of Inertia	$I_x = \frac{H^3(6A^2 + 6AB + B^2)}{36(2A + B)}$ $I_y = I_{y_1} - \frac{H(3A^2 + 3AB + B^2)^2}{18(2A + B)}$ $I_p = I_x + I_y$	$I_{x_1} = \frac{H^3(4A + B)}{12}$ $I_{y_1} = \frac{H[2A^2(2A + 3B) + B^2(4A + B)]}{12}$ $I_{p_1} = I_{x_1} + I_{y_1}$
Radius of Gyration	$\rho_x = \frac{0.236H\sqrt{6A^2 + 6AB + B^2}}{2A + B}$ $\rho_y = \sqrt{\frac{2I_y}{H(2A + B)}}$ $\rho_p = \sqrt{\frac{2I_p}{H(2A + B)}}$	$\rho_{x_1} = H\sqrt{\frac{4A + B}{6(2A + B)}}$ $\rho_{y_1} = \sqrt{\frac{2A^2(2A + 3B) + B^2(4A + B)}{6(2A + B)}}$ $\rho_{p_1} = \sqrt{\frac{2I_{p_1}}{H(2A + B)}}$

3.1 Plane Surfaces (Continued)

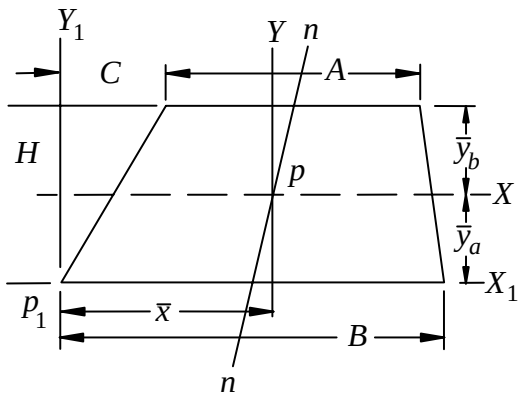
FIGURE 3–1 PLANE SURFACES (CONTINUED)

ISOSCELES TRAPEZOID  17		
Area	$\frac{H(A+B)}{2}$	
Centroid	$\bar{x} = \frac{B}{2}$ $\bar{y}_a = \frac{H(B+2A)}{3(B+A)}$	$\bar{y}_b = \frac{H(A+2B)}{3(A+B)}$
Weight Moment of Inertia	$I_x = \frac{WH^2}{18}\left(1 + \frac{2AB}{(A+B)^2}\right)$ $I_y = \frac{W}{24}(A^2 + B^2)$ $I_p = I_x + I_y$	$I_{x_1} = \frac{WH^2(2A+C)}{6(A+C)}$ $I_{y_1} = \frac{W}{24}(A^2 + 7B^2)$ $I_{p_1} = I_{x_1} + I_{y_1}$
Area Moment of Inertia	$I_x = \frac{H^3(A^2 + 4AB + B^2)}{36(A+B)}$ $I_y = \frac{H(A+B)(A^2+B^2)}{48}$ $I_p = I_x + I_y$	$I_{x_1} = \frac{H^3(A+B)(2A+C)}{12(A+C)}$ $I_{y_1} = \frac{H(A+B)(A^2+7B^2)}{48}$ $I_{p_1} = I_{x_1} + I_{y_1}$
Radius of Gyration	$\rho_x = H\frac{\sqrt{2(B^2 + 4AB + A^2)}}{6(A+B)}$ $\rho_y = \sqrt{\frac{A^2+B^2}{24}}$ $\rho_p = \sqrt{\frac{2I_p}{H(A+B)}}$	$\rho_{x_1} = \sqrt{\frac{H^2(2A+C)}{6(A+C)}}$ $\rho_{y_1} = \sqrt{\frac{A^2+7b^2}{24}}$ $\rho_{p_1} = \sqrt{\frac{2I_{p_1}}{H(A+B)}}$

3.1 Plane Surfaces (Continued)

FIGURE 3–1 PLANE SURFACES (CONTINUED)

OBLIQUE TRAPEZOID



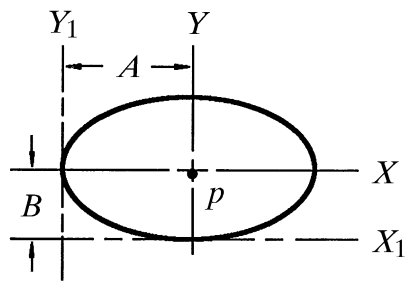
The centroid is on line n–n connecting midpoints of sides A and B and is located at the intersection of line n–n and the Y axis.

(18)

Area	$\dfrac{H(A+B)}{2}$
Centroid	$\bar{x} = \dfrac{1}{3}\left[A + B + C - \dfrac{A(B-C)}{(A+B)}\right]$ $\quad \bar{y}_a = \dfrac{H(B+2A)}{3(A+B)}$ $\quad \bar{y}_b = \dfrac{H(2B+A)}{3(A+B)}$
Weight Moment of Inertia	$I_x = \dfrac{WH^2}{18}\left(1 + \dfrac{2AB}{(A+B)^2}\right)$ $I_{x_1} = \dfrac{WH^2}{6}\left(\dfrac{3A+B}{A+B}\right)$
Area Moment of Inertia	$I_x = \dfrac{H^3(A^2 + 4AB + B^2)}{36(A+B)}$ $I_{x_1} = \dfrac{H^3(B+3A)}{12}$
Radius of Gyration	$\rho_x = H\dfrac{\sqrt{2(A^2 + 4AB + B^2)}}{6(A+B)} = H\sqrt{2(A^2 + 4AB + B^2)}$ $\rho_{x_1} = H\sqrt{\dfrac{3A+B}{6(A+B)}}$

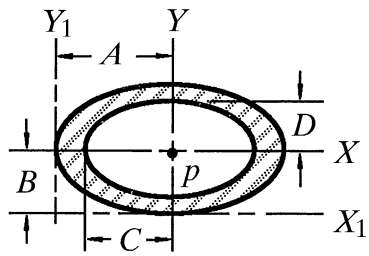
3.1 Plane Surfaces (Continued)

FIGURE 3–1 PLANE SURFACES (CONTINUED)

ELLIPSE  (19)		
Area	πAB	
Centroid	At point p	
Weight Moment of Inertia	$I_x = \frac{WB^2}{4}$ $I_y = \frac{WA^2}{4}$ $I_p = \frac{W(A^2 + B^2)}{4}$	$I_{x_1} = 1.25WB^2$ $I_{y_1} = 1.25WA^2$
Area Moment of Inertia	$I_x = \frac{\pi AB^3}{4}$ $I_y = \frac{\pi A^3 B}{4}$ $I_p = \frac{\pi AB(A^2 + B^2)}{4}$	$I_{x_1} = 1.25\pi AB^3$ $I_{y_1} = 1.25\pi A^3 B$
Radius of Gyration	$\rho_x = \frac{B}{2}$ $\rho_y = \frac{A}{2}$ $\rho_p = \sqrt{\frac{A^2 + B^2}{2}}$	$\rho_{x_1} = B\sqrt{\frac{5}{4}}$ $\rho_{y_1} = A\sqrt{\frac{5}{4}}$

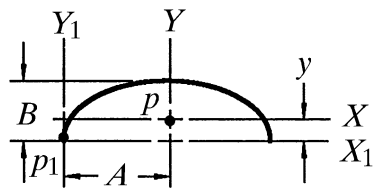
3.1 Plane Surfaces (Continued)

FIGURE 3–1 PLANE SURFACES (CONTINUED)

HOLLOW ELLIPSE  20		
Area	$\pi(AB - CD)$	
Centroid	At point p	
Weight Moment of Inertia	$I_x = \frac{W}{4}\left(\frac{AB^3 - CD^3}{AB - CD}\right)$ $I_y = \frac{W}{4}\left(\frac{A^3B - C^3D}{AB - CD}\right)$ $I_p = I_x + I_y$	$I_{x_1} = \frac{W}{4}\left(\frac{AB^3 - CD^3}{AB - CD}\right) + WB^2$ $I_{y_1} = \frac{W}{4}\left(\frac{A^3B - C^3D}{AB - CD}\right) + WA^2$
Area Moment of Inertia	$I_x = \frac{\pi}{4}(AB^3 - CD^3)$ $I_y = \frac{\pi}{4}(A^3B - C^3D)$ $I_p = I_x + I_y$	$I_{x_1} = \frac{\pi}{4}(AB^3 - CD^3) - \pi B^2(AB - CD)$ $I_{y_1} = \frac{\pi}{4}(A^3B - C^3D) + \pi A^2(AB - CD)$
Radius of Gyration	$\rho_x = \sqrt{\frac{AB^3 - CD^3}{4(AB - CD)}}$ $\rho_y = \sqrt{\frac{A^3B - C^3D}{4(AB - CD)}}$ $\rho_p = \sqrt{\frac{I_p}{\pi(AB - CD)}}$	$\rho_{x_1} = \sqrt{\frac{I_{x_1}}{\pi(AB - CD)}}$ $\rho_{y_1} = \sqrt{\frac{I_{y_1}}{\pi(AB - CD)}}$

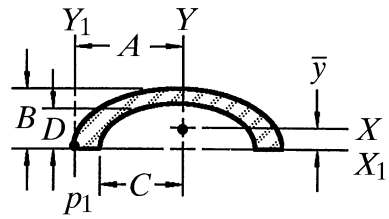
3.1 Plane Surfaces (Continued)

FIGURE 3–1 PLANE SURFACES (CONTINUED)

SEMI–ELLIPSE		
(21) 		
Area	$\frac{\pi AB}{2}$	
Centroid	$y = \frac{4B}{3\pi}$	
Weight Moment of Inertia	$I_x = \left(\frac{1}{4} - \frac{16}{9\pi^2}\right)WB^2$	$I_{x_1} = 0.25WB^2$
	$I_y = 0.25WA^2$	$I_{y_1} = 1.25WA^2$
	$I_p = \frac{W(A^2 + 0.2795B^2)}{4}$	$I_{p_1} = \frac{W(5A^2 + B^2)}{4}$
Area Moment of Inertia	$I_x = AB^3\left(\frac{\pi}{8} - \frac{8}{9\pi}\right)$	$I_{x_1} = \frac{\pi}{8}AB^3$
	$I_y = 0.3927A^3B$	$I_{y_1} = \frac{5}{8}\pi A^3B$
	$I_p = AB(0.3927A^2 + 0.1098B^2)$	$I_{p_1} = AB(1.9635A^2 + 0.3927B^2)$
Radius of Gyration	$\rho_x = B\sqrt{\frac{1}{4} - \frac{16}{9\pi^2}}$	$\rho_{x_1} = \frac{B}{2}$
	$\rho_y = \frac{A}{2}$	$\rho_{y_1} = \frac{A\sqrt{5}}{2}$
	$\rho_p = \frac{1}{2}\sqrt{A^2 + B^2\left(1 - \frac{64}{9\pi^2}\right)}$	$\rho_{p_1} = \sqrt{\frac{2I_{p_1}}{\pi AB}}$

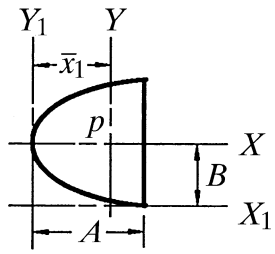
3.1 Plane Surfaces (Continued)

FIGURE 3–1 PLANE SURFACES (CONTINUED)

HOLLOW SEMIELLIPSE  22	
Area	$\pi\dfrac{(AB - CD)}{2}$
Centroid	$\bar{y} = \dfrac{4}{3\pi}\left[\dfrac{AB^2 - CD^2}{AB - CD}\right]$
Weight Moment of Inertia	$I_x = \dfrac{W}{4}\left(\dfrac{AB^3 - CD^3}{AB - CD}\right) - W\left[\dfrac{4}{3\pi}\left(\dfrac{AB^2 - CD^2}{AB - CD}\right)\right]^2$ $\quad I_{x_1} = \dfrac{W}{4}\left(\dfrac{AB^3 - CD^3}{AB - CD}\right)$ $I_y = \dfrac{W}{4}\left(\dfrac{A^3B - C^3D}{AB - CD}\right)$ $\quad I_{y_1} = \dfrac{W}{4}\left(\dfrac{A^3B - C^3D}{AB - CD}\right) + WA^2$ $I_p = I_x + I_y$
Area Moment of Inertia	$I_x = \dfrac{\pi}{8}(AB^3 - CD^3) - \left(\dfrac{8}{9\pi}\right)\dfrac{(AB^2 - CD^2)^2}{AB - CD}$ $\quad I_{x_1} = \dfrac{\pi}{8}(AB^3 - CD^3)$ $I_y = \dfrac{\pi}{8}(A^3B - C^3D)$ $\quad I_{y_1} = \dfrac{\pi}{8}(A^3B - C^3D) + \dfrac{\pi A^2(AB - CD)}{2}$ $I_p = I_x + I_y$ $\quad I_{p_1} = I_{x_1} + I_{y_1}$
Radius of Gyration	$\rho_x = \sqrt{\dfrac{2I_x}{\pi(AB - CD)}}$ $\quad \rho_{x_1} = \sqrt{\dfrac{AB^3 - CD^3}{4(AB - CD)}}$ $\rho_y = \sqrt{\dfrac{2I_y}{\pi(AB - CD)}}$ $\quad \rho_{y_1} = \sqrt{\dfrac{2I_{y_1}}{\pi(AB - CD)}}$ $\rho_p = \sqrt{\dfrac{2I_p}{\pi(AB - CD)}}$ $\quad \rho_{p_1} = \sqrt{\dfrac{2I_{p_1}}{\pi(AB - CD)}}$

3.1 Plane Surfaces (Continued)

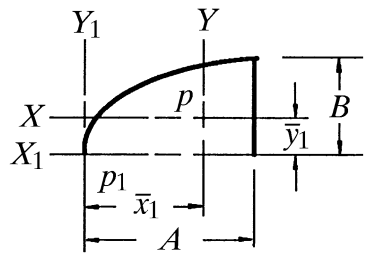
FIGURE 3–1 PLANE SURFACES (CONTINUED)

PARABOLIC SEGMENT (23) 		
Area	$\frac{4}{3}AB$	
Centroid	$\bar{x}_1 = 0.6A$	
Weight Moment of Inertia	$I_x = 0.2WB^2$ $I_y = \frac{12}{175}WA^2$ $I_p = I_x + I_y$	$I_{x_1} = 1.2WB^2 = 6I_x$ $I_{y_1} = \frac{3}{7}WA^2 = 6.25I_y$
Area Moment of Inertia	$I_x = \left(\frac{4}{15}\right)AB^3$ $I_y = \frac{16A^3B}{175}$ $I_p = I_x + I_y$	$I_{x_1} = 1.6AB^3 = 6I_x$ $I_{y_1} = \left(\frac{4}{7}\right)A^3B = 6.25I_y$
Radius of Gyration	$\rho_y = \frac{B}{\sqrt{5}}$ $\rho_y = A\sqrt{\frac{12}{175}}$ $\rho_p = \sqrt{\frac{3I_p}{4AB}}$	$\rho_{x_1} = B\sqrt{\frac{6}{5}}$ $\rho_{y_1} = A\sqrt{\frac{3}{7}}$

3.1 Plane Surfaces (Continued)

FIGURE 3–1 PLANE SURFACES (CONTINUED)

PARABOLIC HALF–SEGMENT



(24)

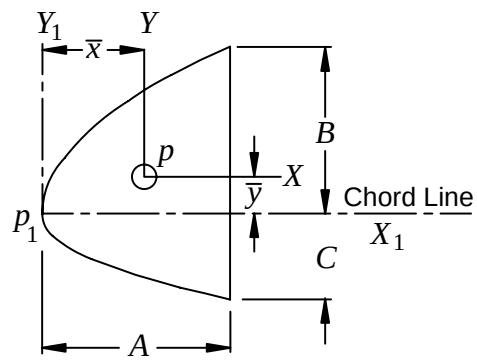
Area	$\frac{2AB}{3}$	
Centroid	$\bar{x}_1 = 0.6A$	$\bar{y}_1 = 0.375B$
Weight Moment of Inertia	$I_x = 0.0594WB^2$ $I_y = 0.0686WA^2$ $I_p = I_x + I_y$	$I_{x_1} = 0.2WB^2 = 3.37I_x$ $I_{y_1} = 0.4286WA^2 = 6.25I_y$ $I_{p_1} = I_{x_1} + I_{y_1}$
Area Moment of Inertia	$I_x = \left(\frac{19}{480}\right)AB^3$ $I_y = \left(\frac{8}{175}\right)A^3B$ $I_p = I_x + I_y$	$I_{x_1} = \frac{2}{15}AB^3 = 3.37I_x$ $I_{y_1} = \frac{2}{7}A^3B = 6.25I_y$ $I_{p_1} = I_{x_1} + I_{y_1}$
Radius of Gyration	$\rho_x = B\sqrt{\frac{19}{320}}$ $\rho_y = A\sqrt{\frac{12}{175}}$ $\rho_p = \sqrt{\frac{3I_p}{2AB}}$	$\rho_{x_1} = \frac{B}{\sqrt{5}}$ $\rho_{y_1} = A\sqrt{\frac{3}{7}}$ $\rho_{p_1} = \sqrt{\frac{3I_p}{2AB}}$

3.1 Plane Surfaces (Continued)

FIGURE 3–1 PLANE SURFACES (CONTINUED)

NOSE RIB

Based on a combination of two parabolic half segments with B>C

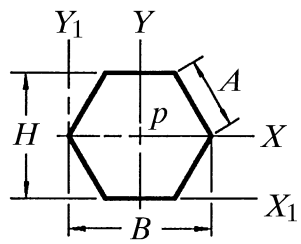
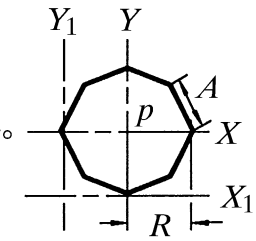


(25)

Area	$\frac{2}{3}A(B + C)$	
Centroid	$\bar{x} = 0.6A$	$\bar{y} = 0.375(B - C)$
Weight Moment of Inertia	$I_x = \frac{W}{320}(19B^2 + 26BC + 19C^2)$ $I_y = 0.0686WA^2$ $I_p = I_x + I_y$	$I_{x_1} = \frac{W}{5}(B^2 - BC + C^2)$ $I_{y_1} = 0.4286WA^2 = 6.25I_y$ $I_{p_1} = I_{x_1} + I_{y_1}$
Area Moment of Inertia	$I_x = \frac{A(B + C)}{480}(19B^2 + 26BC + 19C^2)$ $I_y = \left(\frac{8}{175}\right)A^3(B + C)$ $I_p = I_x + I_y$	$I_{x_1} = \left(\frac{2}{15}\right)A(B^3 + C^3)$ $I_{y_1} = \left(\frac{2}{7}\right)A^3(B + C) = 6.25I_y$ $I_{p_1} = I_{x_1} + I_{y_1}$
Radius of Gyration	$\rho_x = \sqrt{\frac{3I_x}{2A(B + C)}} = \sqrt{\frac{19B^2 + 26BC + 19C^2}{320}}$ $\rho_y = \sqrt{\frac{3I_y}{2A(B + C)}} = 2A\sqrt{\frac{3}{175}}$	$\rho_{x_1} = \sqrt{\frac{B^2 - BC + C^2}{5}}$

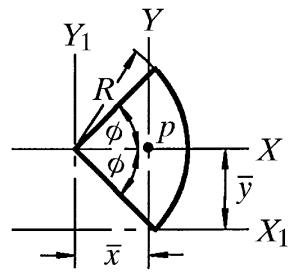
3.1 Plane Surfaces (Continued)

FIGURE 3–1 PLANE SURFACES (CONTINUED)

REGULAR HEXAGON		REGULAR OCTAGON
$B = 2A$ and $H = A\sqrt{3}$  (26)		$A = 2R \cos 22.5°$  (27)
Area	$0.866H^2 = 1.5\sqrt{3}A^2 = 0.375\sqrt{3}B^2$	$2.8286R^2$
Centroid	At point p	At point p
Weight Moment of Inertia	$I_x = I_y = 0.0694WH^2 = 0.208WA^2 = 0.0521WB^2$ $I_{x_1} = 0.3194WH^2 = 0.0521WB^2$ $I_p = 0.1388WH^2 = 0.416WA^2 = 0.1042WB^2$ $I_{y_1} = 1.208WA^2 = 0.4028WH^2 = 0.3021WB^2$	$I_x = I_y = 0.2256WR^2$ $I_{x_1} = I_{y_1} = 0.2256WR^2$ $I_p = 0.4512WR^2$
Area Moment of Inertia	$I_x = I_y = 0.0601H^4$ $I_{x_1} = 0.2766H^4$ $I_{y_1} = 0.3488H^4$ $I_p = 0.1203H^4$	$I_x = I_y = 0.6381R^4$ $I_{x_1} = I_{y_1} = 3.4667R^4$ $I_p = 1.2763R^4$
Radius of Gyration	$\rho_x = \rho_y = 0.2635H = 0.4564A = 0.2282B$ $\rho_{x_1} = 0.5651H$ $\rho_{y_1} = 0.6346H$ $\rho_p = 0.3727H$	$\rho_x = \rho_y = 0.4748R$ $\rho_{x_1} = \rho_{y_1} = 1.1072R$ $\rho_p = 0.6717R$

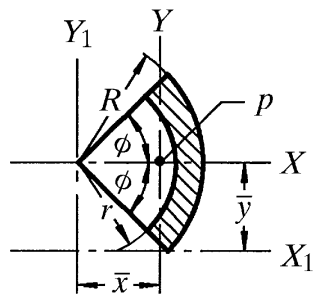
3.1 Plane Surfaces (Continued)

FIGURE 3–1 PLANE SURFACES (CONTINUED)

CIRCULAR SECTOR ϕ in radians 28		
Area	$R^2\phi$	
Centroid	$\bar{x} = \frac{2}{3}\left[\frac{R\sin\phi}{\phi}\right]$	$\bar{y} = R\sin\phi$
Weight Moment of Inertia	$I_x = \frac{WR^2}{4\phi}(\phi - \sin\phi\cos\phi)$ $I_y = \frac{WR^2}{4\phi}\left(\phi - \frac{16\sin^2\phi}{9\phi} + \frac{\sin 2\phi}{2}\right)$ $I_p = I_x + I_y$	$I_{x_1} = I_x + WR^2\sin^2\phi$ $I_{y_1} = \frac{WR^2}{4\phi}(\phi + \sin\phi\cos\phi)$
Area Moment of Inertia	$I_x = \frac{R^4}{4}(\phi - \sin\phi\cos\phi)$ $I_y = \frac{R^4}{4}\left(\phi - \frac{16\sin^2\phi}{9\phi} + \frac{\sin 2\phi}{2}\right)$ $I_p = \frac{R^4}{4}\left(2\phi - \frac{16\sin^2\phi}{9\phi}\right)$	$I_{x_1} = \frac{R^4}{4}(\phi - \sin\phi\cos\phi) + R^4\phi\sin^2\phi$ $I_{y_1} = \frac{R^4}{4}(\phi + \sin\phi\cos\phi)$
Radius of Gyration	$\rho_x = \frac{R}{2}\sqrt{1 - \frac{\sin\phi\cos\phi}{\phi}}$ $\rho_y = \frac{R}{2}\sqrt{1 + \frac{\sin\phi\cos\phi}{\phi} - \frac{16\sin^2\phi}{9\phi^2}}$ $\rho_p = \frac{R}{2}\sqrt{2 - \frac{16\sin^2\phi}{9\phi^2}}$	$\rho_{x_1} = \sqrt{\frac{I_{x_1}}{R^2\phi}}$ $\rho_{y_1} = \frac{R}{2}\sqrt{1 + \frac{\sin\phi\cos\phi}{\phi}}$

3.1 Plane Surfaces (Continued)

FIGURE 3–1 PLANE SURFACES (CONTINUED)

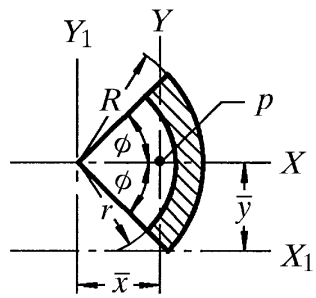
HOLLOW CIRCULAR SECTOR ϕ in radians  29	
Area	$(R^2 - r^2)\phi$
Centroid	$\bar{x} = \dfrac{2\sin\phi(R^3 - r^3)}{3\phi(R^2 - r^2)}$ $\qquad$ $\bar{y} = R\sin\phi$
Weight Moment of Inertia	$I_x = \dfrac{W}{4}(R^2 + r^2)\left(1 - \dfrac{\sin\phi\cos\phi}{\phi}\right)$ $I_y = \dfrac{W}{4}(R^2 + r^2)\left(1 + \dfrac{\sin\phi\cos\phi}{\phi}\right) - W\left[\dfrac{2\sin\phi(R^3 - r^3)}{3\phi(R^2 - r^2)}\right]^2$ $I_p = I_x + I_y$ $I_{x_1} = \dfrac{W}{4}(R^2 + r^2)\left(1 - \dfrac{\sin\phi\cos\phi}{\phi}\right) + WR^2\sin^2\phi$ $I_{y_1} = \dfrac{W}{4}(R^2 + r^2)\left(1 + \dfrac{\sin\phi\cos\phi}{\phi}\right)$
Area Moment of Inertia	$I_x = \dfrac{\phi}{4}(R^4 - r^4)\left(1 - \dfrac{\sin\phi\cos\phi}{\phi}\right)$ $I_y = \dfrac{\phi}{4}(R^4 - r^4)\left(1 + \dfrac{\sin\phi\cos\phi}{\phi}\right) - \dfrac{1}{\phi(R^2 - r^2)}\left[\dfrac{2\sin\phi(R^3 - r^3)}{3}\right]^2$ or $I_y = I_{y_1} - \dfrac{1}{\phi(R^2 - r^2)}\left[\dfrac{2\sin\phi(R^3 - r^3)}{3}\right]^2$ $I_{x_1} = \dfrac{\phi}{4}(R^4 - r^4)\left(1 - \dfrac{\sin\phi\cos\phi}{\phi}\right) + \phi(R^4 - r^2R^2)\sin^2\phi$ $I_{y_1} = \dfrac{\phi}{4}(R^4 - r^4)\left(1 + \dfrac{\sin\phi\cos\phi}{\phi}\right)$ $I_p = I_x + I_y$

3.1 Plane Surfaces (Continued)

FIGURE 3–1 PLANE SURFACES (CONTINUED)

HOLLOW CIRCULAR SECTOR (CONTINUED)

ϕ in radians

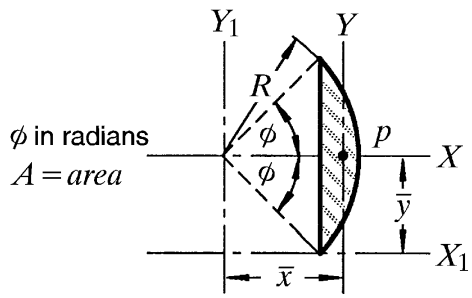


(29)

Radius of Gyration	$\rho_x = \sqrt{\left(\frac{R^2 + r^2}{4}\right)\left(1 - \frac{\sin\phi\cos\phi}{\phi}\right)}$ $\quad \rho_{x_1} = \sqrt{\frac{I_{x_1}}{(R^2 - r^2)\phi}}$ $\rho_y = \sqrt{\frac{I_y}{(R^2 - r^2)\phi}}$ $\quad \rho_{y_1} = \sqrt{\left(\frac{R^2 + r^2}{4}\right)\left(1 + \frac{\sin\phi\cos\phi}{\phi}\right)}$ $\rho_p = \sqrt{\frac{I_p}{(R^2 - r^2)\phi}}$

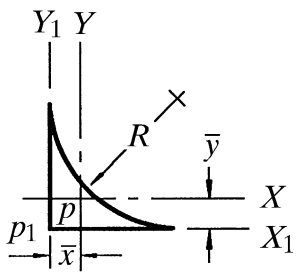
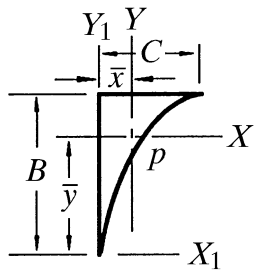
3.1 Plane Surfaces (Continued)

FIGURE 3–1 PLANE SURFACES (CONTINUED)

CIRCULAR SEGMENT  (30)	
Area	$\frac{R^2}{2}(2\phi - \sin 2\phi)$
Centroid	$\bar{x} = \frac{4R\sin^3\phi}{3(2\phi - \sin 2\phi)}$ $\quad$ $\bar{y} = R\sin\phi$
Weight Moment of Inertia	$I_x = \frac{WR^2}{4}\left[1 - \frac{2\sin^3\phi\cos\phi}{3[\phi - \sin\phi\cos\phi]}\right]$ $\quad$ $I_{x_1} = I_x + WR^2\sin^2\phi$ $I_p = I_x + I_y$ $\quad$ $I_{y_1} = \frac{WR^2}{4}\left[1 + \frac{2\sin^3\phi\cos\phi}{\phi - \sin\phi\cos\phi}\right]$ $I_y = I_{y_1} - W\left[\frac{2R\sin^3\phi}{3(\phi - \sin\phi\cos\phi)}\right]^2$
Area Moment of Inertia	$I_x = \frac{AR^2}{4}\left[1 - \frac{2}{3}\left(\frac{\sin^3\phi\cos\phi}{\phi - \sin\phi\cos\phi}\right)\right]$ $\quad$ $I_{x_1} = I_x + \frac{R^4}{2}(2\phi - \sin 2\phi)(\sin^2\phi)$ $I_y = \frac{AR^2}{4}\left[1 + \frac{2\sin^3\phi\cos\phi}{\phi - \sin\phi\cos\phi}\right] - \frac{4R^6\sin^6\phi}{9A}$ $\quad$ $I_{y_1} = \frac{AR^2}{4}\left[1 + \frac{2\sin^3\phi\cos\phi}{\phi - \sin\phi\cos\phi}\right]$ $I_p = I_x + I_y$
Radius of Gyration	$\rho_x = \sqrt{\frac{R^2}{4}\left(1 - \frac{2\sin^3\phi\cos\phi}{3(\phi - \sin\phi\cos\phi)}\right)}$ $\quad$ $\rho_y = \sqrt{\frac{2I_y}{R^2(2\phi - \sin 2\phi)}}$ $\rho_p = \sqrt{\frac{2I_p}{R^2(2\phi - \sin 2\phi)}}$

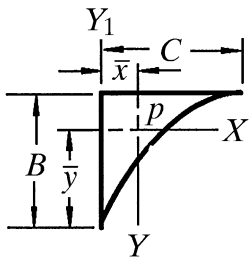
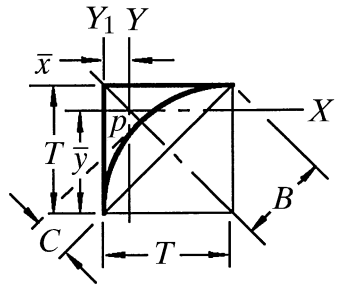
3.1 Plane Surfaces (Continued)

FIGURE 3–1 PLANE SURFACES (CONTINUED)

CIRCULAR COMPLEMENT (90° ARC)		ELLIPTIC COMPLEMENT
 31		 32
Area	$0.2146R^2$	$0.2146BC = \left(1 - \frac{\pi}{4}\right)BC$
Centroid	$\bar{x} = \bar{y} = 0.2233R$	$\bar{x} = 0.2236C$ $\bar{y} = 0.7764B$
Weight Moment of Inertia	$I_x = I_y = 0.035WR^2$ $I_{x_1} = I_{y_1} = 0.085WR^2$ $I_p = 0.070WR^2$ $I_{p_1} = 0.170WR^2$	$I_x = 0.03516B^2W$ $I_y = 0.03516C^2W$ $I_p = 0.03516(B^2 + C^2)W$
Area Moment of Inertia	$I_x = I_y = 0.0075R^4$ $I_{x_1} = I_{y_1} = 0.0182R^4$ $I_p = 0.0150R^4$ $I_{p_1} = 0.0365R^4$	$I_x = 0.007545B^3C$ $I_y = 0.007545BC^3$ $I_p = 0.007545BC(B^2 + C^2)$
Radius of Gyration	$\rho_x = \rho_y = 0.187R$ $\rho_{x_1} = \rho_{y_1} = 0.292R$ $\rho_p = 0.265R$ $\rho_{p_1} = 0.412R$	$\rho_x = 0.18751B$ $\rho_y = 0.18751C$ $\rho_p = \sqrt{0.03516(B^2 + C^2)}$

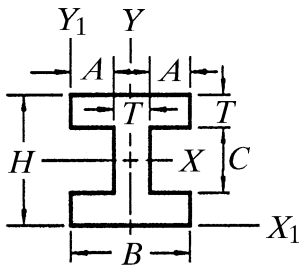
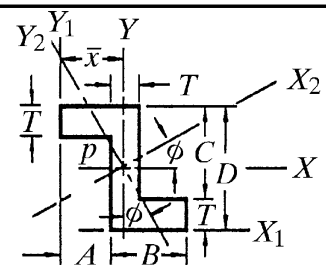
3.1 Plane Surfaces (Continued)

FIGURE 3–1 PLANE SURFACES (CONTINUED)

COMPLEMENT OF HALF PARABOLA		PARABOLIC FILLET IN RIGHT ANGLE
 33		 34
Area	$\frac{BC}{3}$	$\frac{T^2}{6}$
Centroid	$\bar{x} = \frac{1}{4}C$ $\bar{y} = \frac{7}{10}B$	$\bar{x} = \frac{1}{5}T$ $B = \frac{T}{\sqrt{2}}$ $\bar{y} = \frac{4}{5}T$ $C = \frac{T}{2\sqrt{2}}$
Weight Moment of Inertia	$I_x = \frac{37}{700}WB^2$ $I_y = \frac{3}{80}WC^2$ $I_p = \frac{W}{20}\left(\frac{37}{35}B^2 + \frac{3C^2}{4}\right)$	$I_x = I_y = \frac{11}{350}WT^2$ $I_p = \frac{22}{350}WT^2$
Area Moment of Inertia	$I_x = \frac{37}{2100}B^3C$ $I_y = \frac{1}{80}BC^3$ $I_p = \frac{BC}{20}\left(\frac{37}{105}B^2 + \frac{C^2}{4}\right)$	$I_x = I_y = \frac{11}{2100}T^4$ $I_p = \frac{22}{2100}T^4$
Radius of Gyration	$\rho_x = 0.230B$ $\rho_y = 0.194C$ $\rho_p = \sqrt{\frac{3I_p}{BC}}$	$\rho_x = \rho_y = T\sqrt{33/1050}$ $\rho_p = T\sqrt{33/525}$

3.1 Plane Surfaces (Continued)

FIGURE 3–1 PLANE SURFACES (CONTINUED)

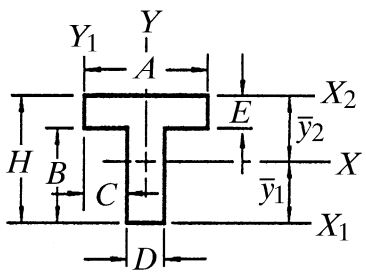
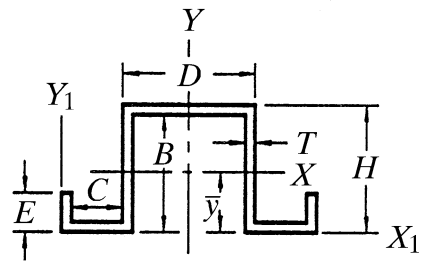
	SYMMETRICAL I SECTION	EQUAL–LEG Z SECTION*
	 35	 36
Area	$BH - 2AC$	$T(D + 2A)$
Centroid	– – –	– – –
Weight Moment of Inertia	$I_x = \frac{W(BH^3 - 2AC^3)}{12(BH - 2AC)}$ $I_{x_1} = \frac{W(BH^3 - 2AC^3)}{12(BH - 2AC)} + \frac{WH^2}{4}$ $I_y = \frac{WT(2B^3 + CT^2)}{12(BH - 2AC)}$	$I_x = \frac{W[BD^3 - A(D - 2T)^3]}{12T(D + 2A)}$ $I_y = \frac{W[D(B + A)^3 - 2A^3C - 6AB^2C]}{12T(D + 2A)}$ $I_{xy} = \frac{I_x - I_y}{2} \tan 2\phi$ $\tan 2\phi = \pm \frac{ABCT}{I_x - I_y}$ $I_{x_2y_2} = \left(\frac{W}{T(D + 2A)}\right)\left[\frac{I_x + I_y}{2} \pm \sqrt{\left(\frac{I_x - I_y}{2}\right)^2 + I_{xy}{}^2}\right]$
Area Moment of Inertia	$I_x = \frac{BH^3 - 2AC^3}{12}$ $I_{x_1} = \frac{2BH^3 - AC(C^2 + 3H^2)}{6}$ $I_y = \frac{T}{12}(2B^3 + CT^2)$	$I_x = \frac{BD^3 - A(D - 2T)^3}{12}$ $I_y = \frac{D(B + A)^3 - 2A^3C - 6AB^2C}{12}$ $I_p = I_x + I_y$ $I_{xy} = -\frac{I_x - I_y}{2} \tan 2\phi$ $\tan 2\phi = \pm \frac{ABCT}{I_x - I_y}$ $I_{x_2y_2} = \frac{I_x + I_y}{2} \pm \sqrt{\left(\frac{I_x - I_y}{2}\right)^2 + I_{xy}^2}$
Radius of Gyration	$\rho_x = \sqrt{\frac{BH^3 - 2AC^3}{12(BH - 2AC)}}$	$\rho_x = \sqrt{\frac{I_x}{T(D + 2A)}}$ $\rho_y = \sqrt{\frac{I_y}{T(D + 2A)}}$

* For equal–leg Z–sections : (a) The formulas are valid only for Z–sections whose web is parallel to one of the major axes. No tilt is allowed. (b) ϕ is measured positive counterclockwise from the positive X–axis.

SECTION PROPERTIES

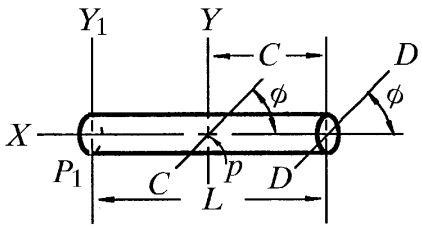
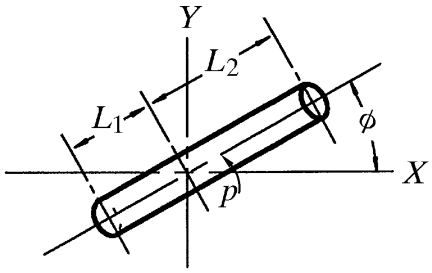
3.1 Plane Surfaces (Continued)

FIGURE 3–1 PLANE SURFACES (CONCLUDED)

SYMMETRICAL T SECTION		SYMMETRICAL HAT SECTION
 37		 38
Area	$AE + BD$	$T(D + 2B + 2C + 2E)$
Centroid	$\bar{y}_2 = \frac{1}{2}\left(\frac{DH^2 + 2CE^2}{DH + 2CE}\right) \quad \bar{y}_1 = H - \bar{y}_2$	$\bar{y} = \frac{2(B^2 + E^2) + T(2C + D) + 2BD}{2[D + 2(B + C + E)]}$
Weight Moment of Inertia	$I_x = \frac{W(\text{AREA MOMENT OF INERTIA, } I_x)}{\text{AREA}}$ $I_y = \frac{W(EA^3 + BD^3)}{12(EA + BD)}$	$I_x = \frac{W(\text{AREA MOMENT OF INERTIA, } I_x)}{\text{AREA}}$ $I_y = \frac{W(\text{AREA MOMENT OF INERTIA, } I_y)}{\text{AREA}}$
Area Moment of Inertia	$I_y = \frac{EA^3 + BD^3}{12}$ $I_{x_2} = \frac{DH^3}{3} + \frac{2CE^3}{3}$ $I_x = I_{x_2} - (AE + BD)\bar{y}_2^2$ $I_{x_1} = I_{x_2} + (AE + BD)(\bar{y}_1^2 - \bar{y}_2^2)$	$I_x = T\left\{\frac{1}{3}[2(E^3 + B^3) + T^2(2C + D)] + BD(B + T) - \bar{y}^2[2(E + B + C) + D]\right\}$ $I_y = \frac{T}{2}\left[\frac{D^3}{6} + \frac{C^3}{3} + C(C + D)^2 + BD(D - 2T) + \frac{T^2}{3}(E + 4B) + E(D + 2C + T)^2\right]$
Radius of Gyration	$\rho_x = \sqrt{\frac{I_x}{AE + BD}}$ $\rho_y = \sqrt{\frac{EA^3 + BD^3}{12(EA + BD)}}$	$\rho_x = \sqrt{\frac{I_x}{T(D + 2B + 2C + 2E)}}$ $\rho_y = \sqrt{\frac{I_y}{T(D + 2B + 2C + 2E)}}$

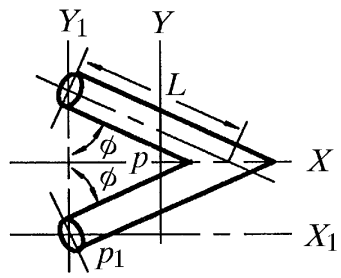
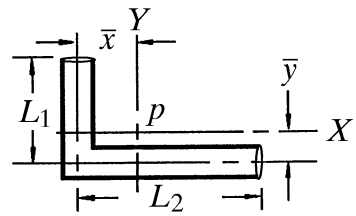
3.2 Thin Rods Formed into Various Shapes

FIGURE 3–2 THIN RODS FORMED INTO VARIOUS SHAPES (CONTINUED ON NEXT PAGE)

	STRAIGHT ROD	INCLINED ROD NOT THROUGH C. G. AXIS
	 ①	 ②
Length	L	$L_1 + L_2$
Weight Moment of Inertia	$I_x =$ Negligible $I_y = I_p = \frac{WL^2}{12}$ $I_{y_1} = I_{p_1} = \frac{WL^2}{3}$ $I_c = \frac{WL^2 \sin^2\phi}{12}$ $I_D = \frac{WL^2 \sin^2\phi}{3}$	$I_x = \frac{W}{3}\left(\sin^2\phi\right)\left(L_1^2 - L_1L_2 + L_2^2\right)$ $I_y = \frac{W}{3}\left(\cos^2\phi\right)\left(L_1^2 - L_1L_2 + L_2^2\right)$ $I_p = I_x + I_y$ NOTE: Assume: Wt. of $L_1 = W_1$ Wt. of $L_2 = W_2$ Then: $W_1 = \frac{L_1}{L_2}W_2,\ W = W_1 + W_2$ $W_1 + W_2 = W_2\left(\frac{L_1}{L_2} + 1\right)$
Radius of Gyration	$\rho_x =$ Negligible $\rho_y = \frac{L}{2\sqrt{3}}$ $\rho_{y_1} = \frac{L}{\sqrt{3}}$ $\rho_c = \frac{L\sin\phi}{2\sqrt{3}}$ $\rho_D = \frac{L\sin\phi}{\sqrt{3}}$	$\rho_x = \sin\phi\sqrt{\frac{L_1^2 - L_1L_2 + L_2^2}{3}}$ $\rho_y = \cos\phi\sqrt{\frac{L_1^2 - L_1L_2 + L_2^2}{3}}$ $\rho_p = \sqrt{\frac{L_1^2 - L_1L_2 + L_2^2}{3}}$

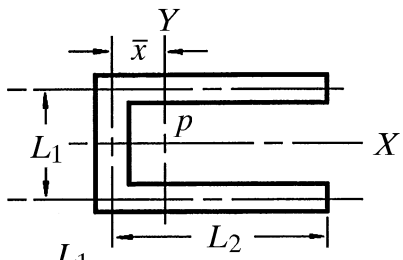
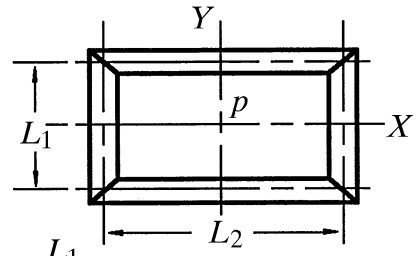
3.2 Thin Rods Formed into Various Shapes (Continued)

FIGURE 3–2 THIN RODS FORMED INTO VARIOUS SHAPES (CONTINUED)

	"V" ROD	"L" ROD
	 ③	 $W_1 + W_2 = W_2\left(\frac{L_1}{L_2} + 1\right) = W$ ④
Length	$2L$	$L_1 + L_2$
Centroid	– – –	$\bar{x} = \frac{L_2^2}{2(L_1 + L_2)}$ $\bar{y} = \frac{L_1^2}{2(L_1 + L_2)}$
Weight Moment of Inertia	$I_x = \frac{WL^2}{3}\cos^2\phi$ $I_{x_1} = \frac{4}{3}WL^2\cos^2\phi$ $I_y = \frac{WL^2}{12}\sin^2\phi$ $I_{y_1} = \frac{WL^2}{3}\sin^2\phi$ $I_p = I_x + I_y$ $I_{p_1} = I_{x_1} + I_{y_1}$	$I_x = \left(\frac{W}{12}\right)\frac{L_1^3(L_1 + 4L_2)}{(L_1 + L_2)^2}$ $I_y = \left(\frac{W}{12}\right)\frac{L_2^3(4L_1 + L_2)}{(L_1 + L_2)}$ $I_p = I_x + I_y$
Radius of Gyration	$\rho_x = \frac{L\cos\phi}{\sqrt{3}}$ $\rho_{x_1} = 1.15L\cos\phi$ $\rho_y = \frac{L\sin\phi}{2\sqrt{3}}$ $\rho_{y_1} = \frac{L\sin\phi}{\sqrt{3}}$ $\rho_p = \frac{L}{2}\sqrt{\frac{4\cos^2\phi + \sin^2\phi}{3}}$ $\rho_{p_1} = L\sqrt{\frac{4\cos^2\phi + \sin^2\phi}{3}}$	$\rho_x = \frac{0.289L_1}{L_1 + L_2}\sqrt{L_1(L_1 + 4L_2)}$ $\rho_y = \frac{0.289L_2}{L_1 + L_2}\sqrt{L_2(4L_1 + L_2)}$ $\rho_p = \sqrt{\frac{I_p}{W}}$

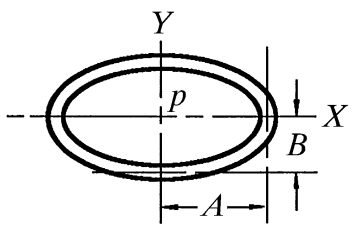
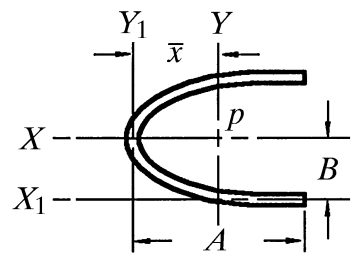
3.2 Thin Rods Formed into Various Shapes (Continued)

FIGURE 3–2 THIN RODS FORMED INTO VARIOUS SHAPES (CONTINUED)

	"U" ROD	RECTANGULAR ROD
	 $W_1 = \frac{L_1}{L_2} W_2$ $W = W_1 + 2W_2 = W_2\left(\frac{L_1}{L_2} + 2\right)$ 5	 $W_1 = \frac{L_1}{L_2} W_2$ $W = 2W_1 + 2W_2 = 2W_2\left(\frac{L_1}{L_2} + 1\right)$ 6
Length	$L_1 + 2L_2$	$2(L_1 + L_2)$
Centroid	$\bar{x} = \frac{L_2^2}{L_1 + 2L_2}$	– – –
Weight Moment of Inertia	$I_x = \frac{WL_1^2(L_1 + 6L_2)}{12(L_1 + 2L_2)}$ $I_y = \frac{WL_2^3(2L_1 + L_2)}{3(L_1 + 2L_2)^2}$ $I_p = I_x + I_y$	$I_x = \frac{WL_1^2(L_1 + 3L_2)}{12(L_1 + L_2)}$ $I_y = \frac{WL_2^2(3L_1 + L_2)}{12(L_1 + L_2)}$ $I_p = I_x + I_y$
Radius of Gyration	$\rho_x = \left(\frac{L_1}{2\sqrt{3}}\right)\sqrt{\frac{L_1 + 6L_2}{L_1 + 2L_2}}$ $\rho_y = \frac{0.577L_2}{L_1 + 2L_2}\sqrt{L_2(2L_1 + L_2)}$ $\rho_p = \sqrt{\frac{I_p}{W}}$	$\rho_x = \left(\frac{L_1}{2\sqrt{3}}\right)\sqrt{\frac{L_1 + 3L_2}{L_1 + L_2}}$ $\rho_y = \left(\frac{L_2}{2\sqrt{3}}\right)\sqrt{\frac{3L_1 + L_2}{L_1 + L_2}}$ $\rho_p = \sqrt{\frac{I_p}{W}}$

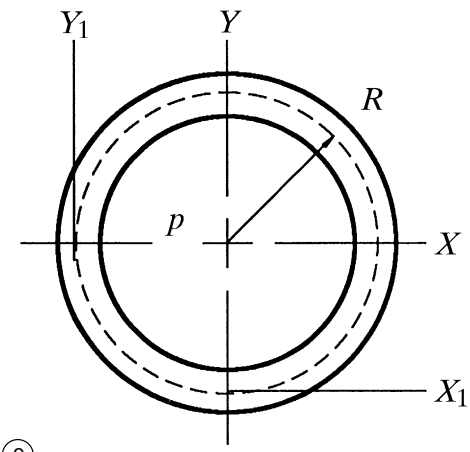
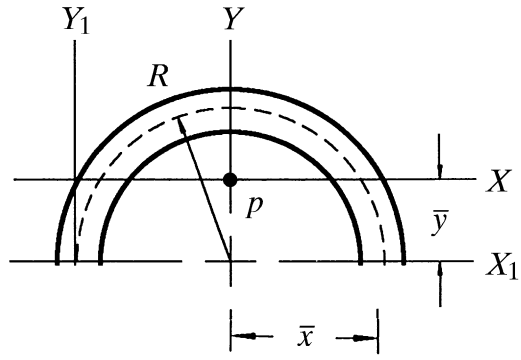
3.2 Thin Rods Formed into Various Shapes (Continued)

FIGURE 3–2 THIN RODS FORMED INTO VARIOUS SHAPES (CONTINUED)

	ELLIPTIC ROD	PARABOLIC ROD
	 $A > B$ (7)	 (8)
Length	$\pi(A+B)\dfrac{64-3R^4}{64-16R^2}$ Where $R=\dfrac{A-B}{A+B}$	$L=\sqrt{4A^2+B^2}+\dfrac{B^2}{2A}\ell n\left(\dfrac{2A+\sqrt{4A^2+B^2}}{B}\right)$
Centroid	– – –	$\bar{x}=\dfrac{\sqrt{(4A^2+B^2)^3}}{8AL}-\dfrac{B^2}{16A}$
Weight Moment of Inertia	$I_x=\dfrac{WB^2(55A^4+10A^2B^2-B^4)}{2(45A^4+22A^2B^2-3B^4)}$ $I_y=\dfrac{WA^2(35A^4+34A^2B^2-5B^4)}{2(45A^4+22A^2B^2-3B^4)}$ $I_p=I_x+I_y$	$I_x=\dfrac{WB^2}{8A^2}\left[\dfrac{\sqrt{(B^2+4A^2)^3}}{L}-\dfrac{B^2}{2}\right]$ $I_{x_1}=\dfrac{WB^2}{8A^2}\left[\dfrac{\sqrt{(B^2+4A^2)^3}}{L}-\dfrac{B^2}{2}\right]+WB^2$ $I_y=\dfrac{W\sqrt{(4A^2+B^2)^3}}{12L}-\dfrac{I_x}{8}-W(\bar{x})^2$ $I_{y_1}=\dfrac{W\sqrt{(4A^2+B^2)^3}}{12L}-\dfrac{I_x}{8}$ $I_p=I_x+I_y$
Radius of Gyration	$\rho_x=\sqrt{\dfrac{I_x}{W}}$ $\rho_y=\sqrt{\dfrac{I_y}{W}}$ $\rho_p=\sqrt{\dfrac{I_p}{W}}$	$\rho_x=\sqrt{\dfrac{I_x}{W}}$ $\rho_{x_1}=\sqrt{\dfrac{I_{x_1}}{W}}$ $\rho_y=\sqrt{\dfrac{I_y}{W}}$ $\rho_{y_1}=\sqrt{\dfrac{I_{y_1}}{W}}$ $\rho_p=\sqrt{\dfrac{I_p}{W}}$

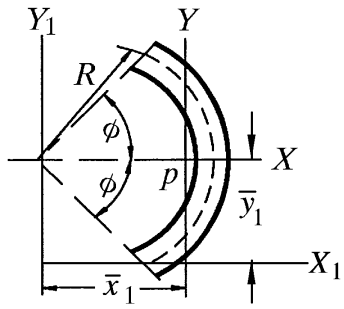
3.2 Thin Rods Formed into Various Shapes (Continued)

FIGURE 3–2 THIN RODS FORMED INTO VARIOUS SHAPES (CONTINUED)

	CIRCULAR ROD	SEMI–CIRCULAR ROD
	 9	 10
Length	$2\pi R$	πR
Centroid	– – –	$\bar{y} = 0.6358R$
Weight Moment of Inertia	$I_x = I_y = \frac{WR^2}{2}$ $I_z = WR^2$ $I_{x_1} = I_{y_1} = \frac{3WR^2}{2}$ $I_p = WR^2$	$I_x = 0.0947WR^2$ $I_{x_1} = \frac{WR^2}{2}$ $I_y = \frac{WR^2}{2}$ $I_{y_1} = \frac{3WR^2}{2}$ $I_p = 0.5947WR^2$ $I_{p_1} = 2WR^2$
Radius of Gyration	$\rho_x = \rho_y = 0.707R$ $\rho_{x_1} = \rho_{y_1} = 1.225R$ $\rho_o = R$	$\rho_x = 0.308R$ $\rho_{x_1} = 0.707R$ $\rho_y = 0.707R$ $\rho_{y_1} = 1.225R$ $\rho_p = 0.771R$ $\rho_{p_1} = 1.414R$

3.2 Thin Rods Formed into Various Shapes (Continued)

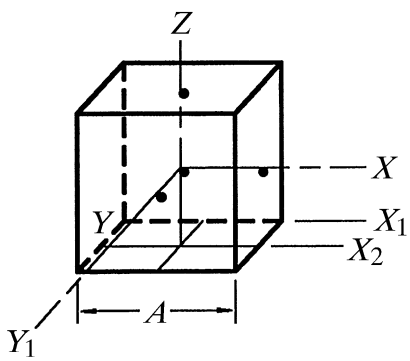
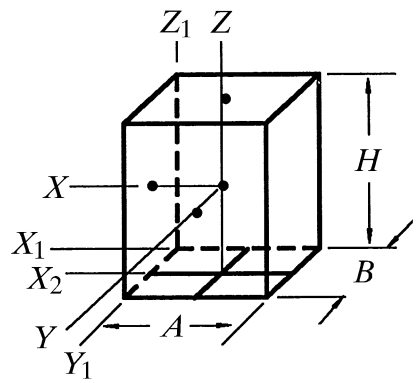
FIGURE 3–2 THIN RODS FORMED INTO VARIOUS SHAPES (CONTINUED)

SEGMENT OF A CIRCULAR ROD ϕ in Radians  (11)	
Length	$2R\phi$
Centroid	$\bar{x}_1 = \dfrac{R\sin\phi}{\phi}$ $\qquad \bar{y}_1 = R\sin\phi$
Weight Moment of Inertia	$I_x = WR^2\left(\dfrac{1}{2} - \dfrac{\sin\phi\cos\phi}{2\phi}\right)$ $I_{x_1} = WR^2\left(\dfrac{1}{2} - \dfrac{\sin\phi\cos\phi}{2\phi} + \sin^2\phi\right)$ $I_y = WR^2\left(\dfrac{1}{2} + \dfrac{\sin\phi\cos\phi}{2\phi} - \dfrac{\sin^2\phi}{\phi^2}\right)$ $I_{y_1} = WR^2\left(\dfrac{1}{2} + \dfrac{\sin\phi\cos\phi}{2\phi}\right)$ $I_p = WR^2\left(1 - \dfrac{\sin^2\phi}{\phi^2}\right)$
Radius of Gyration	$\rho_x = R\sqrt{\dfrac{1}{2} - \dfrac{\sin\phi\cos\phi}{2\phi}}$ $\rho_{x_1} = R\sqrt{\dfrac{1}{2} - \dfrac{\sin\phi\cos\phi}{2\phi} + \sin^2\phi}$ $\rho_y = R\sqrt{\dfrac{1}{2} + \dfrac{\sin\phi\cos\phi}{2\phi} - \dfrac{\sin^2\phi}{\phi^2}}$ $\rho_{y_1} = R\sqrt{\dfrac{1}{2} + \dfrac{\sin\phi\cos\phi}{2\phi}}$ $\rho_p = R\sqrt{1 - \dfrac{\sin^2\phi}{\phi^2}}$

SECTION PROPERTIES

3.3 Solids

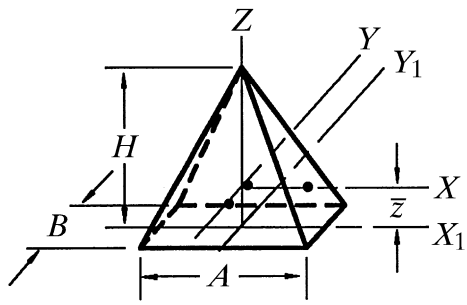
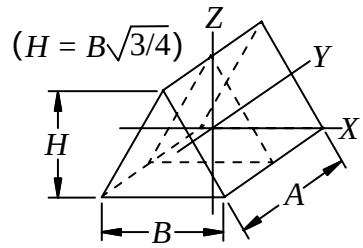
FIGURE 3–3 SOLIDS (CONTINUED ON NEXT PAGE)

	CUBE	RECTANGULAR PRISM
	 ①	 ②
Volume	A^3	ABH
Surface Area	$6A^2$	$2(AH + BH + AB)$
Centroid	– – –	– – –
Weight Moment of Inertia	$I_x = I_y = I_z = \frac{WA^2}{6}$ $I_{x_1} = I_{y_1} = I_{z_1} = \frac{2WA^2}{3}$ $I_{x_2} = \frac{5WA^2}{12}$	$I_x = \frac{W}{12}(B^2 + H^2)$ $I_{x_1} = \frac{W}{3}(B^2 + H^2)$ $I_{x_2} = \frac{W}{12}(B^2 + 4H^2)$ $I_y = \frac{W}{12}(A^2 + H^2)$ $I_{y_1} = \frac{W}{3}(A^2 + H^2)$ $I_z = \frac{W}{12}(A^2 + B^2)$ $I_{z_1} = \frac{W}{3}(A^2 + B^2)$
Radius of Gyration	$\rho_x = \rho_y = \rho_z = \frac{A}{\sqrt{6}}$ $\rho_{x_1} = \rho_{y_1} = \rho_{z_1} = 0.816A$ $\rho_{x_2} = 0.646A$	$\rho_x = \frac{1}{2}\sqrt{\frac{B^2 + H^2}{3}}$ $\rho_{x_1} = \sqrt{\frac{B^2 + H^2}{3}}$ $\rho_y = \frac{1}{2}\sqrt{\frac{A^2 + H^2}{3}}$ $\rho_{x_2} = \frac{1}{2}\sqrt{\frac{B^2 + 4H^2}{3}}$ $\rho_{y_1} = \sqrt{\frac{A^2 + H^2}{3}}$ $\rho_z = \frac{1}{2}\sqrt{\frac{A^2 + B^2}{3}}$ $\rho_{z_1} = \sqrt{\frac{A^2 + B^2}{3}}$

SECTION PROPERTIES

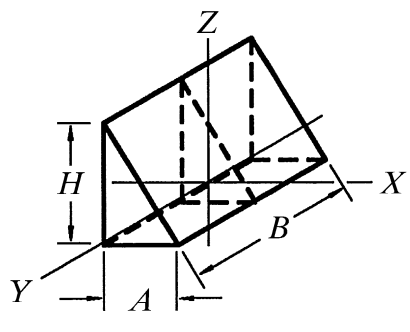
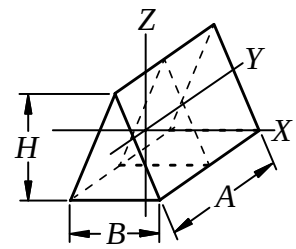
3.3 Solids (Continued)

FIGURE 3–3 SOLIDS (CONTINUED)

RIGHT RECTANGULAR PYRAMID		EQUILATERAL TRIANGULAR PRISM
 ③		 ④
Volume	$\frac{ABH}{3}$	$\frac{\sqrt{3}}{4}B^2A$
Surface Area	$AB + A\sqrt{H^2 + 0.25B^2} + B\sqrt{H^2 + 0.25A^2}$	$3AB + BH$
Centroid	$\bar{z} = \frac{H}{4}$	$\bar{z} = \frac{H}{3}$
Weight Moment of Inertia	$I_x = \frac{W}{20}\left(B^2 + \frac{3H^2}{4}\right)$ $I_{x_1} = \frac{W}{20}(B^2 + 2H^2)$ $I_y = \frac{W}{20}\left(A^2 + \frac{3}{4}H^2\right)$ $I_{y_1} = \frac{W}{20}(A^2 + 2H^2)$ $I_z = \frac{W}{20}(A^2 + B^2)$	$I_x = I_z = \frac{W}{24}(B^2 + 2A^2)$ $I_y = \frac{WB^2}{12}$
Radius of Gyration	$\rho_x = 0.224\sqrt{B^2 + \frac{3H^2}{4}}$ $\rho_{x_1} = 0.224\sqrt{B^2 + 2H^2}$ $\rho_y = 0.224\sqrt{A^2 + \frac{3H^2}{4}}$ $\rho_{y_1} = 0.224\sqrt{A^2 + 2H^2}$ $\rho_z = 0.224\sqrt{A^2 + B^2}$	$\rho_x = \rho_z = 0.204\sqrt{B^2 + 2A^2}$ $\rho_y = 0.289B = \frac{B}{2\sqrt{3}}$

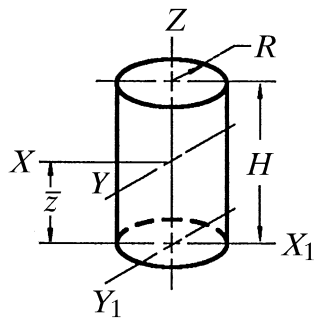
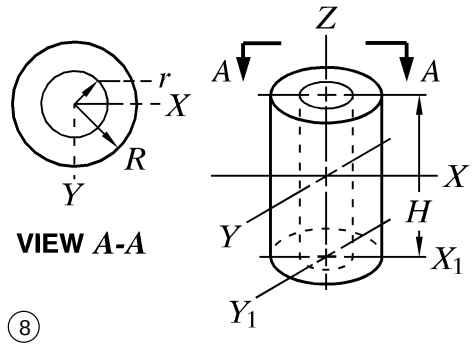
3.3 Solids (Continued)

FIGURE 3–3 SOLIDS (CONTINUED)

RIGHT ANGLED WEDGE		ISOSCELES WEDGE
		
Volume	$\frac{ABH}{2}$	$\frac{ABH}{2}$
Surface Area	$AH + B\left(H + A + \sqrt{H^2 + A^2}\right)$	$B(A + H) + A\sqrt{(4H^2 + B^2)}$
Centroid	$\bar{x} = \frac{A}{3}$ $\bar{y} = \frac{B}{2}$ $\bar{z} = \frac{H}{3}$	$\bar{z} = \frac{H}{3}$
Weight Moment of Inertia	$I_x = \frac{W}{36}(2H^2 + 3B^2)$ $I_y = \frac{W}{18}(A^2 + H^2)$ $I_z = \frac{W}{36}(2A^2 + 3B^2)$	$I_x = \frac{W}{36}(2H^2 + 3A^2)$ $I_y = \frac{W}{72}(4H^2 + 3B^2)$ $I_z = \frac{W}{24}(2A^2 + B^2)$
Radius of Gyration	$\rho_x = 0.167\sqrt{2H^2 + 3B^2}$ $\rho_y = 0.236\sqrt{A^2 + H^2}$ $\rho_z = 0.167\sqrt{2A^2 + 3B^2}$	$\rho_x = 0.167\sqrt{2H^2 + 3A^2}$ $\rho_y = 0.118\sqrt{4H^2 + 3B^2}$ $\rho_z = 0.204\sqrt{2A^2 + B^2}$

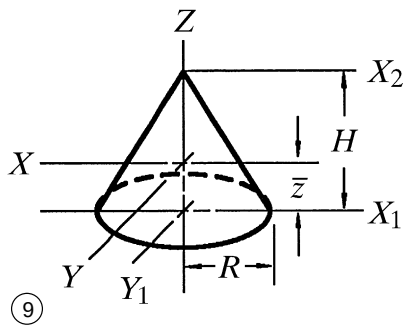
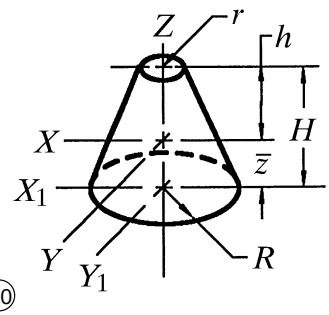
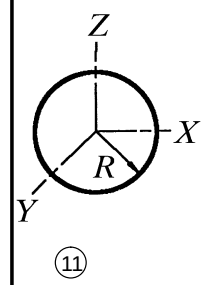
3.3 Solids (Continued)

FIGURE 3–3 SOLIDS (CONTINUED)

	RIGHT CIRCULAR CYLINDER	HOLLOW RIGHT CIRCULAR CYLINDER
	 7	 8
Volume	$\pi R^2 H$	$\pi H(R^2 - r^2)$
Surface Area	$2\pi R(R + H)$	$2\pi(R + r)(R - r + H)$
Centroid	– – –	– – –
Weight Moment of Inertia	$I_x = I_y = \frac{W}{12}(3R^2 + H^2)$ $I_{x_1} = I_{y_1} = \frac{W}{12}(3R^2 + 4H^2)$ $I_z = \frac{WR^2}{2}$	$I_x = I_y = \frac{W}{12}[3(R^2 + r^2) + H^2]$ $I_{x_1} = I_{y_1} = W\left(\frac{R^2 + r^2}{4} + \frac{H^2}{3}\right)$ $I_z = \frac{W}{2}(R^2 + r^2)$
Radius of Gyration	$\rho_x = \rho_y = \frac{1}{2}\sqrt{\frac{3R^2 + H^2}{3}}$ $\rho_{x_1} = \rho_{y_1} = \frac{1}{2}\sqrt{\frac{3R^2 + 4H^2}{3}}$ $\rho_z = 0.707R$	$\rho_x = \rho_y = \frac{1}{2}\sqrt{\frac{3(R^2 + r^2) + H^2}{3}}$ $\rho_z = 0.707\sqrt{R^2 + r^2}$ $\rho_{x_1} = \rho_{y_1} = \sqrt{\frac{R^2 + r^2}{4} + \frac{H^2}{3}}$

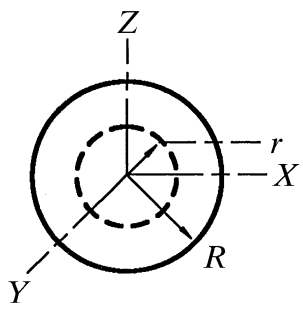
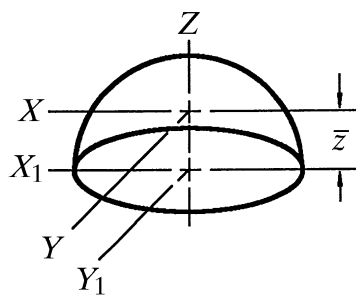
3.3 Solids (Continued)

FIGURE 3–3 SOLIDS (CONTINUED)

	RIGHT CIRCULAR CONE	FRUSTUM OF A CONE	SPHERE
			
Volume	$\frac{\pi R^2 H}{3}$	$\frac{\pi H}{3}(R^2 + Rr + r^2)$	$\frac{4}{3}\pi R^3$
Surface Area	$\pi R\left(R + \sqrt{H^2 + R^2}\right)$	$\pi\left[R^2 + r^2 + (R + r)\sqrt{H^2 + (R - r)^2}\right]$	$4\pi R^2$
Centroid	$\bar{z} = \frac{H}{4}$	$\bar{z} = \frac{H}{4}\left[\frac{R^2 + 2Rr + 3r^2}{R^2 + Rr + r^2}\right]$	At geometric center
Weight Moment of Inertia	$I_x = I_y = \frac{3W}{20}\left(R^2 + \frac{H^2}{4}\right)$ $I_{x_1} = I_{y_1} = \frac{W}{20}(3R^2 + 2H^2)$ $I_{x_2} = \frac{3W}{20}(R^2 + 4H^2)$ $I_z = \frac{3W}{10}R^2$	$I_z = \frac{3W}{10}\left(\frac{R^5 - r^5}{R^3 - r^3}\right)$ $I_{x_1} = I_{y_1} = \frac{W}{20}\left(3R^2\frac{H}{h} + \frac{2H^3}{h}\right.$ $\left. - 14H^2 - 4hH + 3R^2 - 3r^2\right)$ $I_x = I_y = I_{x_1} - W(\bar{z})^2$	$I_x = I_y = I_z$ $= \frac{2}{5}WR^2$
Radius of Gyration	$\rho_x = \rho_y = 0.387\sqrt{R^2 + \frac{H^2}{4}}$ $\rho_{x_1} = \rho_{y_1} = 0.224\sqrt{3R^2 + 2H^2}$ $\rho_{x_2} = 0.387\sqrt{R^2 + 4H^2}$ $\rho_z = 0.548R$	$\rho_z = 0.548\sqrt{\frac{R^5 - r^5}{R^3 - r^3}}$	$\rho_x = \rho_y = \rho_z$ $= 0.633R$

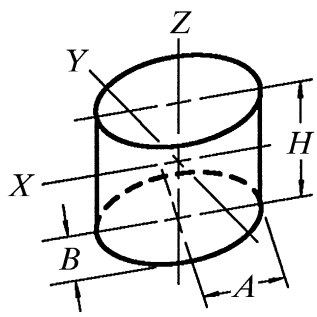
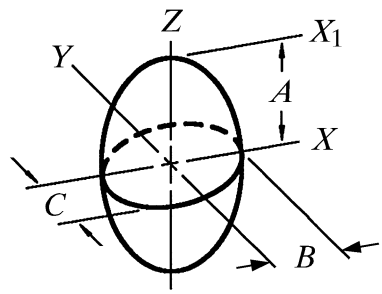
3.3 Solids (Continued)

FIGURE 3–3 SOLIDS (CONTINUED)

HOLLOW SPHERE		HEMISPHERE
 12		 13
Volume	$\frac{4}{3}\pi(R^3 - r^3)$	$\frac{2}{3}\pi R^3$
Surface Area	$4\pi R^2$	$3\pi R^2$
Centroid	At geometric center	$\bar{z} = \frac{3}{8}R$
Weight Moment of Inertia	$I_x = I_y = I_z = \frac{2}{5}W\left(\frac{R^5 - r^5}{R^3 - r^3}\right)$	$I_x = I_y = 0.26WR^2$ $I_{x_1} = I_{y_1} = I_z = 0.4WR^2$
Radius of Gyration	$\rho_x = \rho_y = \rho_z = 0.633\sqrt{\left(\frac{R^5 - r^5}{R^3 - r^3}\right)}$	$\rho_x = \rho_y = 0.51R$ $\rho_{x_1} = \rho_{y_1} = \rho_z = R\sqrt{\frac{2}{5}}$

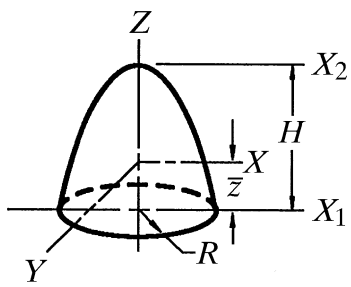
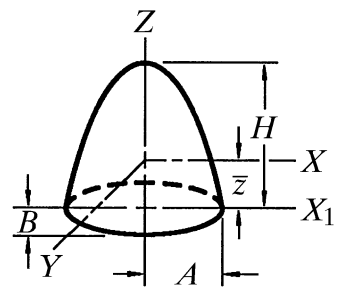
3.3 Solids (Continued)

FIGURE 3–3 SOLIDS (CONTINUED)

ELLIPTICAL CYLINDER		ELLIPSOID
 (14)		 (15) $A > B > C$
Volume	πABH	$\frac{4}{3}\pi ABC$
Surface Area	$\pi\left\{2AB + \left[(A+B) + \frac{(A-B)^2}{4(A+B)}\right]H\right\}$	No closed solution
Centroid	– – –	– – –
Weight Moment of Inertia	$I_x = \frac{W}{12}(3B^2 + H^2)$ $I_y = \frac{W}{12}(3A^2 + H^2)$ $I_z = \frac{W}{4}(A^2 + B^2)$	$I_x = \frac{W}{5}(A^2 + C^2)$ $I_{x_1} = \frac{W}{5}(6A^2 + C^2)$ $I_y = \frac{W}{5}(A^2 + B^2)$ $I_{y_1} = \frac{W}{5}(6A^2 + B^2)$ $I_z = \frac{W}{5}(B^2 + C^2)$
Radius of Gyration	$\rho_x = \frac{1}{2}\sqrt{\frac{3B^2 + H^2}{3}}$ $\rho_y = \frac{1}{2}\sqrt{\frac{3A^2 + H^2}{3}}$ $\rho_z = \frac{\sqrt{A^2 + B^2}}{2}$	$\rho_x = 0.447\sqrt{A^2 + C^2}$ $\rho_{x_1} = 0.447\sqrt{6A^2 + C^2}$ $\rho_y = 0.447\sqrt{A^2 + B^2}$ $\rho_{y_1} = 0.447\sqrt{6A^2 + B^2}$ $\rho_z = 0.447\sqrt{B^2 + C^2}$

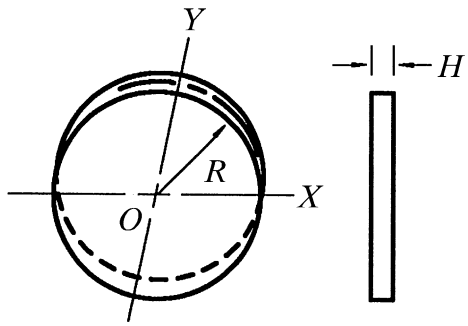
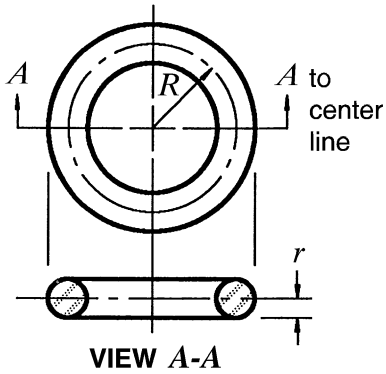
3.3 Solids (Continued)

FIGURE 3–3 SOLIDS (CONTINUED)

PARABOLOID OF REVOLUTION		ELLIPTIC PARABOLOID
 (16)		 (17)
Volume	$\frac{\pi R^2 H}{2}$	$\frac{\pi ABH}{2}$
Surface Area	$\frac{\pi R}{6H^2}[(4H^2 + R^2)^{3/2} - R^3]$	No closed solution
Centroid	$\bar{z} = \frac{H}{3}$	$\bar{z} = \frac{H}{3}$
Weight Moment of Inertia	$I_x = I_y = \frac{W}{18}(3R^2 + H^2)$ $I_{x_1} = \frac{W}{6}(R^2 + H^2)$ $I_{x_2} = \frac{W}{6}(R^2 + 3H^2)$ $I_z = \frac{WR^2}{3}$	$I_x = \frac{W}{18}(3B^2 + H^2)$ $I_{x_1} = \frac{W}{6}(B^2 + H^2)$ $I_y = \frac{W}{18}(3A^2 + H^2)$ $I_{y_1} = \frac{W}{6}(A^2 + H^2)$ $I_z = \frac{W}{6}(A^2 + B^2)$
Radius of Gyration	$\rho_x = \rho_y = 0.236\sqrt{3R^2 + H^2}$ $\rho_{x_1} = \sqrt{\frac{R^2 + H^2}{6}}$ $\rho_{x_2} = \sqrt{\frac{R^2 + 3H^2}{6}}$ $\rho_z = \frac{R}{\sqrt{3}}$	$\rho_x = 0.236\sqrt{3B^2 + H^2}$ $\rho_{x_1} = \sqrt{\frac{B^2 + H^2}{6}}$ $\rho_y = 0.236\sqrt{3A^2 + H^2}$ $\rho_{y_1} = \sqrt{\frac{A^2 + H^2}{6}}$ $\rho_z = \sqrt{\frac{A^2 + B^2}{6}}$

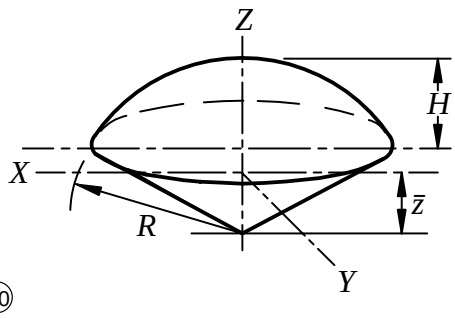
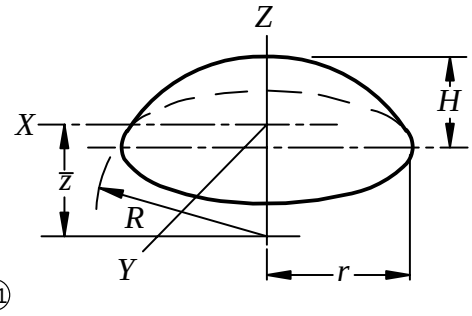
3.3 Solids (Continued)

FIGURE 3–3 SOLIDS (CONTINUED)

THIN CIRCULAR LAMINA		TORUS
 **NOTE:** Y axis is perpendicular to plane H = thickness 18		 19
Volume	$V = \pi R^2 H$	$2\pi^2 r^2 R$
Surface Area	$2\pi R(R + H)$	$4\pi^2 Rr$
Centroid	At 0 = geometric center	Centroid is at geometric center
Weight Moment of Inertia	$I_x = \frac{WR^2}{4}$ $I_y = \frac{WR^2}{2}$	$I_x = I_z = \frac{W}{8}(4R^2 + 5r^2)$ $I_y = \frac{W}{4}(4R^2 + 3r^2)$
Radius of Gyration	$\rho_x = \frac{R}{2}$ $\rho_y = \frac{R}{\sqrt{2}}$	$\rho_x = \rho_z = 0.354\sqrt{4R^2 + 5r^2}$ $\rho_y = \frac{\sqrt{4R^2 + 3r^2}}{2}$

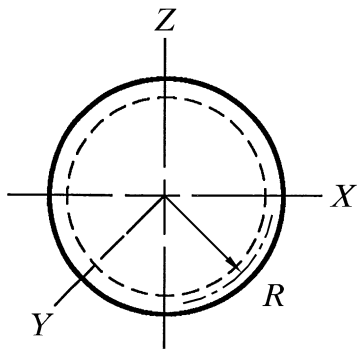
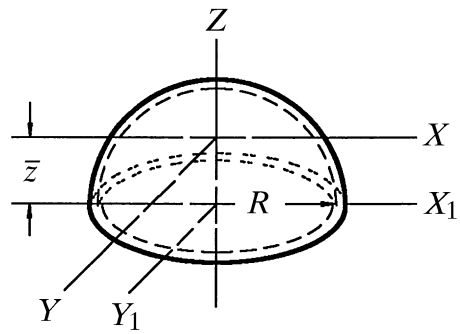
3.3 Solids (Continued)

FIGURE 3–3 SOLIDS (CONTINUED)

SPHERICAL SECTOR		SPHERICAL SEGMENT
		 **NOTE:** The bottom surface of the segment is flat.
Volume	$\frac{2}{3}\pi R^2 H$	$\frac{\pi H}{6}(3r^2 + H^2) = \pi H^2\left(R - \frac{H}{3}\right)$
Surface Area	Curved surface : $2\pi RH$ Cone : $\pi R\sqrt{2RH - H^2}$ Total : $\pi R\left(2H + \sqrt{2RH - H^2}\right)$	Curved surface : $2\pi RH$ Base : $2\pi RH - \pi H^2$ Total : $\pi H(4R - H)$
Centroid	$\bar{z} = \frac{3}{8}(2R - H)$	$\bar{z} = \frac{3}{4}\frac{(2R - H)^2}{(3R - H)}$
Weight Moment of Inertia	$I_z = \frac{WH}{5}(3R - H)$	$I_z = \frac{2HW}{3(R - H)}\left(R^2 - \frac{3}{4}RH + \frac{3}{20}H^2\right)$
Radius of Gyration	$\rho_z = 0.447\sqrt{(3R - H)(H)}$	$\rho_z = \sqrt{\frac{2H}{3R - H}\left(R^2 - \frac{3}{4}RH + \frac{3}{20}H^2\right)}$

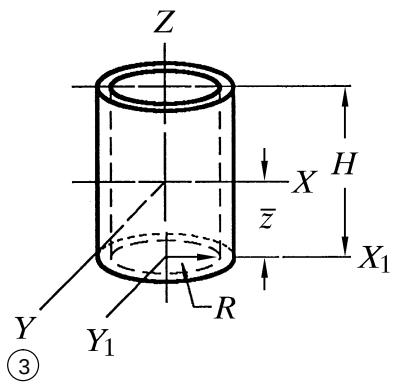
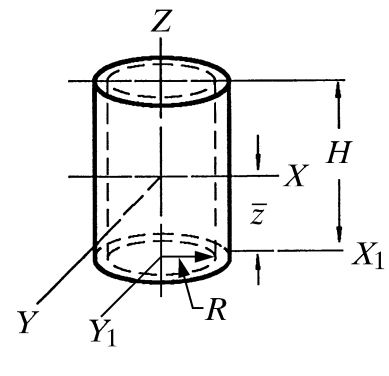
3.4 Thin Shells

FIGURE 3–4 THIN SHELLS

SPHERICAL SHELL		HEMISPHERICAL SHELL
 ①		 ②
Surface Area	$4\pi R^2$	$2\pi R^2$
Centroid	At geometric center	$\bar{z} = \frac{3R}{8}$
Weight Moment of Inertia	$I_x = I_y = I_z = \frac{2}{3}WR^2$	$I_x = I_y = \frac{5}{12}WR^2$ $I_z = I_{x_1} = I_{y_1} = \frac{2}{3}WR^2$
Radius of Gyration	$\rho_x = \rho_y = \rho_z = 0.816R$	$\rho_x = \rho_y = 0.646R$ $\rho_z = \rho_{x_1} = \rho_{y_1} = 0.816R$

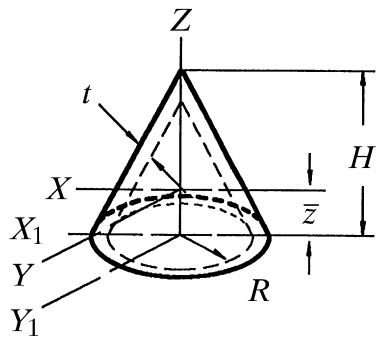
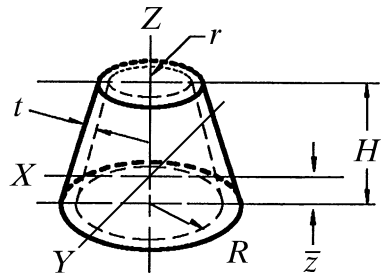
3.4 Thin Shells (Continued)

FIGURE 3–4 THIN SHELLS (CONTINUED)

	LATERAL CYLINDER SHELL	TOTAL CYLINDRICAL SHELL
		
Surface Area	$2\pi RH$	$2\pi R(R + H)$
Centroid	– – –	– – –
Weight Moment of Inertia	$I_x = I_y = \frac{W}{2}\left(R^2 + \frac{H^2}{6}\right)$ $I_{x_1} = I_{y_1} = \frac{W}{6}(3R^2 + 2H^2)$ $I_z = WR^2$	$I_x = I_y = \frac{W}{12(R + H)}[3R^2(R + 2H) + H^2(3R + H)]$ $I_{x_1} = I_{y_1} = \frac{W}{12(R + H)}[3R^2(R + 2H)$ $+ 2H^2(3R + 2H)]$ $I_z = \frac{WR^2}{2}\left(\frac{R + 2H}{R + H}\right)$
Radius of Gyration	$\rho_x = \rho_y = 0.707\sqrt{R^2 + \frac{H^2}{6}}$ $\rho_{x_1} = \rho_{y_1} = \sqrt{\frac{3R^2 + 2H^2}{6}}$ $\rho_z = R$	$\rho_x = \rho_y = \frac{1}{2}\sqrt{\frac{3R^2(R + 2H) + H^2(3R + H)}{3(R + H)}}$ $\rho_{x_1} = \rho_{y_1} = \frac{1}{2}\sqrt{\frac{3R^2(R + 2H) + 2H^2(3R + 2H)}{3(R + H)}}$ $\rho_z = 0.707R\sqrt{\frac{R + 2H}{R + H}}$

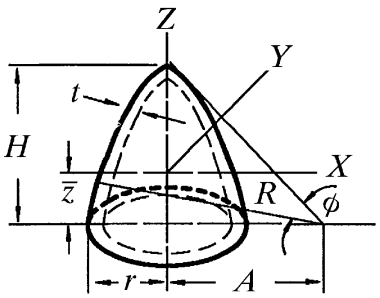
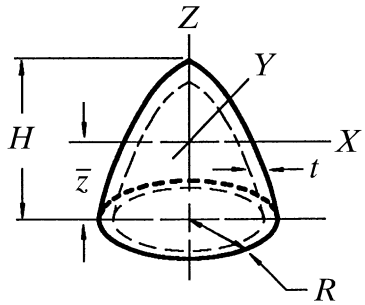
3.4 Thin Shells (Continued)

FIGURE 3–4 THIN SHELLS (CONTINUED)

CIRCULAR CONICAL SHELL		FRUSTUM OF CIRCULAR CONICAL SHELL
 5		 6
Volume	$\pi R t \sqrt{R^2 + H^2}$	$\pi(R + r)t\sqrt{H^2 + (R - r)^2}$
Surface Area	$\pi R \sqrt{R^2 + H^2}$	$\pi(R + r)\sqrt{H^2 + (R - r)^2}$
Centroid	$\bar{z} = \frac{H}{3}$	$\bar{z} = \frac{H}{3}\left(\frac{2r + R}{r + R}\right)$
Weight Moment of Inertia	$I_x = I_y = \frac{W}{4}(R^2 + \frac{2}{9}H^2)$ $I_z = \frac{WR^2}{2}$ $I_{x_1} = I_{y_1} = \frac{W}{12}(3R^2 + 2H^2)$	$I_x = I_y = \frac{W}{4}(R^2 + r^2) + \frac{WH^2}{18}\left[1 + \frac{2Rr}{(R + r)^2}\right]$ $I_z = \frac{W}{2}(R^2 + r^2)$
Radius of Gyration	$\rho_x = \rho_y = \frac{\sqrt{9R^2 + 2H^2}}{6}$ $\rho_{x_1} = \rho_{y_1} = \frac{1}{2}\sqrt{\frac{3R^2 + 2H^2}{3}}$ $\rho_z = 0.707R$	$\rho_x = \rho_y = \sqrt{\frac{(R^2 + r^2)}{4} + \frac{H^2}{18}\left[1 + \frac{2Rr}{(R + r)^2}\right]}$ $\rho_z = 0.707\sqrt{R^2 + r^2}$

3.4 Thin Shells (Continued)

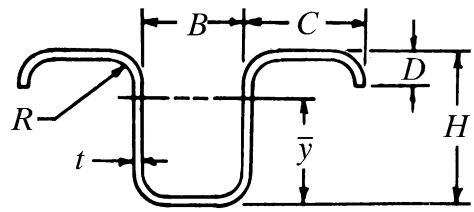
FIGURE 3–4 THIN SHELLS (CONCLUDED)

	OGIVE OF REVOLUTION SHELL	PARABOLIC SHELL
	 ϕ in radians 7	 $P = (4H^2 + R^2)^{3/2}$ 8
Volume	$2\pi R^2 t \sin\phi - 2\pi R t + \phi A$	$\frac{\pi R t}{6H^2}[P - R^3]$
Surface Area	$2\pi R^2 \sin\phi - 2\pi R \phi A$	$\frac{\pi R}{6H^2}[P - R^3]$
Centroid	$\frac{RA(1 - \cos\phi) - (R^2/4)(1 - \cos 2\phi)}{(\phi A - R\sin\phi)}$	$\bar{z} = H - \frac{P(6H^2 - R^2) + R^5}{10H(P - R^3)}$
Weight Moment of Inertia	$I_x = I_y = \frac{-W}{8(\phi A - R\sin\phi)}\left[\frac{K_1 + K_2}{3} + R^2\frac{[4A(1 - \cos\phi) - R(1 - \cos 2\phi)]^2}{2(\phi A - R\sin\phi)}\right]$ with $K_1 = 4R(2R^2 + 9A^2) - \pi A(9R^2 - 6A^2)$ $K_2 = 2R^2[-3A(2\phi - \sin 2\phi) + 4R\sin^3\phi]$	$I_x = I_y = \frac{W}{28H^2}\left[\frac{P(12H^4 + 6R^2H^2 - R^4) + R^7}{P - R^3}\right] - \frac{W}{100H^2}\left[\frac{P(6H^2 - R^2) + R^5}{P - R^3}\right]^2$ $I_z = \frac{WR^2(H - \bar{z})}{H}$
Radius of Gyration	$\rho_x = \rho_y = \sqrt{\frac{I_x}{W}} = \sqrt{\frac{I_y}{W}}$	$\rho_x = \rho_y = \sqrt{\frac{I_x}{W}} = \sqrt{\frac{I_y}{W}}$ $\rho_z = R\sqrt{I - \frac{\bar{z}}{H}}$

4 PROPERTIES OF TYPICAL SECTIONS

4.1 Hat Section

FIGURE 4–1 BAC1498 HAT SECTION (CONTINUED ON NEXT PAGE)



Section BAC1498	H in	B in	C in	D in	t in	R in	Area in^2	$\bar{y}$ in	I_x in^4	ρ in
–45	1.00	0.750	0.694	0.200	0.032	0.125	0.1295	0.5853	0.0186	0.3787
–46	1.00	0.750	0.710	0.200	0.040	0.125	0.1615	0.5847	0.0229	0.3764
–47	1.00	0.750	0.732	0.250	0.051	0.125	0.2102	0.5885	0.0287	0.3698
–49	1.00	0.812	0.818	0.350	0.064	0.156	0.2821	0.5936	0.0365	0.3595
–50	1.00	0.812	0.834	0.350	0.072	0.156	0.3166	0.5930	0.0404	0.3572
–87	1.25	0.686	0.726	0.200	0.032	0.125	0.1455	0.7398	0.0313	0.4638
–88	1.25	0.750	0.710	0.200	0.040	0.125	0.1815	0.7220	0.0393	0.4654
–89	1.25	0.750	0.732	0.250	0.051	0.125	0.2357	0.7277	0.0498	0.4595
–90	1.25	0.812	0.818	0.350	0.064	0.156	0.3141	0.7372	0.0637	0.4504
–91	1.25	0.812	0.834	0.350	0.072	0.156	0.3526	0.7367	0.0708	0.4482
–95	1.25	0.760	0.740	0.200	0.032	0.130	0.1484	0.7309	0.0326	0.4686
–96	1.25	0.760	0.740	0.200	0.040	0.130	0.1837	0.7268	0.0399	0.4661
–97	1.25	0.760	0.740	0.250	0.050	0.130	0.2320	0.7281	0.0491	0.4601
–98	1.25	1.000	0.880	0.300	0.056	0.160	0.2885	0.7227	0.0633	0.4683
–99	1.25	1.000	0.880	0.300	0.063	0.160	0.3222	0.7195	0.0700	0.4661
–100	1.25	1.000	0.880	0.300	0.071	0.160	0.3602	0.7159	0.0774	0.4636
–101	1.25	1.000	0.880	0.350	0.080	0.160	0.4100	0.7159	0.0857	0.4572
–102	1.25	1.000	0.880	0.350	0.090	0.160	0.4565	0.7114	0.0941	0.4540
–107	1.19	0.760	0.740	0.200	0.040	0.130	0.1789	0.6939	0.0354	0.4449
–108	1.19	0.760	0.740	0.200	0.032	0.130	0.1445	0.6980	0.0289	0.4474
–109	1.25	1.000	0.870	0.300	0.050	0.160	0.2582	0.7235	0.0571	0.4701
–110	1.25	0.760	0.750	0.250	0.056	0.130	0.2592	0.7272	0.0545	0.4584
–112	1.19	0.760	0.740	0.200	0.050	0.130	0.2210	0.6889	0.0431	0.4418
–113	1.25	1.000	0.880	0.250	0.045	0.160	0.2300	0.7229	0.0519	0.4751

SECTION PROPERTIES

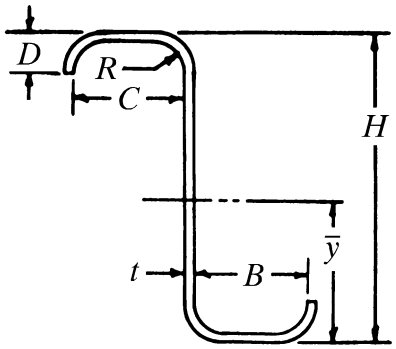
4.1 Hat Section (Continued)

FIGURE 4–1 BAC1498 HAT SECTION (CONCLUDED)

Section BAC1498	H in	B in	C in	D in	t in	R in	Area in^2	$\bar{y}$ in	I_x in^4	ρ in
–114	1.25	0.760	0.740	0.250	0.063	0.130	0.2880	0.7216	0.0599	0.4561
–124	1.25	0.820	1.040	0.300	0.063	0.160	0.3311	0.7735	0.0687	0.4556
–125	1.25	0.820	1.040	0.300	0.071	0.160	0.3701	0.7700	0.0761	0.4533
–126	1.25	0.800	1.080	0.350	0.080	0.190	0.4198	0.7774	0.0822	0.4425
–127	1.25	0.780	1.100	0.350	0.090	0.190	0.4693	0.7792	0.0901	0.4382
–128	1.25	0.760	1.120	0.350	0.100	0.220	0.5105	0.7801	0.0947	0.4307
–134	1.25	1.000	0.740	0.300	0.056	0.160	0.2728	0.6940	0.0591	0.4656
–135	1.25	1.000	0.740	0.300	0.063	0.160	0.3046	0.6906	0.0654	0.4632
–136	1.19	0.760	0.740	0.200	0.045	0.130	0.2001	0.6914	0.0393	0.4434
–137	1.18	0.760	0.740	0.250	0.063	0.130	0.2792	0.6831	0.0519	0.4312
–138	1.25	1.000	0.765	0.260	0.032	0.190	0.1565	0.7044	0.0351	0.4738
–139	1.25	1.000	0.770	0.260	0.036	0.190	0.1757	0.7035	0.0392	0.4726
–140	1.25	1.000	0.780	0.260	0.040	0.190	0.1952	0.7037	0.0434	0.4715
–141	1.25	1.000	0.790	0.300	0.045	0.190	0.2229	0.7077	0.0487	0.4674
–142	1.25	1.000	0.800	0.300	0.050	0.190	0.2473	0.7074	0.0537	0.4659
–143	1.25	1.000	0.810	0.300	0.056	0.190	0.2763	0.7066	0.0595	0.4642
–144	1.25	1.000	0.825	0.300	0.063	0.190	0.3104	0.7064	0.0663	0.4622
–145	1.25	1.000	0.840	0.340	0.071	0.190	0.3547	0.7093	0.0741	0.4570
–146	1.25	1.000	0.860	0.340	0.080	0.190	0.3990	0.7092	0.0824	0.4545
–147	1.25	1.000	0.880	0.340	0.090	0.190	0.4477	0.7086	0.0914	0.4517
–149	1.25	0.760	0.740	0.250	0.071	0.130	0.3216	0.7175	0.0662	0.4536

4.2 Zee Section

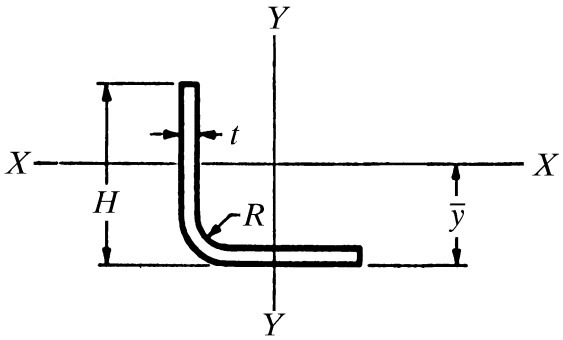
FIGURE 4–2 BAC1517 SHAPES



BAC1517	H in	B in	C in	D in	t in	R in	Area in^2	$\bar{y}$ in	I_x in^4	ρ in
–496	2.460	0.900	0.900	0.330	0.081	0.190	0.3656	1.230	0.318	0.933
–705	2.491	0.900	0.900	0.330	0.051	0.250	0.2281	1.246	0.203	0.944
–706	2.504	0.900	0.900	0.330	0.064	0.250	0.2865	1.252	0.255	0.944
–708	2.491	1.060	0.740	0.330	0.051	0.250	0.2281	1.246	0.201	0.940
–709	2.504	1.090	0.710	0.330	0.064	0.250	0.2865	1.252	0.253	0.939
–1004	2.500	0.900	0.900	0.500	0.072	0.250	0.3460	1.250	0.302	0.934
–1062	2.490	0.900	0.900	0.330	0.056	0.250	0.2526	1.245	0.225	0.944
–1080	2.500	0.900	0.900	0.500	0.081	0.250	0.3891	1.250	0.338	0.932
–1147	2.505	0.900	0.900	0.330	0.045	0.250	0.2034	1.253	0.184	0.951
–1256	2.500	0.900	0.900	0.500	0.090	0.250	0.4314	1.250	0.373	0.929

4.3 Equal Leg Formed Angle

FIGURE 4–3 EQUAL LEG FORMED ANGLE (CONTINUED ON NEXT PAGE)



H in	t in	R in	Area in^2	Devel. Width in	$\bar{y}$ in	I_x in^4
1/2	0.020	3/32	0.019	0.94	0.139	0.0005
	0.025	3/32	0.023	0.93	0.141	0.0006
	0.032	3/32	0.030	0.92	0.144	0.0007
9/16	0.025	3/32	0.026	1.06	0.162	0.0008
	0.032	3/32	0.033	1.05	0.159	0.0011
	0.040	1/8	0.041	1.02	0.165	0.0013
5/8	0.032	3/32	0.037	1.17	0.175	0.0015
	0.040	1/8	0.046	1.15	0.181	0.0018
	0.051	5/32	0.057	1.12	0.188	0.0022
11/16	0.032	3/32	0.042	1.30	0.191	0.0020
	0.040	1/8	0.051	1.27	0.196	0.0024
	0.051	5/32	0.063	1.24	0.203	0.0029
3/4	0.032	3/32	0.045	1.42	0.207	0.0026
	0.040	1/8	0.056	1.40	0.212	0.0031
	0.051	5/32	0.070	1.37	0.218	0.0039
	0.064	3/16	0.086	1.34	0.227	0.0047
	0.072	7/32	0.095	1.32	0.233	0.0051
13/16	0.040	1/8	0.061	1.52	0.228	0.0040
	0.051	5/32	0.076	1.49	0.234	0.0080
	0.064	3/16	0.094	1.47	0.242	0.0060
	0.072	7/32	0.104	1.44	0.246	0.0066
	0.081	1/4	0.115	1.42	0.254	0.0072

SECTION PROPERTIES

4.3 Equal Leg Formed Angle (Continued)

FIGURE 4–3 EQUAL LEG FORMED ANGLE (CONTINUED)

H in	t in	R in	Area in^2	Devel. Width in	$\bar{y}$ in	I_x in^4
7/8	0.040	1/8	0.066	1.65	0.242	0.0051
	0.051	5/32	0.083	1.62	0.249	0.0063
	0.064	3/16	0.102	1.59	0.257	0.0076
	0.072	7/32	0.113	1.57	0.264	0.0084
	0.081	1/4	0.125	1.54	0.270	0.0091
1	0.040	1/8	0.076	1.90	0.274	0.0077
	0.051	5/32	0.095	1.87	0.280	0.0096
	0.064	3/16	0.118	1.84	0.288	0.0116
	0.072	7/32	0.131	1.82	0.294	0.0128
	0.081	1/4	0.145	1.80	0.300	0.0141
	0.091	9/32	0.161	1.77	0.307	0.0154
1–1/8	0.040	1/8	0.086	2.15	0.305	0.0109
	0.051	5/32	0.108	2.12	0.311	0.0138
	0.064	3/16	0.134	2.09	0.319	0.0169
	0.072	7/32	0.149	2.07	0.325	0.0185
	0.081	1/4	0.166	2.04	0.331	0.0221
	0.091	9/32	0.184	2.02	0.337	0.0226
	0.102	11/32	0.203	1.99	0.346	0.0247
1–1/4	0.040	1/8	0.096	2.40	0.337	0.0154
	0.051	5/32	0.121	2.37	0.343	0.0190
	0.064	3/16	0.150	2.34	0.350	0.0234
	0.072	7/32	0.167	2.32	0.356	0.0258
	0.081	1/4	0.186	2.29	0.362	0.0285
	0.091	9/32	0.206	2.27	0.368	0.0317
	0.102	11/32	0.229	2.24	0.376	0.0346
1–3/8	0.040	1/8	0.106	2.65	0.367	0.0206
	0.051	5/32	0.134	2.62	0.374	0.0256
	0.064	3/16	0.166	2.59	0.381	0.0315
	0.072	7/32	0.185	2.57	0.387	0.0349
	0.081	1/4	0.206	2.55	0.392	0.0387
	0.091	9/32	0.229	2.52	0.399	0.0426
	0.102	11/32	0.254	2.49	0.407	0.0468

SECTION PROPERTIES

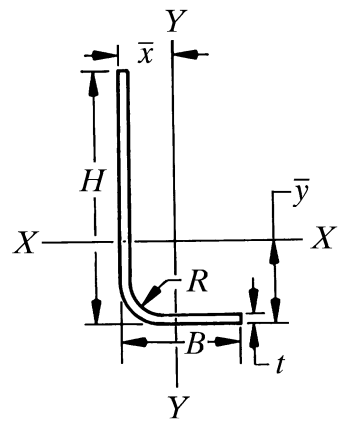
4.3 Equal Leg Formed Angle (Continued)

FIGURE 4–3 EQUAL LEG FORMED ANGLE (CONCLUDED)

H in	t in	R in	Area in^2	Devel. Width in	$\bar{y}$ in	I_x in^4
1–1/2	0.051	5/32	0.146	2.88	0.405	0.0334
	0.064	3/16	0.182	2.84	0.413	0.0411
	0.072	7/32	0.203	2.82	0.418	0.0459
	0.081	1/4	0.226	2.80	0.423	0.0509
	0.091	9/32	0.252	2.77	0.430	0.0563
	0.102	11/32	0.280	2.74	0.438	0.0615
1–5/8	0.051	5/32	0.150	3.12	0.435	0.0430
	0.064	3/16	0.198	3.10	0.443	0.0528
	0.072	7/32	0.221	3.07	0.448	0.0586
	0.081	1/4	0.248	3.06	0.453	0.0652
	0.091	9/32	0.275	3.02	0.461	0.0727
	0.102	11/32	0.305	2.99	0.468	0.0796
1–3/4	0.051	5/32	0.172	3.37	0.467	0.0537
	0.064	3/16	0.214	3.34	0.475	0.0666
	0.072	7/32	0.239	3.32	0.479	0.0739
	0.081	1/4	0.267	3.30	0.483	0.0823
	0.091	9/32	0.297	3.27	0.493	0.0911
	0.102	11/32	0.341	3.24	0.500	0.1008
1–7/8	0.051	5/32	0.184	3.62	0.498	0.0652
	0.064	3/16	0.230	3.50	0.506	0.0820
	0.072	7/32	0.257	3.57	0.512	0.0917
	0.081	1/4	0.287	3.55	0.518	0.1018
	0.091	9/32	0.320	3.52	0.523	0.1125
	0.102	11/32	0.356	3.49	0.529	0.1252
2	0.051	5/32	0.197	3.87	0.529	0.0809
	0.064	3/16	0.246	3.85	0.537	0.0998
	0.072	7/32	0.275	3.82	0.543	0.1114
	0.081	1/4	0.307	3.79	0.548	0.1241
	0.091	9/32	0.342	3.77	0.555	0.1382
	0.102	11/32	0.382	3.74	0.562	0.1527

4.4 Unequal Leg Formed Angle

FIGURE 4–4 UNEQUAL LEG FORMED ANGLE (CONTINUED ON NEXT PAGE)



H in	B in	t in	R in	Area in^2	Devel. Width in	$\bar{x}$ in	$\bar{y}$ in	I_y in^4	I_x in^4
9/16	1/4	0.020	1/16	0.015	0.76	0.048	0.211	0.0001	0.0005
	1/4	0.032	3/32	0.023	0.73	0.054	0.221	0.0001	0.0008
	7/16	0.025	3/32	0.023	0.93	0.111	0.176	0.0004	0.0008
	7/16	0.032	3/32	0.029	0.92	0.113	0.179	0.0005	0.0010
	7/16	0.040	1/8	0.036	0.90	0.118	0.186	0.0006	0.0012
5/8	1/4	0.020	1/16	0.016	0.82	0.045	0.240	0.0001	0.0007
	1/4	0.032	3/32	0.025	0.76	0.051	0.250	0.0001	0.0010
	1/2	0.032	3/32	0.034	1.05	0.125	0.194	0.0008	0.0013
	1/2	0.040	1/8	0.041	1.02	0.134	0.200	0.0009	0.0016
	1/2	0.051	5/32	0.051	1.00	0.141	0.208	0.0012	0.0020
11/16	1/4	0.020	1/16	0.018	0.89	0.043	0.269	0.0001	0.0009
	1/4	0.032	3/32	0.028	0.86	0.049	0.279	0.0001	0.0013
	1/4	0.040	1/8	0.033	0.84	0.052	0.266	0.0001	0.0016
	7/16	0.032	3/32	0.034	1.05	0.101	0.232	0.0006	0.0017
	7/16	0.040	1/8	0.041	1.02	0.106	0.239	0.0007	0.0020
	7/16	0.051	5/32	0.051	1.00	0.113	0.248	0.0008	0.0024
	9/16	0.032	3/32	0.038	1.17	0.144	0.209	0.0011	0.0018
	9/16	0.040	1/8	0.046	1.15	0.149	0.215	0.0014	0.0022
	9/16	0.051	5/32	0.057	1.12	0.156	0.222	0.0017	0.0027

4.4 Unequal Leg Formed Angle (Continued)

FIGURE 4–4 UNEQUAL LEG FORMED ANGLE (CONTINUED)

H in	B in	t in	R in	Area in^2	Devel. Width in	$\bar{x}$ in	$\bar{y}$ in	I_y in^4	I_x in^4
3/4	5/16	0.020	1/16	0.020	1.01	0.056	0.281	0.0001	0.0012
	5/16	0.032	3/32	0.031	0.98	0.061	0.291	0.0002	0.0018
	5/16	0.040	1/8	0.038	0.96	0.066	0.299	0.0003	0.0022
	5/16	0.051	5/32	0.048	0.93	0.072	0.309	0.0003	0.0027
	1/2	0.032	3/32	0.037	1.17	0.117	0.246	0.0008	0.0022
	1/2	0.040	1/8	0.046	1.15	0.122	0.253	0.0010	0.0027
	1/2	0.051	5/32	0.057	1.12	0.128	0.262	0.0012	0.0033
	1/2	0.064	3/16	0.070	1.09	0.135	0.271	0.0015	0.0040
	1/2	0.072	7/32	0.077	1.07	0.141	0.279	0.0016	0.0042
	5/8	0.032	3/32	0.041	1.29	0.160	0.224	0.0016	0.0024
	5/8	0.040	1/8	0.051	1.27	0.165	0.230	0.0019	0.0030
	5/8	0.051	5/32	0.063	1.24	0.172	0.238	0.0023	0.0036
	5/8	0.064	3/16	0.078	1.22	0.179	0.246	0.0028	0.0043
	5/8	0.072	7/32	0.086	1.19	0.186	0.253	0.0030	0.0047
7/8	5/16	0.020	1/16	0.023	1.14	0.051	0.340	0.0001	0.0018
	5/16	0.032	3/32	0.035	1.11	0.056	0.350	0.0002	0.0028
	5/16	0.040	1/8	0.043	1.09	0.061	0.358	0.0003	0.0034
	5/16	0.051	5/32	0.054	1.06	0.066	0.368	0.0003	0.0041
	1/2	0.040	1/8	0.051	1.27	0.112	0.308	0.0011	0.0041
	1/2	0.051	5/32	0.064	1.25	0.118	0.316	0.0013	0.0050
	1/2	0.064	3/16	0.076	1.22	0.126	0.326	0.0015	0.0061
	1/2	0.072	7/32	0.086	1.19	0.130	0.334	0.0017	0.0065
	1/2	0.081	1/4	0.095	1.17	0.135	0.342	0.0018	0.0071
	1/2	0.091	9/32	0.104	1.15	0.142	0.351	0.0020	0.0077
	5/8	0.040	1/8	0.056	1.40	0.152	0.283	0.0020	0.0045
	5/8	0.051	5/32	0.070	1.37	0.158	0.289	0.0024	0.0056
	5/8	0.064	3/16	0.086	1.34	0.165	0.300	0.0029	0.0066
	5/8	0.072	7/32	0.095	1.32	0.171	0.306	0.0032	0.0072
	5/8	0.081	1/4	0.105	1.29	0.177	0.313	0.0035	0.0079
	5/8	0.091	9/32	0.115	1.27	0.183	0.322	0.0038	0.0085
	3/4	0.040	1/8	0.061	1.52	0.196	0.261	0.0033	0.0048
	3/4	0.051	5/32	0.076	1.50	0.202	0.268	0.0041	0.0059
	3/4	0.064	3/16	0.094	1.47	0.210	0.276	0.0050	0.0072
	3/4	0.072	7/32	0.104	1.44	0.215	0.283	0.0054	0.0079
	3/4	0.081	1/4	0.115	1.42	0.222	0.290	0.0059	0.0086
	3/4	0.091	9/32	0.127	1.40	0.228	0.297	0.0065	0.0093

SECTION PROPERTIES

4.4 Unequal Leg Formed Angle (Continued)

FIGURE 4–4 UNEQUAL LEG FORMED ANGLE (CONTINUED)

H in	B in	t in	R in	Area in^2	Devel. Width in	$\bar{x}$ in	$\bar{y}$ in	I_y in^4	I_x in^4
1	5/16	0.020	1/16	0.025	1.26	0.047	0.399	0.0002	0.0026
	5/16	0.032	3/32	0.039	1.23	0.052	0.409	0.0002	0.0041
	5/16	0.040	1/8	0.048	1.21	0.057	0.418	0.0003	0.0049
	5/16	0.051	5/32	0.060	1.18	0.062	0.426	0.0003	0.0060
	1/2	0.040	1/8	0.056	1.40	0.103	0.364	0.0011	0.0059
	1/2	0.051	5/32	0.070	1.37	0.189	0.373	0.0013	0.0073
	1/2	0.064	3/16	0.086	1.34	0.116	0.384	0.0016	0.0088
	1/2	0.072	7/32	0.095	1.32	0.121	0.391	0.0018	0.0095
	1/2	0.081	1/4	0.105	1.29	0.126	0.400	0.0018	0.0103
	1/2	0.091	9/32	0.115	1.27	0.132	0.408	0.0020	0.0111
	5/8	0.040	1/8	0.061	1.52	0.141	0.336	0.0021	0.0065
	5/8	0.051	5/32	0.076	1.50	0.147	0.344	0.0025	0.0080
	5/8	0.064	3/16	0.094	1.47	0.154	0.353	0.0031	0.0096
	5/8	0.072	7/32	0.104	1.44	0.159	0.360	0.0034	0.0105
	5/8	0.081	1/4	0.115	1.42	0.165	0.368	0.0037	0.0115
	5/8	0.091	9/32	0.127	1.40	0.171	0.377	0.0040	0.0124
	3/4	0.040	1/8	0.066	1.65	0.183	0.312	0.0035	0.0069
	3/4	0.051	5/32	0.082	1.62	0.188	0.319	0.0043	0.0086
	3/4	0.064	3/16	0.102	1.59	0.195	0.328	0.0052	0.0104
	3/4	0.072	7/32	0.113	1.57	0.201	0.335	0.0057	0.0114
	3/4	0.081	1/4	0.125	1.55	0.207	0.342	0.0063	0.0125
	3/4	0.091	9/32	0.138	1.52	0.214	0.348	0.0067	0.0136
	7/8	0.040	1/8	0.071	1.77	0.227	0.282	0.0053	0.0073
	7/8	0.051	5/32	0.089	1.75	0.233	0.299	0.0066	0.0091
	7/8	0.064	3/16	0.110	1.72	0.241	0.307	0.0080	0.0111
	7/8	0.072	7/32	0.122	1.69	0.247	0.313	0.0088	0.0122
	7/8	0.081	1/4	0.135	1.67	0.252	0.320	0.0097	0.0134
	7/8	0.091	9/32	0.150	1.64	0.259	0.327	0.0106	0.0146

4.4 Unequal Leg Formed Angle (Continued)

FIGURE 4–4 UNEQUAL LEG FORMED ANGLE (CONTINUED)

H in	B in	t in	R in	Area in^2	Devel. Width in	$\bar{x}$ in	$\bar{y}$ in	I_y in^4	I_x in^4
1–1/8	5/16	0.040	1/8	0.053	1.34	0.053	0.478	0.0003	0.0068
	5/16	0.051	5/32	0.067	1.31	0.050	0.489	0.0003	0.0083
	5/16	0.064	3/16	0.082	1.28	0.065	0.501	0.0004	0.0094
	5/8	0.040	1/8	0.066	1.65	0.132	0.391	0.0021	0.0089
	5/8	0.064	3/16	0.102	1.50	0.143	0.408	0.0032	0.0133
	5/8	0.081	1/4	0.125	1.55	0.155	0.426	0.0038	0.0160
	5/8	0.102	11/32	0.152	1.49	0.168	0.444	0.0045	0.0187
	3/4	0.040	1/8	0.071	1.77	0.171	0.364	0.0036	0.0096
	3/4	0.064	3/16	0.110	1.72	0.184	0.382	0.0054	0.0144
	3/4	0.081	1/4	0.135	1.67	0.194	0.396	0.0065	0.0179
	3/4	0.102	11/32	0.165	1.62	0.208	0.414	0.0077	0.0205
	7/8	0.040	1/8	0.076	1.90	0.213	0.342	0.0059	0.0101
	7/8	0.064	3/16	0.118	1.84	0.226	0.357	0.0083	0.0154
	7/8	0.081	1/4	0.145	1.79	0.238	0.371	0.0101	0.0186
	7/8	0.102	11/32	0.178	1.74	0.251	0.388	0.0120	0.0221
	1	0.040	1/8	0.081	2.02	0.258	0.322	0.0080	0.0106
	1	0.064	3/16	0.125	1.97	0.272	0.337	0.0121	0.0162
	1	0.081	1/4	0.156	1.92	0.283	0.350	0.0148	0.0198
	1	0.102	11/32	0.190	1.87	0.297	0.365	0.0176	0.0235
1–1/4	5/16	0.040	1/8	0.058	1.46	0.050	0.539	0.0003	0.0091
	5/16	0.051	5/32	0.073	1.43	0.056	0.550	0.0004	0.0111
	5/16	0.064	3/16	0.090	1.40	0.062	0.562	0.0004	0.0133
	5/16	0.072	7/32	0.099	1.38	0.066	0.572	0.0005	0.0144
	5/8	0.040	1/8	0.071	1.77	0.125	0.448	0.0024	0.0119
	5/8	0.064	3/16	0.110	1.72	0.136	0.465	0.0033	0.0179
	5/8	0.081	1/4	0.135	1.67	0.146	0.482	0.0039	0.0214
	5/8	0.102	11/32	0.165	1.62	0.159	0.501	0.0048	0.0252
	3/4	0.040	1/8	0.076	1.90	0.161	0.419	0.0039	0.0127
	3/4	0.064	3/16	0.118	1.84	0.174	0.438	0.0056	0.0193
	3/4	0.081	1/4	0.145	1.79	0.184	0.450	0.0067	0.0235
	3/4	0.102	11/32	0.178	1.74	0.197	0.468	0.0079	0.0277

SECTION PROPERTIES

4.4 Unequal Leg Formed Angle (Continued)

FIGURE 4–4 UNEQUAL LEG FORMED ANGLE (CONTINUED)

H in	B in	t in	R in	Area in^2	Devel. Width in	$\bar{x}$ in	$\bar{y}$ in	I_y in^4	I_x in^4
1–1/4	7/8	0.040	1/8	0.081	2.02	0.201	0.395	0.0059	0.0135
	7/8	0.064	3/16	0.126	1.97	0.214	0.411	0.0086	0.0205
	7/8	0.072	7/32	0.140	1.94	0.282	0.320	0.0093	0.0232
	7/8	0.081	1/4	0.156	1.92	0.225	0.424	0.0105	0.0249
	7/8	0.102	11/32	0.190	1.87	0.238	0.441	0.0125	0.0297
	1	0.040	1/8	0.086	2.15	0.244	0.373	0.0084	0.0141
	1	0.064	3/16	0.134	2.09	0.257	0.388	0.0126	0.0216
	1	0.081	1/4	0.166	2.04	0.268	0.401	0.0153	0.0263
	1	0.102	11/32	0.203	1.99	0.282	0.417	0.0183	0.0315
	1–1/8	0.040	1/8	0.091	2.27	0.289	0.353	0.0116	0.0147
	1–1/8	0.064	3/16	0.142	2.22	0.303	0.368	0.0175	0.0226
	1–1/8	0.081	1/4	0.176	2.17	0.314	0.380	0.0211	0.0276
	1–1/8	0.102	11/32	0.216	2.12	0.328	0.395	0.0257	0.0331
1–3/8	5/16	0.040	1/8	0.063	1.50	0.048	0.600	0.0003	0.0118
	5/16	0.051	5/32	0.079	1.56	0.053	0.611	0.0004	0.0146
	5/16	0.064	3/16	0.098	1.53	0.060	0.624	0.0004	0.0174
	5/16	0.072	7/32	0.108	1.51	0.064	0.634	0.0005	0.0189
	3/4	0.040	1/8	0.081	2.02	0.152	0.475	0.0040	0.0165
	3/4	0.064	3/16	0.126	1.97	0.165	0.492	0.0057	0.0251
	3/4	0.081	1/4	0.156	1.92	0.174	0.508	0.0069	0.0303
	3/4	0.102	11/32	0.190	1.87	0.187	0.426	0.0082	0.0361
	7/8	0.040	1/8	0.086	2.15	0.191	0.448	0.0060	0.0175
	7/8	0.064	3/16	0.134	2.09	0.203	0.464	0.0088	0.0267
	7/8	0.081	1/4	0.166	2.04	0.214	0.479	0.0108	0.0324
	7/8	0.102	11/32	0.201	1.98	0.227	0.500	0.0129	0.0384
	1	0.040	1/8	0.091	2.27	0.232	0.425	0.0087	0.0183
	1	0.064	3/16	0.142	2.22	0.244	0.440	0.0130	0.0281
	1	0.081	1/4	0.176	2.17	0.255	0.454	0.0158	0.0343
	1	0.102	11/32	0.215	2.10	0.270	0.473	0.0189	0.0409

4.4 Unequal Leg Formed Angle (Continued)

FIGURE 4–4 UNEQUAL LEG FORMED ANGLE (CONTINUED)

H in	B in	t in	R in	Area in^2	Devel. Width in	$\bar{x}$ in	$\bar{y}$ in	I_y in^4	I_x in^4
1–3/8	1–1/8	0.040	1/8	0.096	2.40	0.275	0.403	0.0118	0.0190
	1–1/8	0.064	3/16	0.150	2.34	0.288	0.419	0.0181	0.0293
	1–1/8	0.081	1/4	0.186	2.30	0.299	0.431	0.0220	0.0359
	1–1/8	0.102	11/32	0.227	2.23	0.314	0.449	0.0266	0.0431
	1–1/4	0.040	1/8	0.101	2.52	0.320	0.384	0.0157	0.0198
	1–1/4	0.064	3/16	0.158	2.47	0.334	0.399	0.0242	0.0305
	1–1/4	0.081	1/4	0.196	2.42	0.345	0.411	0.0297	0.0373
	1–1/4	0.102	11/32	0.241	2.37	0.359	0.425	0.0359	0.0452
1–1/2	5/16	0.040	1/8	0.068	1.71	0.046	0.661	0.0003	0.0151
	5/16	0.051	5/32	0.086	1.66	0.051	0.672	0.0004	0.0186
	5/16	0.064	3/16	0.106	1.65	0.058	0.685	0.0004	0.0223
	5/16	0.072	7/32	0.117	1.63	0.062	0.696	0.0005	0.0242
	3/4	0.051	5/32	0.108	2.12	0.150	0.539	0.0048	0.0260
	3/4	0.072	7/32	0.149	2.07	0.161	0.556	0.0065	0.0351
	3/4	0.091	9/32	0.184	2.02	0.172	0.573	0.0078	0.0423
	3/4	0.102	11/32	0.202	1.98	0.180	0.587	0.0084	0.0456
	7/8	0.051	5/32	0.115	2.25	0.187	0.510	0.0075	0.0275
	7/8	0.072	7/32	0.158	2.19	0.199	0.526	0.0101	0.0374
	7/8	0.091	9/32	0.195	2.14	0.209	0.542	0.0122	0.0452
	7/8	0.102	11/32	0.214	2.10	0.217	0.555	0.0132	0.0490
	1	0.051	5/32	0.121	2.37	0.227	0.485	0.0109	0.0290
	1	0.072	7/32	0.167	2.32	0.238	0.499	0.0147	0.0396
	1	0.091	9/32	0.206	2.27	0.250	0.515	0.0179	0.0479
	1	0.102	11/32	0.227	2.23	0.258	0.527	0.0195	0.0521
	1–1/8	0.051	5/32	0.127	2.50	0.268	0.461	0.0152	0.0303
	1–1/8	0.072	7/32	0.176	2.44	0.280	0.476	0.0205	0.0413
	1–1/8	0.091	9/32	0.218	2.39	0.292	0.490	0.0250	0.0503
	1–1/8	0.192	11/32	0.240	2.35	0.301	0.501	0.0274	0.0584
	1–1/4	0.051	5/32	0.134	2.62	0.312	0.441	0.0202	0.0314
	1–1/4	0.072	7/32	0.185	2.57	0.325	0.455	0.0276	0.0429
	1–1/4	0.091	9/32	0.229	2.52	0.336	0.468	0.0338	0.0525
	1–1/4	0.102	11/32	0.254	2.49	0.343	0.476	0.0370	0.0576

SECTION PROPERTIES

4.4 Unequal Leg Formed Angle (Continued)

FIGURE 4–4 UNEQUAL LEG FORMED ANGLE (CONTINUED)

H in	B in	t in	R in	Area in^2	Devel. Width in	$\bar{x}$ in	$\bar{y}$ in	I_y in^4	I_x in^4
1–1/2	1–3/8	0.051	5/32	0.140	2.75	0.358	0.422	0.0264	0.0325
	1–3/8	0.072	7/32	0.194	2.69	0.371	0.435	0.0360	0.0444
	1–3/8	0.091	9/32	0.240	2.65	0.383	0.448	0.0439	0.0544
	1–3/8	0.102	11/32	0.268	2.60	0.392	0.458	0.0482	0.0595
1–5/8	7/8	0.051	5/32	0.121	2.37	0.178	0.566	0.0076	0.0343
	7/8	0.072	7/32	0.167	2.32	0.190	0.582	0.0103	0.0465
	7/8	0.091	9/32	0.207	2.27	0.200	0.599	0.0125	0.0564
	7/8	0.102	11/32	0.227	2.23	0.208	0.617	0.0136	0.0612
	1	0.051	5/32	0.127	2.50	0.217	0.539	0.0111	0.0361
	1	0.072	7/32	0.176	2.45	0.228	0.555	0.0151	0.0491
	1	0.091	9/32	0.218	2.40	0.239	0.570	0.0184	0.0593
	1	0.102	11/32	0.240	2.35	0.247	0.532	0.0200	0.0651
	1–1/8	0.051	5/32	0.134	2.62	0.257	0.514	0.0155	0.0377
	1–1/8	0.072	7/32	0.185	2.57	0.268	0.529	0.0210	0.0514
	1–1/8	0.091	9/32	0.229	2.52	0.279	0.544	0.0257	0.0628
	1–1/8	0.102	11/32	0.253	2.48	0.283	0.555	0.0281	0.0685
	1–1/4	0.051	5/32	0.140	2.75	0.299	0.492	0.0207	0.0391
	1–1/4	0.072	7/32	0.194	2.69	0.311	0.506	0.0283	0.0555
	1–1/4	0.091	9/32	0.240	2.64	0.322	0.529	0.0347	0.0655
	1–1/4	0.102	11/32	0.265	2.60	0.331	0.531	0.0379	0.0716
	1–3/8	0.051	5/32	0.146	2.87	0.343	0.472	0.0271	0.0404
	1–3/8	0.072	7/32	0.203	2.82	0.356	0.486	0.0369	0.0554
	1–3/8	0.091	9/32	0.252	2.77	0.367	0.499	0.0452	0.0679
	1–3/8	0.102	11/32	0.278	2.73	0.377	0.509	0.0496	0.0744
	1–1/2	0.051	5/32	0.153	3.00	0.389	0.453	0.0343	0.0417
	1–1/2	0.072	7/32	0.212	2.95	0.401	0.466	0.0471	0.0571
	1–1/2	0.091	9/32	0.263	2.90	0.413	0.479	0.0579	0.0702
	1–1/2	0.102	11/32	0.291	2.85	0.423	0.489	0.0633	0.0770
1–3/4	3/8	0.051	5/32	0.102	2.00	0.057	0.771	0.0006	0.0306
	3/8	0.064	3/16	0.126	1.97	0.063	0.784	0.0008	0.0370
	3/8	0.072	7/32	0.140	1.95	0.068	0.794	0.0009	0.0405
	3/8	0.081	1/4	0.155	1.92	0.073	0.804	0.0009	0.0441

SECTION PROPERTIES

4.4 Unequal Leg Formed Angle (Continued)

FIGURE 4–4 UNEQUAL LEG FORMED ANGLE (CONTINUED)

H in	B in	t in	R in	Area in^2	Devel. Width in	$\bar{x}$ in	$\bar{y}$ in	I_y in^4	I_x in^4
1–3/4	7/8	0.051	5/32	0.127	2.50	0.171	0.622	0.0078	0.0419
	7/8	0.072	7/32	0.176	2.45	0.182	0.639	0.0105	0.0570
	7/8	0.091	9/32	0.218	2.40	0.192	0.655	0.0127	0.0693
	7/8	0.102	11/32	0.240	2.35	0.199	0.669	0.0139	0.0753
	1	0.051	5/32	0.134	2.62	0.208	0.593	0.0113	0.0441
	1	0.072	7/32	0.185	2.57	0.218	0.610	0.0154	0.0601
	1	0.091	9/32	0.229	2.52	0.229	0.625	0.0188	0.0733
	1	0.102	11/32	0.253	2.48	0.236	0.634	0.0206	0.0805
	1–1/8	0.051	5/32	0.140	1.75	0.247	0.568	0.0158	0.0460
	1–1/8	0.072	7/32	0.194	2.69	0.257	0.583	0.0215	0.0629
	1–1/8	0.091	9/32	0.240	2.64	0.268	0.598	0.0263	0.0770
	1–1/8	0.102	11/32	0.265	2.60	0.277	0.609	0.0288	0.0841
	1–1/4	0.051	5/32	0.146	2.87	0.287	0.544	0.0212	0.0478
	1–1/4	0.072	7/32	0.203	2.82	0.299	0.559	0.0290	0.0655
	1–1/4	0.091	9/32	0.252	2.77	0.310	0.573	0.0356	0.0803
	1–1/4	0.102	11/32	0.278	2.73	0.318	0.584	0.0389	0.0878
	1–3/8	0.051	5/32	0.153	3.00	0.330	0.522	0.0277	0.0495
	1–3/8	0.072	7/32	0.212	2.95	0.342	0.536	0.0379	0.0678
	1–3/8	0.091	9/32	0.263	2.90	0.353	0.550	0.0463	0.0875
	1–3/8	0.102	11/32	0.291	2.85	0.362	0.561	0.0509	0.0913
	1–1/2	0.051	5/32	0.159	3.12	0.375	0.502	0.0351	0.0510
	1–1/2	0.072	7/32	0.221	3.07	0.386	0.516	0.0483	0.0700
	1–1/2	0.091	9/32	0.275	3.02	0.398	0.529	0.0594	0.0861
	1–1/2	0.102	11/32	0.304	2.98	0.407	0.539	0.0650	0.0965
	1–5/8	0.051	5/32	0.166	3.25	0.420	0.484	0.0439	0.0524
	1–5/8	0.072	7/32	0.230	3.19	0.432	0.497	0.0601	0.0720
	1–5/8	0.091	9/32	0.286	3.15	0.444	0.510	0.0742	0.0867
	1–5/8	0.102	11/32	0.316	3.10	0.454	0.519	0.0814	0.0975

4.4 Unequal Leg Formed Angle (Continued)

FIGURE 4–4 UNEQUAL LEG FORMED ANGLE (CONTINUED)

H in	B in	t in	R in	Area in^2	Devel. Width in	$\bar{x}$ in	$\bar{y}$ in	I_y in^4	I_x in^4
1–7/8	3/8	0.051	5/32	0.108	2.12	0.055	0.833	0.0007	0.0381
	3/8	0.064	3/16	0.134	2.09	0.062	0.845	0.0008	0.0450
	3/8	0.072	7/32	0.149	2.07	0.066	0.856	0.0009	0.0493
	3/8	0.081	1/4	0.166	2.05	0.071	0.866	0.0009	0.0528
	15/16	0.051	5/32	0.137	2.68	0.181	0.664	0.0096	0.0519
	15/16	0.072	7/32	0.190	2.63	0.192	0.681	0.0130	0.0707
	15/16	0.091	9/32	0.235	2.58	0.202	0.696	0.0159	0.0863
	15/16	0.102	11/32	0.259	2.54	0.210	0.710	0.0173	0.0940
	1–1/8	0.051	5/32	0.146	2.87	0.237	0.622	0.0161	0.0555
	1–1/8	0.072	7/32	0.203	2.82	0.247	0.638	0.0220	0.0759
	1–1/8	0.091	9/32	0.252	2.77	0.258	0.652	0.0269	0.9931
	1–1/8	0.102	11/32	0.278	2.73	0.266	0.665	0.0294	0.1017
	1–5/16	0.051	5/32	0.156	3.06	0.297	0.585	0.0248	0.0587
	1–5/16	0.072	7/32	0.217	3.01	0.308	0.600	0.0330	0.0805
	1–5/16	0.091	9/32	0.269	2.96	0.319	0.615	0.0417	0.0989
	1–5/16	0.102	11/32	0.297	2.92	0.327	0.625	0.0457	0.1036
	1–1/2	0.051	5/32	0.166	3.25	0.361	0.553	0.0358	0.0615
	1–1/2	0.072	7/32	0.230	3.19	0.373	0.567	0.0493	0.0846
	1–1/2	0.091	9/32	0.286	3.15	0.384	0.580	0.0607	0.1041
	1–1/2	0.102	11/32	0.316	3.10	0.393	0.591	0.0666	0.1144
	1–11/16	0.051	5/32	0.175	3.43	0.428	0.524	0.0497	0.0640
	1–11/16	0.072	7/32	0.244	3.38	0.440	0.538	0.0683	0.0882
	1–11/16	0.091	9/32	0.303	3.33	0.452	0.550	0.0843	0.1087
	1–11/16	0.102	11/32	0.336	3.29	0.461	0.560	0.0928	0.1197
2	1/2	0.051	5/32	0.121	2.37	0.074	0.848	0.0016	0.0490
	1/2	0.064	3/16	0.150	2.34	0.080	0.860	0.0019	0.0598
	1/2	0.072	7/32	0.167	2.32	0.084	0.870	0.0021	0.0658
	1/2	0.081	1/4	0.186	2.29	0.089	0.880	0.0023	0.0723
	1–1/16	0.051	5/32	0.150	2.95	0.210	0.691	0.0139	0.0647
	1–1/16	0.072	7/32	0.208	2.88	0.220	0.707	0.0190	0.0885
	1–1/16	0.091	9/32	0.257	2.83	0.231	0.722	0.0232	0.1084
	1–1/16	0.102	11/32	0.285	2.79	0.238	0.735	0.0254	0.1185

SECTION PROPERTIES

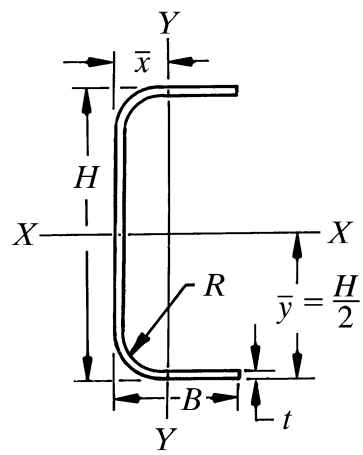
4.4 Unequal Leg Formed Angle (Continued)

FIGURE 4–4 UNEQUAL LEG FORMED ANGLE (CONCLUDED)

H in	B in	t in	R in	Area in^2	Devel. Width in	$\bar{x}$ in	$\bar{y}$ in	I_y in^4	I_x in^4
2	1–1/4	0.051	5/32	0.159	3.12	0.266	0.651	0.0220	0.0687
	1–1/4	0.072	7/32	0.221	3.07	0.277	0.666	0.0301	0.0942
	1–1/4	0.091	9/32	0.274	3.02	0.289	0.680	0.0370	0.1137
	1–1/4	0.102	11/32	0.304	2.98	0.296	0.692	0.0406	0.1270
	1–7/16	0.051	5/32	0.169	3.31	0.327	0.615	0.0325	0.0722
	1–7/16	0.072	7/32	0.235	3.26	0.330	0.630	0.0446	0.0992
	1–7/16	0.091	9/32	0.291	3.20	0.351	0.643	0.0548	0.1222
	1–7/16	0.102	11/32	0.323	3.16	0.358	0.654	0.0604	0.1343
	1–5/8	0.051	5/32	0.178	3.50	0.392	0.584	0.0457	0.0754
	1–5/8	0.072	7/32	0.248	3.44	0.403	0.597	0.0627	0.1038
	1–5/8	0.091	9/32	0.308	3.39	0.416	0.609	0.0777	0.1260
	1–5/8	0.102	11/32	0.342	3.35	0.424	0.620	0.0853	0.1410
	1–13/16	0.051	5/32	0.188	3.68	0.460	0.555	0.0617	0.0782
	1–13/16	0.072	7/32	0.262	3.63	0.472	0.568	0.0850	0.1078
	1–13/16	0.091	9/32	0.325	3.58	0.487	0.580	0.1039	0.1331
	1–13/16	0.102	11/32	0.361	3.54	0.492	0.590	0.1158	0.1468

4.5 Equal Leg Formed Channel

FIGURE 4–5 EQUAL LEG FORMED CHANNEL (CONTINUED ON NEXT PAGE)



H in	B in	t in	R in	Area in^2	Devel. Width in	$\bar{x}$ in	I_y in^4	I_x in^4	ρ_y in	ρ_x in
5/8	1/4	0.032	3/32	0.031	0.97	0.074	0.0002	0.0016	0.075	0.226
	1/4	0.040	1/8	0.037	0.92	0.080	0.0002	0.0016	0.073	0.218
3/4	1/4	0.032	3/32	0.035	1.09	0.068	0.0002	0.0025	0.072	0.267
	1/4	0.040	1/8	0.042	1.05	0.073	0.0002	0.0028	0.071	0.259
7/8	1/4	0.032	3/32	0.039	1.22	0.062	0.0002	0.0037	0.070	0.308
	1/4	0.040	1/8	0.047	1.17	0.068	0.0002	0.0042	0.069	0.299
	3/8	0.032	3/32	0.047	1.47	0.105	0.0006	0.0051	0.115	0.330
	3/8	0.040	1/8	0.057	1.43	0.111	0.0007	0.0059	0.113	0.323
1	1/4	0.032	3/32	0.043	1.34	0.058	0.0002	0.0052	0.068	0.347
	1/4	0.040	1/8	0.052	1.30	0.063	0.0002	0.0059	0.067	0.330
	3/8	0.040	1/8	0.062	1.55	0.103	0.0008	0.0082	0.112	0.385
	3/8	0.051	5/32	0.076	1.49	0.110	0.0009	0.0096	0.110	0.356
	3/8	0.064	3/16	0.092	1.43	0.118	0.0011	0.0110	0.108	0.346
1–1/8	1/4	0.032	3/32	0.047	1.47	0.054	0.0002	0.0071	0.066	0.386
	1/4	0.040	1/8	0.057	1.42	0.059	0.0002	0.0081	0.066	0.377
	1/2	0.040	1/8	0.077	1.92	0.141	0.0018	0.0140	0.154	0.427
	1/2	0.051	5/32	0.095	1.87	0.148	0.0022	0.0167	0.152	0.419
	1/2	0.064	3/16	0.116	1.81	0.157	0.0026	0.0194	0.150	0.410
1–1/4	3/8	0.040	1/8	0.072	1.79	0.092	0.0008	0.0144	0.108	0.447
	3/8	0.051	5/32	0.089	1.75	0.098	0.0010	0.0170	0.106	0.438
	3/8	0.064	3/16	0.108	1.68	0.106	0.0012	0.0196	0.104	0.427

SECTION PROPERTIES

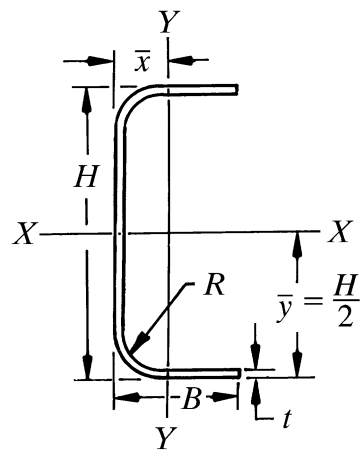
4.5 Equal Leg Formed Channel (Continued)

FIGURE 4–5 EQUAL LEG FORMED CHANNEL (CONCLUDED)

H in	B in	t in	R in	Area in^2	Devel. Width in	$\bar{x}$ in	I_y in^4	I_x in^4	ρ_y in	ρ_x in
1–3/8	3/8	0.040	1/8	0.077	1.92	0.087	0.0009	0.0182	0.106	0.487
	3/8	0.051	5/32	0.095	1.87	0.093	0.0010	0.0217	0.104	0.478
	3/8	0.064	3/16	0.116	1.81	0.100	0.0012	0.0252	0.102	0.467
1–1/2	1/2	0.051	5/32	0.114	2.25	0.128	0.0025	0.0339	0.146	0.544
	1/2	0.064	3/16	0.140	2.18	0.135	0.0029	0.0399	0.145	0.534
	1/2	0.072	7/32	0.154	2.14	0.141	0.0032	0.0427	0.143	0.527
1–3/4	1/2	0.051	5/32	0.125	2.49	0.118	0.0026	0.0496	0.142	0.625
	1/2	0.064	3/16	0.156	2.43	0.125	0.0031	0.0589	0.141	0.615
	1/2	0.072	7/32	0.172	2.39	0.130	0.0033	0.0633	0.140	0.607
2	5/8	0.051	5/32	0.153	2.99	0.147	0.0051	0.0816	0.183	0.731
	5/8	0.064	3/16	0.188	2.93	0.154	0.0061	0.0978	0.181	0.721
	5/8	0.081	1/4	0.230	2.84	0.165	0.0073	0.1147	0.179	0.706
2–1/2	3/4	0.064	3/16	0.236	3.68	0.174	0.0111	0.1946	0.217	0.909
	3/4	0.072	7/32	0.262	3.64	0.178	0.0122	0.2128	0.216	0.901
	3/4	0.091	9/32	0.322	3.54	0.190	0.0147	0.2521	0.214	0.885
3	3/4	0.064	3/16	0.268	4.18	0.157	0.0117	0.3059	0.209	1.069
	3/4	0.091	9/32	0.367	4.04	0.172	0.0156	0.4004	0.206	1.044
	3/4	0.102	11/32	0.404	3.96	0.180	0.0168	0.4279	0.204	1.030

4.5 Equal Leg Formed Channel (Continued)

FIGURE 4–6 EQUAL LEG FORMED CHANNEL (CONTINUED ON NEXT PAGE)



H in	B in	t in	R in	Area in^2	Devel. Width in	$\bar{x}$ in	I_y in^4	I_x in^4
1/2	1/4	0.032	3/32	0.027	0.84	0.083	0.0002	0.0009
	3/8	0.032	3/32	0.035	1.09	0.135	0.0005	0.0014
5/8	3/8	0.032	3/32	0.039	1.22	0.124	0.0005	0.0023
	3/8	0.040	1/8	0.047	1.17	0.130	0.0009	0.0026
	1/2	0.032	3/32	0.047	1.47	0.177	0.0012	0.0030
	1/2	0.040	1/8	0.057	1.42	0.184	0.0014	0.0035
3/4	3/8	0.032	3/32	0.043	1.34	0.113	0.0006	0.0035
	3/8	0.040	1/8	0.052	1.30	0.119	0.0007	0.0041
	1/2	0.032	3/32	0.051	1.59	0.164	0.0013	0.0048
	1/2	0.040	1/8	0.062	1.55	0.170	0.0015	0.0053
	5/8	0.032	3/32	0.059	1.84	0.218	0.0024	0.0056
	5/8	0.040	1/8	0.072	1.80	0.225	0.0029	0.0066
7/8	1/2	0.032	3/32	0.055	1.72	0.153	0.0014	0.0065
	1/2	0.040	1/8	0.067	1.67	0.160	0.0017	0.0077
	1/2	0.051	5/32	0.083	1.82	0.167	0.0020	0.0090
	1/2	0.064	3/16	0.100	1.56	0.177	0.0023	0.0104
	5/8	0.032	3/32	0.063	1.97	0.205	0.0026	0.0080
	5/8	0.040	1/8	0.077	1.92	0.212	0.0031	0.0094
	5/8	0.051	5/32	0.095	1.87	0.220	0.0037	0.0112
	5/8	0.064	3/16	0.116	1.81	0.230	0.0044	0.0131
	3/4	0.040	1/8	0.087	2.17	0.267	0.0051	0.0112
	3/4	0.051	5/32	0.108	2.12	0.275	0.0062	0.0134
	3/4	0.064	3/16	0.132	2.06	0.285	0.0074	0.0157

SECTION PROPERTIES

4.5 Equal Leg Formed Channel (Continued)

FIGURE 4–6 EQUAL LEG FORMED CHANNEL (CONTINUED)

H in	B in	t in	R in	Area in^2	Devel. Width in	$\bar{x}$ in	I_y in^4	I_x in^4
1	1/2	0.040	1/8	0.072	1.80	0.150	0.0017	0.0106
	1/2	0.051	5/32	0.089	1.75	0.157	0.0021	0.0126
	1/2	0.064	3/16	0.108	1.88	0.166	0.0025	0.0146
	5/8	0.040	1/8	0.082	2.05	0.200	0.0033	0.0129
	5/8	0.051	5/32	0.102	1.99	0.208	0.0040	0.0154
	5/8	0.064	3/16	0.124	1.93	0.217	0.0047	0.0180
	3/4	0.040	1/8	0.092	2.30	0.253	0.0054	0.0152
	3/4	0.051	5/32	0.114	2.24	0.261	0.0066	0.0183
	3/4	0.064	3/16	0.140	2.18	0.271	0.0078	0.0215
	7/8	0.040	1/8	0.102	2.25	0.308	0.0082	0.0175
	7/8	0.051	5/32	0.127	2.49	0.317	0.0101	0.0212
	7/8	0.064	3/16	0.156	2.43	0.326	0.0123	0.0233
1–1/8	3/4	0.040	1/8	0.097	2.42	0.241	0.0056	0.0199
	3/4	0.051	5/32	0.121	2.37	0.249	0.0069	0.0241
	3/4	0.064	3/16	0.148	2.31	0.258	0.0083	0.0285
	3/4	0.072	7/32	0.163	2.26	0.265	0.0090	0.0306
	1	0.040	1/8	0.117	2.92	0.350	0.0124	0.0258
	1	0.051	5/32	0.146	2.87	0.358	0.0153	0.0314
	1	0.064	3/16	0.180	2.81	0.368	0.0185	0.0375
	1	0.072	7/32	0.199	2.76	0.375	0.0201	0.0406
1–1/4	5/8	0.040	1/8	0.092	2.30	0.181	0.0035	0.0217
	5/8	0.064	3/16	0.140	2.18	0.196	0.0052	0.0309
	5/8	0.072	7/32	0.154	2.14	0.202	0.0056	0.0331
	7/8	0.040	1/8	0.112	2.80	0.283	0.0090	0.0290
	7/8	0.064	3/16	0.172	2.83	0.299	0.0133	0.0422
	7/8	0.072	7/32	0.190	2.64	0.306	0.0145	0.0456
	1–1/8	0.040	1/8	0.132	3.30	0.391	0.0178	0.0363
	1–1/8	0.064	3/16	0.204	3.18	0.409	0.0268	0.0534
	1–1/8	0.072	7/32	0.226	3.14	0.417	0.0293	0.0581

4.5 Equal Leg Formed Channel (Continued)

FIGURE 4–6 EQUAL LEG FORMED CHANNEL (CONTINUED)

H in	B in	t in	R in	Area in^2	Devel. Width in	$\bar{x}$ in	I_y in^4	I_x in^4
1–3/8	5/8	0.040	1/8	0.097	2.42	0.172	0.0037	0.0271
	5/8	0.064	3/16	0.148	2.31	0.187	0.0054	0.0390
	5/8	0.072	7/32	0.163	2.26	0.193	0.0058	0.0419
	7/8	0.040	1/8	0.117	2.92	0.271	0.0093	0.0361
	7/8	0.064	3/16	0.180	2.81	0.287	0.0139	0.0527
	7/8	0.072	7/32	0.199	2.76	0.294	0.0152	0.0572
	1–1/8	0.051	5/32	0.172	3.37	0.386	0.0229	0.0553
	1–1/8	0.064	3/16	0.212	3.31	0.395	0.0279	0.0665
	1–1/8	0.081	1/4	0.260	3.21	0.410	0.0334	0.0787
1–1/2	3/4	0.051	5/32	0.140	2.74	0.218	0.0077	0.0473
	3/4	0.064	3/16	0.172	2.68	0.226	0.0093	0.0564
	3/4	0.081	1/4	0.210	2.59	0.239	0.0111	0.0648
	1	0.051	5/32	0.165	3.24	0.320	0.0172	0.0607
	1	0.064	3/16	0.204	3.18	0.328	0.0208	0.0729
	1	0.081	1/4	0.250	3.09	0.342	0.0250	0.0862
	1–1/4	0.051	5/32	0.191	3.74	0.427	0.0316	0.0740
	1–1/4	0.064	3/16	0.238	3.68	0.436	0.0386	0.0894
	1–1/4	0.081	1/4	0.291	3.59	0.451	0.0466	0.1066
1–3/4	7/8	0.051	5/32	0.165	3.24	0.249	0.0126	0.0773
	7/8	0.064	3/16	0.204	3.18	0.257	0.0153	0.0930
	7/8	0.072	7/32	0.226	3.14	0.263	0.0168	0.1014
	7/8	0.081	1/4	0.250	3.09	0.270	0.0183	0.1101
	1–1/4	0.064	3/16	0.252	3.93	0.411	0.0410	0.1275
	1–1/4	0.072	7/32	0.280	3.89	0.417	0.0453	0.1394
	1–1/4	0.081	1/4	0.311	3.84	0.424	0.0498	0.1524
	1–1/4	0.091	9/32	0.345	3.79	0.432	0.0546	0.1660
	1–5/8	0.064	3/16	0.300	4.68	0.575	0.0856	0.1613
	1–5/8	0.072	7/32	0.334	4.64	0.582	0.0930	0.1774
	1–5/8	0.081	1/4	0.372	4.59	0.590	0.1027	0.1948
	1–5/8	0.091	9/32	0.413	4.54	0.598	0.1130	0.2130

SECTION PROPERTIES

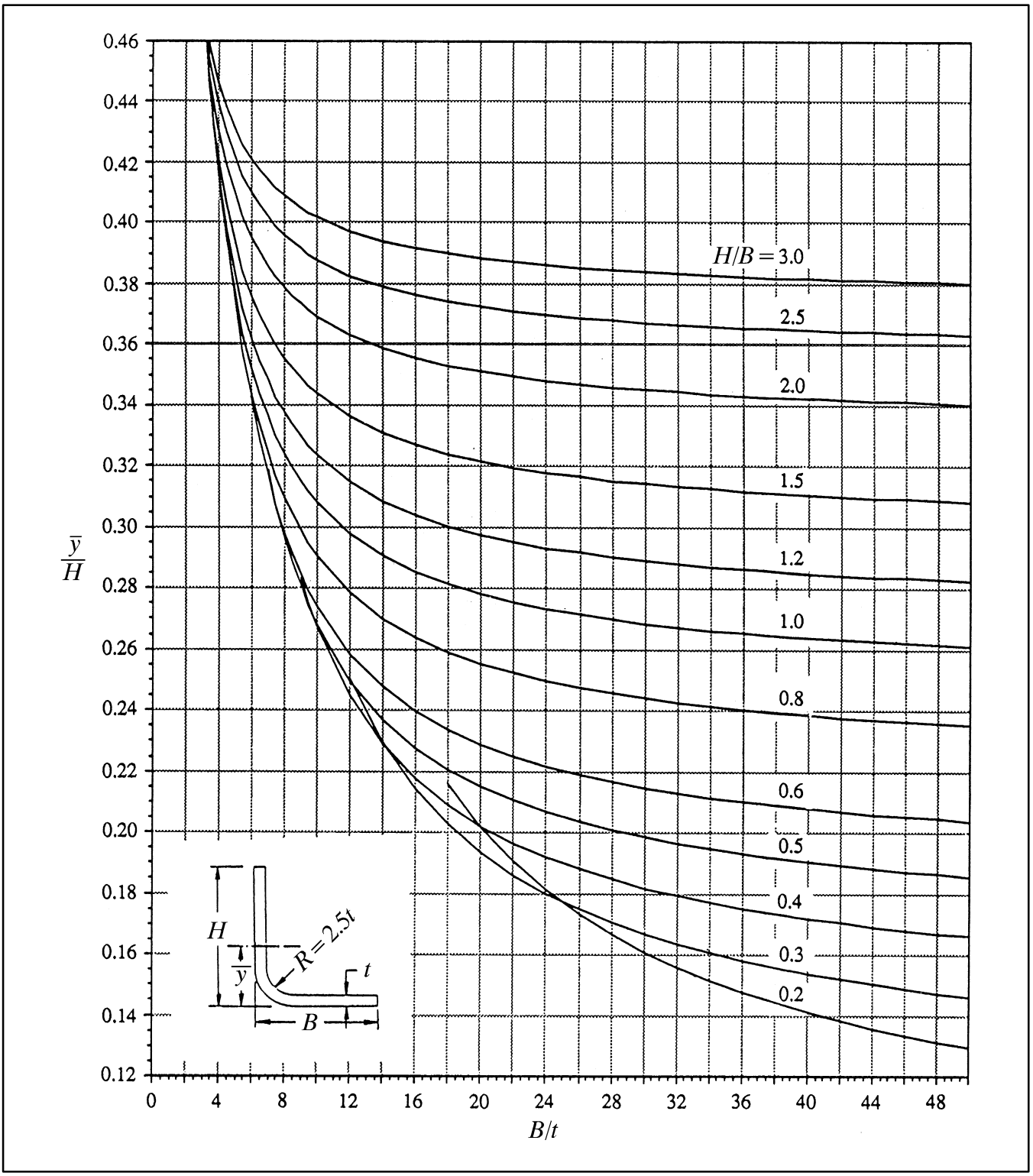
4.5 Equal Leg Formed Channel (Continued)

FIGURE 4–6 EQUAL LEG FORMED CHANNEL (CONCLUDED)

H in	B in	t in	R in	Area in^2	Devel. Width in	$\bar{x}$ in	I_y in^4	I_x in^4
2	1	0.064	3/16	0.235	3.68	0.288	0.0233	0.1438
	1	0.081	1/4	0.291	3.59	0.300	0.0282	0.1706
	1	0.091	9/32	0.322	3.54	0.307	0.0309	0.1855
	1–3/8	0.064	3/16	0.284	4.43	0.440	0.0561	0.1878
	1–3/8	0.081	1/4	0.351	4.34	0.453	0.0685	0.2266
	1–3/8	0.091	9/32	0.390	4.29	0.461	0.0753	0.2478
	1–3/4	0.064	3/16	0.332	5.18	0.603	0.1084	0.2328
	1–3/4	0.081	1/4	0.412	5.09	0.617	0.1329	0.2826
	1–3/4	0.091	9/32	0.458	5.04	0.625	0.1466	0.3100
2–1/4	1	0.064	3/16	0.252	3.95	0.272	0.0243	0.1885
	1–1/4	0.064	3/16	0.284	4.43	0.368	0.0450	0.2288
	1–3/4	0.064	3/16	0.348	5.43	0.576	0.1133	0.3032
2–1/2	1–1/4	0.064	3/16	0.300	4.66	0.350	0.0468	0.2896
	1–1/4	0.072	7/32	0.334	4.64	0.356	0.0519	0.3190
	1–1/4	0.091	9/32	0.413	4.54	0.368	0.0632	0.3842
	1–3/4	0.064	3/16	0.364	5.68	0.552	0.1179	0.3846
	1–3/4	0.072	7/32	0.406	5.64	0.559	0.1309	0.4251
	1–3/4	0.091	9/32	0.504	5.54	0.573	0.1606	0.5136
2–3/4	1–1/4	0.064	3/16	0.316	4.93	0.334	0.0484	0.2752
	1–1/2	0.064	3/16	0.348	5.43	0.430	0.0800	0.4195
	2	0.064	3/16	0.412	6.43	0.635	0.1756	0.5350
3	1–1/2	0.064	3/16	0.364	5.68	0.412	0.0813	0.5130
	1–1/2	0.091	9/32	0.504	5.54	0.430	0.1124	0.6893
	1–1/2	0.102	11/32	0.557	5.46	0.440	0.1231	0.7492
	2	0.072	7/32	0.478	6.64	0.618	0.2016	0.7214
	2	0.091	9/32	0.595	6.54	0.632	0.2488	0.8819
	2	0.102	11/32	0.659	6.46	0.642	0.2734	0.9635

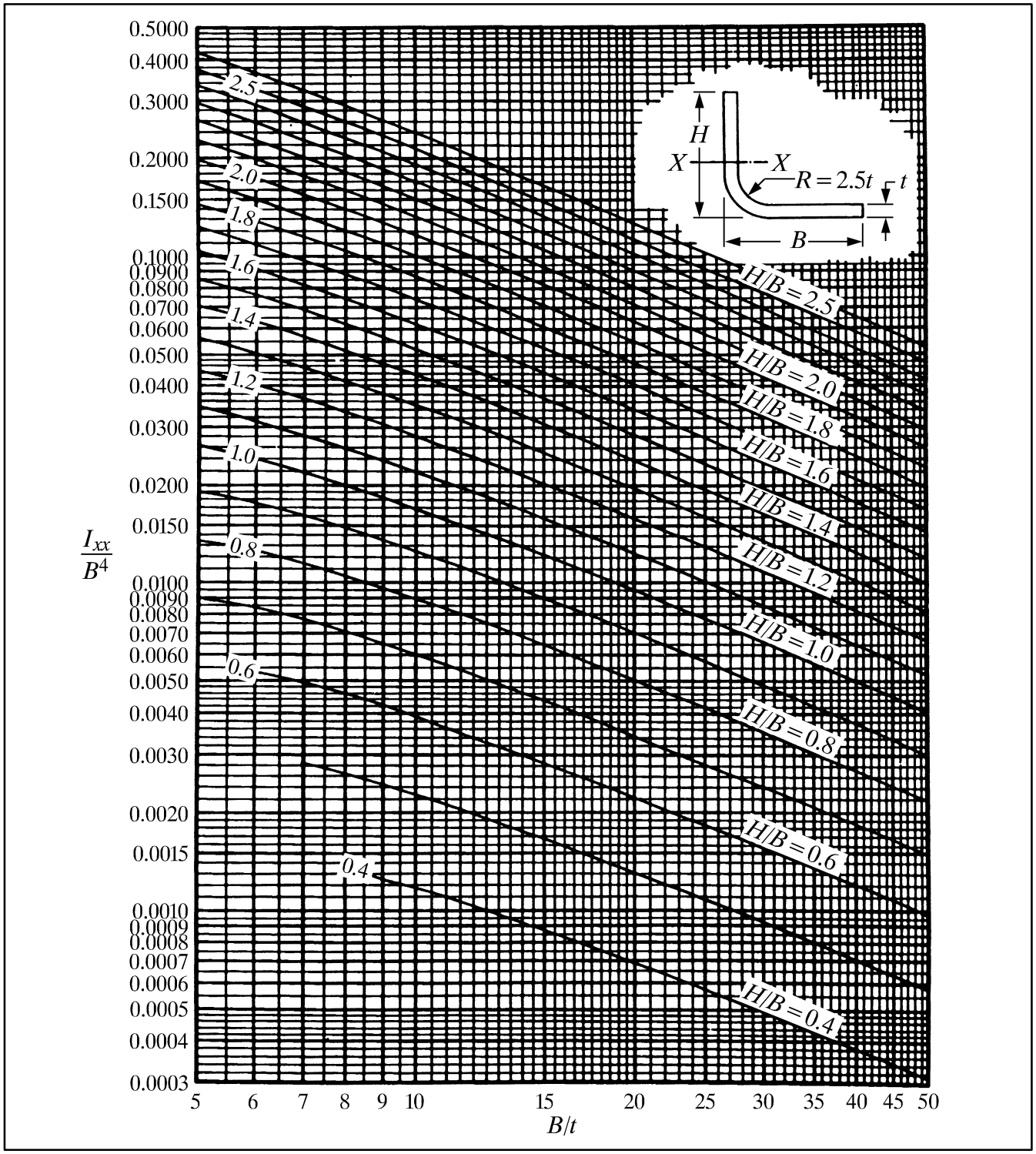
4.6 Centroid of Formed Angle

FIGURE 4–7 CENTROID OF FORMED ANGLE



4.7 Moment of Inertia of Formed Angle

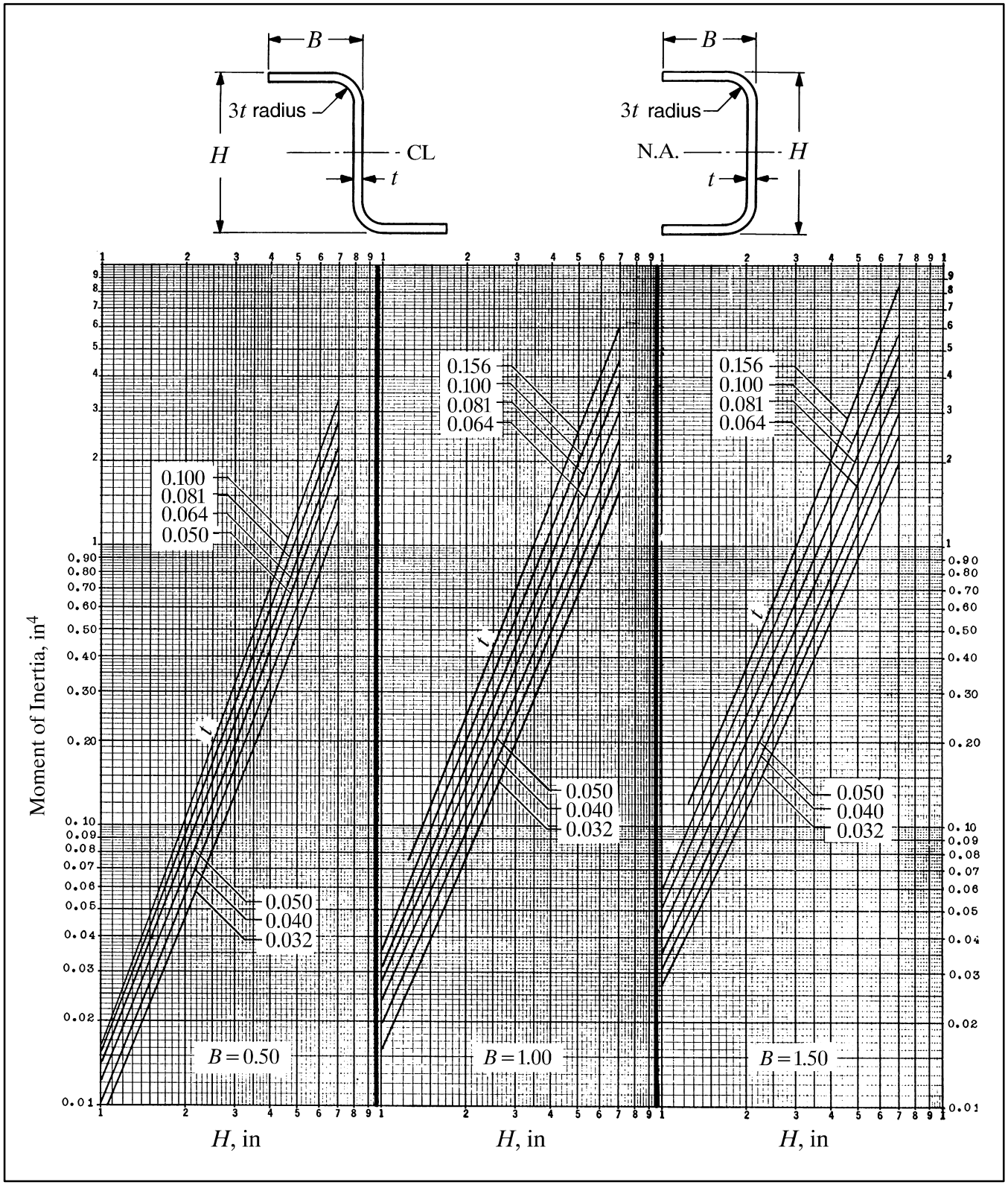
FIGURE 4–8 MOMENT OF INERTIA OF FORMED ANGLE



SECTION PROPERTIES

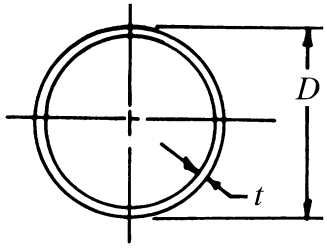
4.8 Moment of Inertia of Formed Channel and Zee Section

FIGURE 4–9 MOMENT OF INERTIA OF FORMED CHANNEL AND ZEE SECTION



4.9 Section Properties of Round Tubing

FIGURE 4–10 SECTION PROPERTIES OF ROUND TUBING (CONTINUED ON NEXT PAGE)



D in	t in	Area in^2	I in^4	ρ in
3/8	0.022	0.0244	0.000382	0.125
	0.028	0.0305	0.000462	0.123
	0.035	0.0374	0.000546	0.121
	0.049	0.0502	0.000682	0.117
	0.058	0.0578	0.000750	0.114
1/2	0.028	0.0415	0.001160	0.167
	0.035	0.0511	0.001390	0.165
	0.049	0.0694	0.001790	0.160
	0.058	0.0805	0.002000	0.158
	0.065	0.0888	0.002150	0.156
5/8	0.028	0.0525	0.002340	0.211
	0.035	0.0649	0.002830	0.209
	0.049	0.0887	0.003700	0.204
	0.058	0.1030	0.004200	0.202
	0.065	0.1140	0.004540	0.199
3/4	0.028	0.0635	0.004140	0.255
	0.035	0.0786	0.005040	0.253
	0.049	0.1080	0.006660	0.248
	0.058	0.1260	0.007600	0.246
	0.065	0.1400	0.008280	0.243
	0.083	0.1740	0.009820	0.238
7/8	0.028	0.0745	0.006690	0.300
	0.035	0.0924	0.008160	0.297
	0.049	0.1270	0.010900	0.293
	0.058	0.1490	0.012500	0.290
	0.065	0.1650	0.013700	0.287

SECTION PROPERTIES

4.9 Section Properties of Round Tubing (Continued)

FIGURE 4–10 SECTION PROPERTIES OF ROUND TUBING (CONTINUED)

D in	t in	Area in^2	I in^4	ρ in
1	0.035	0.1060	0.012400	0.341
	0.049	0.1460	0.016600	0.337
	0.058	0.1720	0.019100	0.334
	0.065	0.1910	0.021000	0.331
	0.083	0.2390	0.025300	0.326
	0.095	0.2700	0.028000	0.322
1–1/8	0.035	0.1200	0.017800	0.386
	0.049	0.1660	0.024000	0.381
	0.058	0.1940	0.027700	0.378
	0.065	0.2160	0.030500	0.375
1–1/4	0.035	0.134	0.0247	0.430
	0.049	0.185	0.0334	0.425
	0.058	0.217	0.0387	0.422
	0.065	0.242	0.0426	0.420
	0.083	0.304	0.0521	0.414
1–3/8	0.035	0.147	0.0331	0.474
	0.049	0.204	0.0449	0.469
	0.058	0.240	0.0521	0.466
	0.065	0.268	0.0575	0.464
	0.083	0.337	0.0706	0.458
1–1/2	0.035	0.161	0.0432	0.518
	0.049	0.223	0.0589	0.513
	0.058	0.263	0.0684	0.510
	0.065	0.293	0.0756	0.508
	0.083	0.369	0.0931	0.502
	0.125	0.540	0.1290	0.488
	0.250	0.982	0.1990	0.451
1–5/8	0.035	0.175	0.0553	0.562
	0.058	0.286	0.0878	0.554

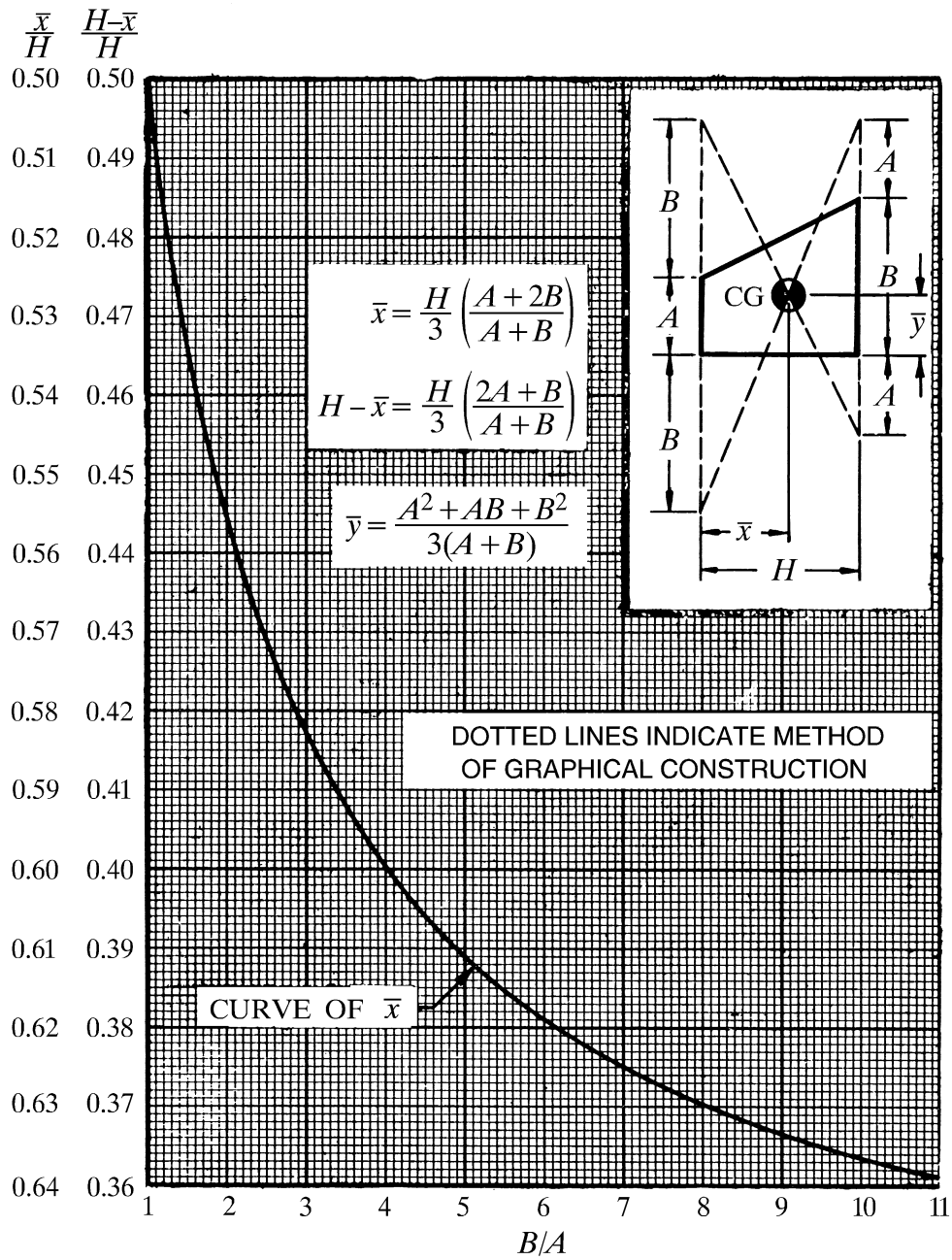
4.9 Section Properties of Round Tubing (Continued)

FIGURE 4–10 SECTION PROPERTIES OF ROUND TUBING (CONCLUDED)

D in	t in	Area in^2	I in^4	ρ in
1–3/4	0.035	0.189	0.0694	0.606
	0.049	0.262	0.0948	0.602
	0.058	0.308	0.1100	0.599
	0.083	0.435	0.1510	0.590
	0.120	0.614	0.2052	0.578
2	0.035	0.216	0.1040	0.695
	0.049	0.300	0.1430	0.690
	0.058	0.354	0.1670	0.687
	0.065	0.395	0.1850	0.685
	0.083	0.500	0.2300	0.678
	0.125	0.736	0.3250	0.664
	0.250	1.374	0.5370	0.625
2–1/4	0.049	0.339	0.2050	0.778
	0.065	0.446	0.2670	0.773
	0.083	0.565	0.3320	0.767
2–1/2	0.035	0.271	0.2060	0.872
	0.049	0.377	0.2830	0.867
	0.065	0.497	0.3690	0.861
	0.083	0.630	0.4610	0.855
	0.120	0.897	0.6370	0.843
	0.125	0.933	0.6590	0.841
	0.250	1.767	1.1320	0.800

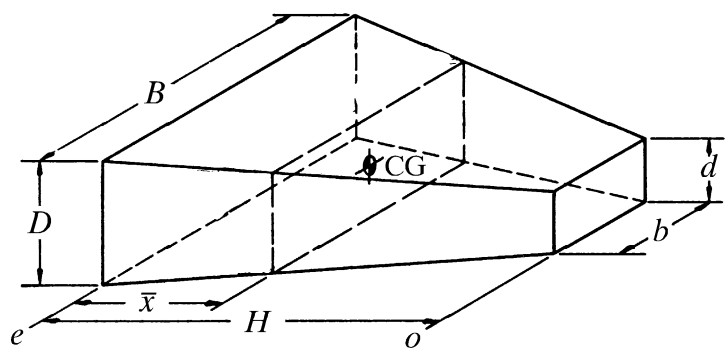
4.10 Centroid of Right Angle Trapezoidal Section

FIGURE 4–11 CENTROID OF A RIGHT ANGLE TRAPEZOID



4.11 Centroid of Trapezoidal Pyramid

FIGURE 4–12 CENTROID OF A TRAPEZOIDAL PYRAMID



$$A_o = bd$$

$$A_e = BD$$

$$A_m = \left(\frac{D+d}{2}\right)\left(\frac{B+b}{2}\right)$$

A_m = area of plane parallel to and midway between ends

$$V = \frac{H}{6}(A_o + A_e + 4A_m)$$

$$\bar{x} = \frac{H(A_o + 2A_m)}{(A_o + A_e + 4A_m)}$$

5 SHEAR CENTER AND WARPING CONSTANT

5.1 Shear Center

a. Definition – The shear center of a beam cross–section is the point through which the shear load must be applied so that the bending of the beam will not be accompanied by twisting. If a beam cross–section has two axes of symmetry, the shear center will be located at the centroid of the cross–section. If the section has only one axis of symmetry, the shear center will lie somewhere on the axis of symmetry.

b. Location – The shear center of a section built up of thin material can be found by locating the resultant of the shearing forces on each individual element of the cross–section, as computed from the shearing stresses produced in the beam when it is loaded so that it does not twist as it bends. The procedure for locating the shear center of a typical section such as that shown in FIGURE 5–1 is outlined in “c” through “f”.

c. General Equations – With a load V at the unknown shear center as shown in FIGURE 5–1, the resultant shear forces are given by summing the shear flow q_i over all elements.

$$R_i = \int_0^{b_i} q_i ds$$

where,

R and s_i values are selected to be continuous and in opposite directions.

$$q_i = \int_0^{S_i} \left[Z_{xy}(xV_y + yV_x) - xV_xZ_x - yV_yZ_y\right]t_i ds + q_o$$

$q_o =$ the value of q_i at $s_i = 0$

x and y are the distances defining the segment (elements) from the centroid.

NOTE: $\bar{x}$, $\bar{y}$, x_o, are positive as shown in FIGURE 5–6.

d. Horizontal Location of Shear Center

$$q_i = \int_0^{S_i} (xZ_{xy} - yZ_y)t_i ds + q_o$$

Step 1: Let $V_y = 1$ and $V_x = 0$; then

5.1 Shear Center (Continued)

Step 2: Select a convenient reference point, P, and sum moments about that point. Appropriate selection of the reference point may reduce the number of resultant forces for which solutions are required. For FIGURE 5–1, the summation of moments about P is

$$\sum M = 0 = R_1 b_2 + R_2 b_3 + V_y(x_o - \bar{x})$$

Expressing x_o in terms of the resultant shearing forces R_i,

$$x_o = +\bar{x} - R_1 b_2 - R_2 b_3$$

Step 3: To solve for x_o, q_i and R_i must be determined from above equations.

Step 4: Solve for x_o by substituting R_i into the expression of Step 2.

e. Vertical Location of Shear Center

Step 5: Let $V_x = 1$ and $V_y = 0$; then

$$q_i = \int_0^{S_i} (yZ_{xy} - xZ_x)t_i ds + q_o$$

Step 6: Using the same reference point, P, chosen in Step 2, sum moments about that point:

$$\sum M = 0 = R_1 b_2 + R_2 b_3 + V_x(\bar{y} - y_o)$$

Expressing y_o in terms of the resultant shearing forces R_i,

$$y_o = \bar{y} + R_1 b_2 + R_2 b_3$$

Step 7: To solve for y_o, q_i and R_i must be determined.

Step 8: Solve for y_o by substituting R_i into the expression of Step 6.

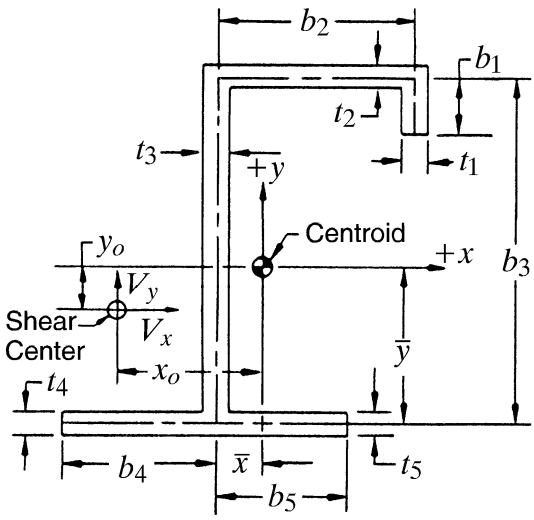
5.1 Shear Center (Continued)

FIGURE 5–1 TYPICAL SECTION RESULTANT SHEAR FORCE AND ORIGIN OF S FOR EACH ELEMENT

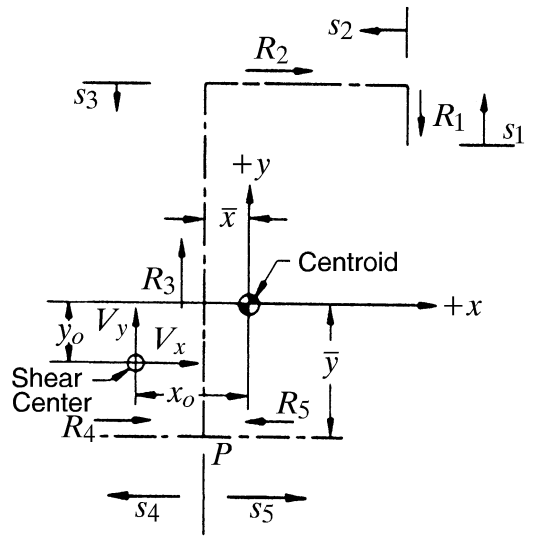
$$Z_x = \frac{I_{xx}}{I_{xx}I_{yy} - I_{xy}^2}$$

$$Z_y = \frac{I_{yy}}{I_{xx}I_{yy} - I_{xy}^2}$$

$$Z_{xy} = \frac{I_{xy}}{I_{xx}I_{yy} - I_{xy}^2}$$



(a)



(b)

NOTE: R_z and s_i are continuous and in opposite directions

SECTION PROPERTIES

5.1 Shear Center (Continued)

f. **Example Problem**

Given: Example dimensions shown in FIGURE 5–2.

FIGURE 5–2 LAYOUT DIMENSIONS AND SECTION PROPERTIES

$A = 0.6012 \text{ in}^2$

$\bar{x} = 0.2015 \text{ in}$

$\bar{y} = 0.9718 \text{ in}$

$I_{xx} = 0.6900 \text{ in}^4$

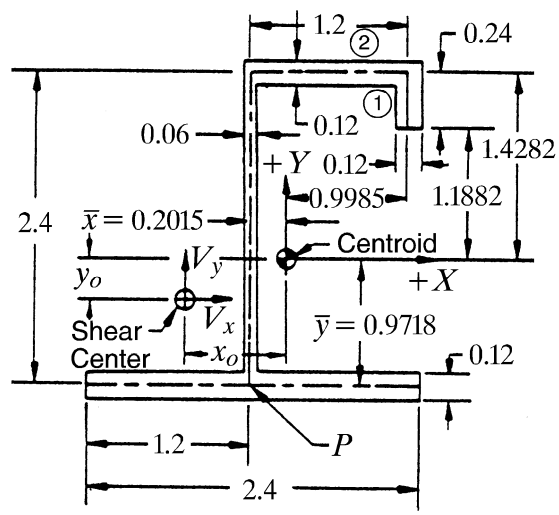
$I_{yy} = 0.2250 \text{ in}^4$

$I_{xy} = 0.1691 \text{ in}^4$

$Z_x = 5.4479 \text{ in}^{-4}$

$Z_y = 1.7765 \text{ in}^{-4}$

$Z_{xy} = 1.3352 \text{ in}^{-4}$



Find: Horizontal location of shear center with respect to centroid.

Solution: Assume a unit vertical load is applied at the shear center. $V_y = 1, \; V_x = 0.$

Step 1:

$$q_i = \int_0^s (1.3352x - 1.7765y)t_i ds + q_o$$

Step 2: For equilibrium the sum of moments about any point in the plane must be zero.

Take moments about point P, and assume clockwise moments as positive.

$$\sum M = 1.2R_1 + 2.4R_2 + V_y(x_o - \bar{x}) = 0$$

whence $x_o = -1.2R_1 - 2.4R_2 + 0.2015$

5.1 Shear Center (Continued)

Step 3: Starting with element 1 :

$$q_1 = \int_0^s (1.3352x - 1.7765y)0.12ds + q_o$$

where $x = 0.9985$, $y = s + 1.1882$, $q_o = 0$ at $(s = 0$ free edge, $q = 0)$

$$q_1 = \int_0^s [1.3352(0.9985) - 1.7765(s + 1.1882)]0.12ds = -0.1066s^2 - 0.0933s$$

$$R_1 = \int_0^{0.24} (-0.1066s^2 - 0.0933s)ds = \frac{-0.1066(0.24)^3}{3} - \frac{0.0933(0.24)^2}{2} = -0.0032$$

Step 4: For element 2 :

$$q_o = -0.1066(0.24)^2 - 0.0933(0.24) = -0.0285$$

$$x = 0.9985 - s, \qquad y = 1.4282$$

$$q_2 = \int_0^s [1.3352(0.9985 - s) - 1.7765(1.4282)]0.12ds - 0.0285$$

$$q_2 = -0.0801s^2 - 0.1445s - 0.0285$$

$$R_2 = \int_0^{1.2} \left(-0.0801s^2 - 0.1445s - 0.0285\right)ds$$

$$= \frac{-0.0801(1.2)^3}{3} - \frac{0.1445(1.2)^2}{2} - 0.0285(1.2) = -0.1844 \text{ units}$$

Step 5:

$$x_o = 1.2(-0.0032) - 2.4(-0.1844) + 0.2015 = +0.6478 \text{ in}$$

Find: Vertical location of shear center with respect to centroid.

Solution: Assume a unit horizontal load applied at the shear center. $V_x = 1$, $V_y = 0$.

Step 1:

$$q_i = \int_0^s (1.3352y - 5.4479x)t_i ds + q_o;$$

SECTION PROPERTIES

5.1 Shear Center (Continued)

Step 2: Take moments about point P, and assume clockwise moments as positive.

$$\sum M = 1.2R_1 + 2.4R_2 + V_x(\bar{y} - y_o) = 0$$

whence $y_o = +1.2R_1 + 2.4R_2 + 0.9718$

Step 3: Starting with element 1 :

$$x = 0.9985, \qquad y = s + 1.1882$$

$$q_1 = \int_0^s [1.3352(s + 1.1882) - 5.4479(0.9985)]0.12ds = 0.0801s^2 - 0.4624s$$

$$R_1 = \int_0^{0.24} (0.0801s^2 - 0.4624s)ds = \frac{0.0801(0.24)^3}{3} - \frac{0.4624(0.24)^2}{2} = -0.0129 \text{ units}$$

Step 4: For element 2 :

$$q_o = 0.0801(0.24)^2 - 0.4624(0.24) = -0.1064$$

$$x = 0.9985 - s, \qquad y = 1.4282$$

$$q_2 = \int_0^s [1.3352(1.4282) - 5.4479(0.9985 - s)]0.12ds - 0.1064$$

$$= 0.3269s^2 - 0.4239s - 0.1064$$

$$R_2 = \int_0^{1.2} (0.3269s^2 - 0.4239s - 0.1064)ds$$

$$= \frac{0.3269(1.2)^3}{3} - \frac{0.4239(1.2)^2}{2} - 0.1064(1.2) = -0.2446 \text{ units}$$

Step 5:

$$y_o = 1.2(-0.0129) + 2.4(-0.2446) + 0.9718 = +0.3693 \text{ in}$$

5.2 Warping Constant

The warping constant Γ is defined by:

$$\Gamma = (\Gamma_1 + \Gamma_2)$$

Where the constant Γ_1 relates to the strain in the middle plane of the walls arising from variation of warping of the cross–section along the length due to twist, and the constant, Γ_2 relates to the strain across the thickness t of the walls.

Two alternative methods of calculating Γ are described in the following paragraphs. Although either method can be used, the Numerical Method may be more expedient for many sections.

Both methods require that the point of rotation or twist (point p) be known. For unrestrained members, the center of twist is the shear center as defined in § 5.1. For members with lateral constraints, the center of twist coincides with the line(s) of action of the constraint(s).

a. General Method of Calculating Γ

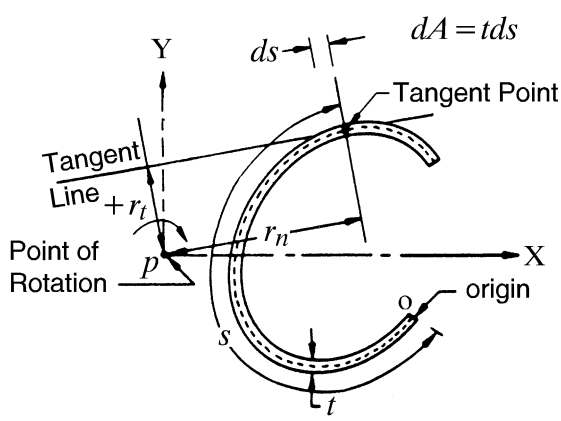
In sections whose individual elements intersect in a common point, such as angles, tees, and cruciforms, Γ_1 vanishes and thus

$$\Gamma = \Gamma_2 = \frac{1}{12}\int_A (r_n t_s)^2 dA \text{ in}^6$$

where r_n = distance from point of rotation to the normal of the line tangent to the middle of the wall at s (see FIGURE 5–3).

A = total cross–section area.

FIGURE 5–3 r_t AND r_n



NOTE: r_t is positive if a force along the tangent line in the direction of s would produce positive rotation about the point of rotation (positive rotation is arbitrarily chosen as clockwise). Thus r_t above is positive.

SECTION PROPERTIES

5.2 Warping Constant (Continued)

In other sections, Γ_2 is generally very small compared with Γ_1, so that

$$\Gamma \approx \Gamma_1 = \int_A w^2 dA - \frac{1}{A}\left[\int_A w dA\right]^2 \text{in}^6$$

where

$$w = \int_0^s r_t ds$$

r_t = distance from the point of rotation to the line tangent to the centerline of the section wall at "s" (see FIGURE 5–3).

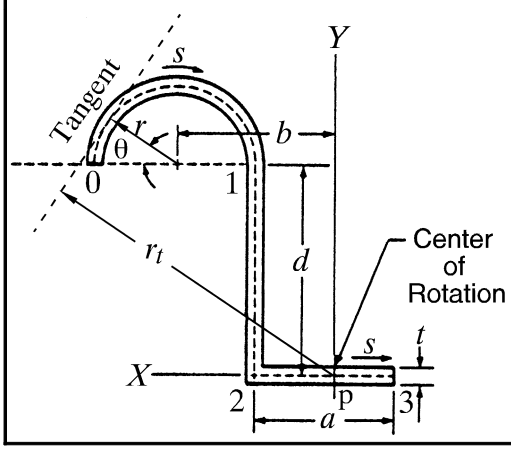
s = the distance from any free end along the middle of the wall.

An example of the use of this general method for Γ_1 is shown in FIGURE 5–4.

5.2 Warping Constant (Continued)

FIGURE 5–4 EXAMPLE OF GENERAL METHOD

Given: Sketch shown below. Note that this example uses a left–hand coordinate system.



NOTE: The center of rotation (point p) is at a fastener attaching the section to other structure.

Find: Determine Γ, using the general method.

Solution:

Step 1: Assume that the center of rotation of the section shown is at the point p. Then Γ is evaluated as follows: Taking the origin of s at point 0, then from 0 to 1.

$$r_t = r + b\cos\theta + d\sin\theta \qquad ds = rd\theta$$

$$w_{0-1} = \int_0^s r_t ds = r\int_0^\theta (r + b\cos\theta + d\sin\theta)d\theta = r(r\theta + b\sin\theta - d\cos\theta + d)$$

Step 2: From 1 to 2, $r_t = -(b - r)$ and

$$w_{1-2} = \left[w_{0-1}\big|_{\theta=\pi}\right] + \int_0^s (r - b)ds = r^2\pi + 2rd + (r - b)s$$

Step 3: From 2 to 3, $r_t = 0$ and

$$w_{2-3} = \left[w_{1-2}\big|_{s=d}\right] = r^2\pi + d(3r - b)$$

$$\Gamma \approx \Gamma_1 = \int_A w^2 dA - \frac{1}{A}\left[\int_A w dA\right]^2 = tr^3\int_0^\pi (r\theta + b\sin\theta - d\cos\theta + d)^2 d\theta$$

$$+ t\int_0^d \left[r^2\pi + 2rd + (r - b)s\right]^2 ds + t\int_0^a \left[r^2\pi + d(3r - b)\right]^2 ds$$

$$- \frac{1}{A}\left\{r^2 t\int_0^\pi (r\theta + b\sin\theta - d\cos\theta + d)d\theta + t\int_0^d \left[r^2\pi + 2rd + (r - b)s\right]ds\right.$$

$$\left. + t\int_0^a \left[r^2\pi + d(3r - b)\right]ds\right\}^2$$

where A = total cross sectional area.

SECTION PROPERTIES

5.2 Warping Constant (Continued)

b. Numerical Method Procedure For Calculating Γ_1

Step 1: Break the section into elements, and number each element and each element end. Number the elements with 1 , 2 , 3 , . . . j and the element ends with 0, 1, 2, . . . i.

Step 2: Start with a free end and proceed from element to element drawing arrows, s, along the tangent line (major axis) of each element.

Step 3: Determine the magnitude of the normal radius r_t from the point of rotation p to the major axis of each element. Point p is the shear center for the section shape with no skin or other structure attached to it. For a section attached to a skin or other rigid structure, point p may be at the fastener or some more distant rotation center of the structural assembly.

Step 4: Determine the sign of r_t for each element by the following rule. If the direction of s causes a clockwise moment at point p, r_t is positive; if the moment is counterclockwise, r_t is negative.

Step 5: Develop the relative strain constants C for each end of each element. There must be strain compatibility between adjacent elements because the end point of one element is the beginning of the next. Starting at a free end, set $C_o = C_i = 0$. Then the other strain constants are determined from

$$C_j = C_i + b_{(j)} r_{t(j)}; \; i = j - 1 \qquad \boxed{1}$$

where:

$b_{(j)}$ = length of element (j)

C_i = strain constant for the ith point (beginning of element (j))

C_j = strain constant for the jth point (end of element (j))

1 If elements are not in continuous series, see element 5 in example problem 2 of FIGURE 5–5.

Therefore, for the first element, where $C_i = 0$, the equation reduces to

$$C_j = b_{(j)} r_{t\,(j)}$$

The development of strain constants then proceeds in this fashion.

5.2 Warping Constant (Continued)

Step 6: Calculate $\bar{I}$ and $\bar{D}$ values

$$\bar{I} = \sum I_{j} \text{ and } \bar{D} = \sum D_{j}$$

where

$$A_{j} = b_{j} t_{j}$$

t_{j} = thickness of element j

$$I_{j} = A_{j} \left(C_j^2 - C_j b_{j} r_{t\,j} + \frac{b^2_{j} r_{t\,j}^2}{3} \right)$$

$$D_{j} = A_{j} \left(C_j - \frac{b_{j} \; r_{t_{j}}}{2} \right)$$

Step 7: Solve for $\Gamma_1 = \bar{I} - \frac{\bar{D}^2}{A}$

Two example problems in FIGURE 5–5 demonstrate the usage of the numerical method. The first, a hat section, illustrates a straightforward problem. The second, I–section with bulbs at two ends, becomes a more complex problem.

5.2 Warping Constant (Continued)

FIGURE 5–5 TWO EXAMPLE PROBLEMS USING NUMERICAL METHOD (CONTINUED ON NEXT PAGE)

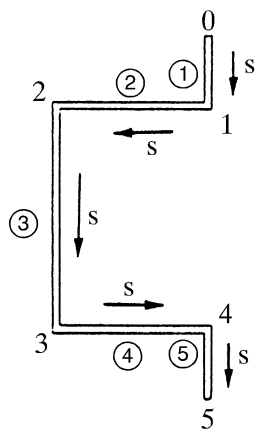
Example Problems

Determine Γ_1 for a hat section and an I–section with bulbs at two ends.

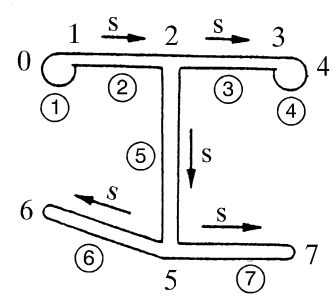
Step 1: Break the section into elements, and number each element and each element end.

Step 2: Draw arrows, s, along the tangent line (major axis) of each element.

Example 1

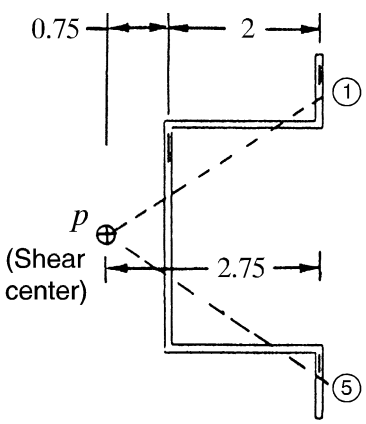


Example 2



Step 3: Determine the magnitude of the normal radius r_t. Distance is normal radius from p to element.

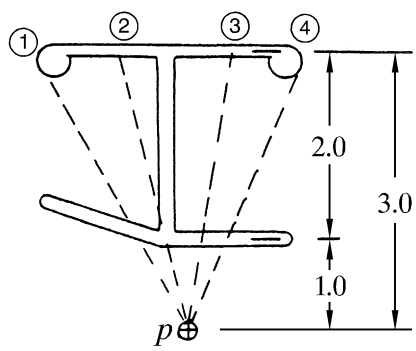
Example 1



For elements 1 and 5

$r_t = 2.75$

Example 2



For elements 1 , 2 , 3 , and 4

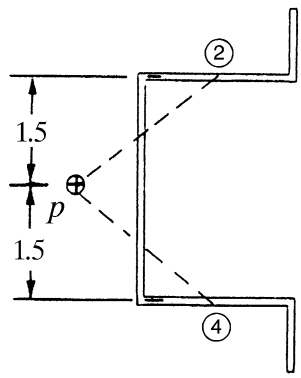
$r_t = 3.0$

NOTE: Point p in this example assumes that the I–section is attached to other structure.

5.2 Warping Constant (Continued)

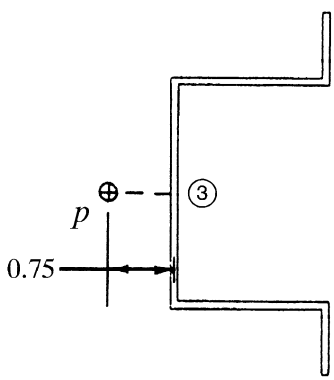
FIGURE 5–5 TWO EXAMPLE PROBLEMS USING NUMERICAL METHOD (CONTINUED)

Example 1 (Continued)



For elements 2 and 4

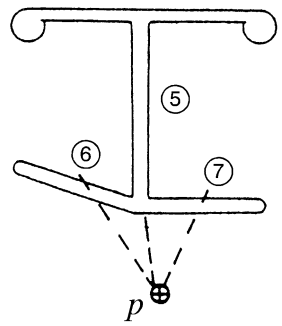
$r_t = 1.5$



For element 3

$r_t = 0.75$

Example 2 (Continued)



For elements 5 , 6 , and 7 see sketch below

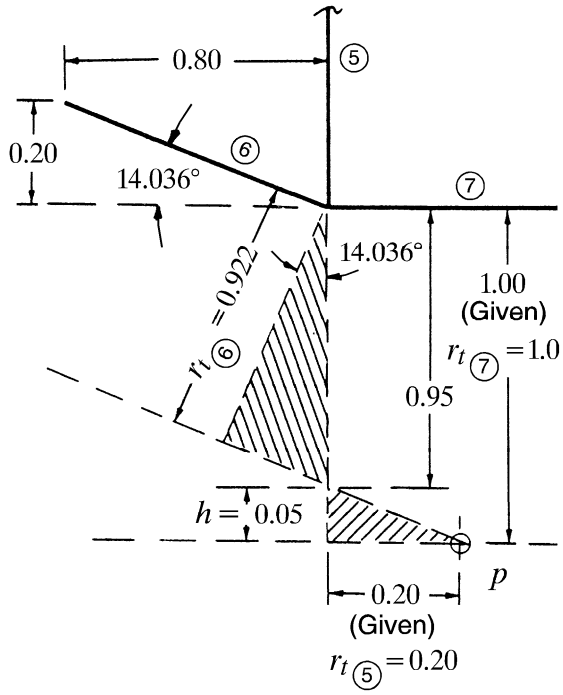
$r_{t\,5} = 0.20$

$r_{t\,6} = 0.922$ (see calculation below)

$r_{t\,7} = 1.0$

$h = \tan 14.036°(0.20) = 0.05$

$r_{t\,6} = \cos 14.036°(1.0 - 0.05) = 0.922$

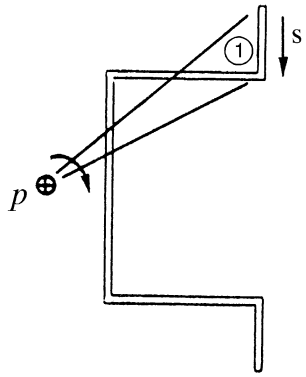


5.2 Warping Constant (Continued)

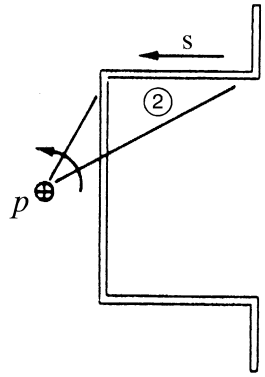
FIGURE 5–5 TWO EXAMPLE PROBLEMS USING NUMERICAL METHOD (CONTINUED)

Step 4: Determine the sign of r_t for each element.

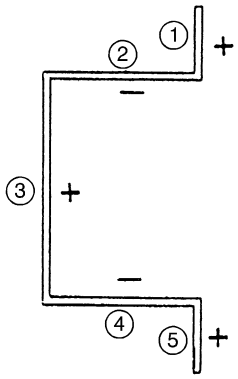
Example 1 (Continued)



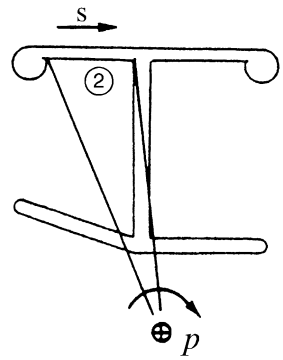
For element 1, $r_{t_{(1)}}$ is positive(+).



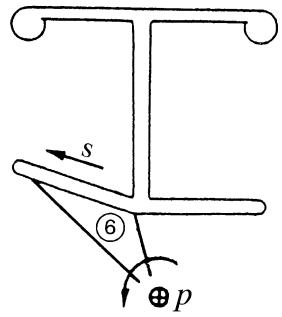
For element 2, $r_{t_{(2)}}$ is negative(−).



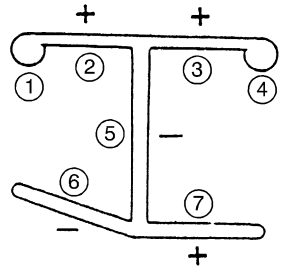
Example 2 (Continued)



For element 2, $r_{t_{(2)}}$ is positive(+).



For element 6, $r_{t_{(6)}}$ is negative(−).



5.2 Warping Constant (Continued)

FIGURE 5–5 TWO EXAMPLE PROBLEMS USING NUMERICAL METHOD (CONTINUED)

Step 5: Develop the relative strain constants C for each end of each element.

NOTE: $t_{(j)}$ = thickness of element

Example 1 (Continued)

Element 1:

$C_i = 0$

$C_j = C_i + b_{(j)}\ r_{t(j)}$

$= 0 + (1)(2.75)$

$= 2.75$

Element 2:

$C_i = 2.75$

$C_j = 2.75 + (2)(-1.50)$

$= -0.25$

Element 3:

$C_i = -0.25$

$C_j = -0.25 + (3)(0.75)$

$= 2.00$

Element 4:

$C_i = 2.00$

$C_j = 2.00 + (2)(-1.50)$

$= -1.00$

Element 5:

$C_i = -1.00$

$C_j = -1.00 + (1)(2.75)$

$= 1.75$

Example 2 (Continued)

Element 1:

$C_i = 0$

$C_j = 0 + b_{(j)}\ r_{t(j)}$

$= 0$

NOTE: For a bulb, $b_{(j)}\ r_{t(j)} = 0$

Element 2:

$C_i = 0$

$C_j = 0 + (0.8)(3)$

$= 2.4$

Element 3:

$C_i = 2.4$

$C_j = 2.4 + (1)(3)$

$= 5.4$

Element 4:

$C_i = 5.4$

$C_j = 5.4 + 0$

$= 5.4$

Element 5:

$C_i = 2.4$

$C_j = 2.4 + (2)(-0.20)$

$= 2.0$

Element 6:

$C_i = 2.0$

$C_j = 2.0 + (0.825)(-0.922)$

$= 1.24$

Element 7:

$C_i = 2.0$

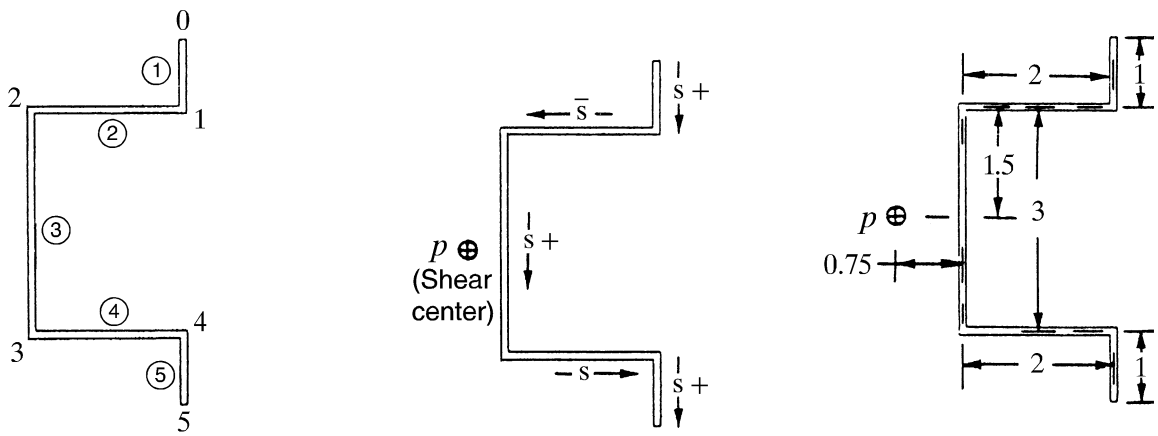
$C_j = 2.0 + (1.0)(1.0)$

$= 3.0$

5.2 Warping Constant (Continued)

FIGURE 5–5 TWO EXAMPLE PROBLEMS USING NUMERICAL METHOD (CONTINUED)

Solving example problem 1 (hat section):



Element and element end numbers — Direction of s; Sign of r_t — Dimensions in inches; all $t = 0.10$ in

Step 1			Step 3		Step 5			Step 6	
$\boxed{1}$	$\boxed{2}$	$\boxed{3}$	$\boxed{4}$	$\boxed{5}$	$\boxed{6}$	$\boxed{7}$	$\boxed{8}$	$\boxed{9}$	$\boxed{10}$
j	i [1]	t_j	b_j	r_{t_j}	$b_j r_{t_j}$	C_j	A_j	D_j	I_j
—	—	—	—	—	$\boxed{4} \times \boxed{5}$	$\boxed{7}\, n - 1 + \boxed{6}\, n$	$\boxed{3} \times \boxed{4}$	$\boxed{8}\left(\boxed{7} - \frac{\boxed{6}}{2}\right)$	$\boxed{8}\left[\boxed{7}^2 - \boxed{6}\left(\boxed{7} - \frac{\boxed{6}}{3}\right)\right]$
1	0	0.10	1.0	2.75	2.75	2.75	0.10	0.1375	0.252
2	1	0.10	2.0	–1.50	–3.00	–0.25	0.20	0.250	0.4625
3	2	0.10	3.0	0.75	2.25	2.00	0.30	0.2625	0.3563
4	3	0.10	2.0	–1.50	–3.00	–1.00	0.20	0.100	0.200
5	4	0.10	1.0	2.75	2.75	1.75	0.10	0.0375	0.077
$\sum$	—	—	—	—	—	—	0.90	0.7875	1.3478

Step 7:

$$\Gamma_1 = \sum \boxed{10} - \frac{\left(\sum \boxed{9}\right)^2}{\sum \boxed{8}} = 1.3478 - \frac{(0.7875)^2}{0.90} = 0.659 \text{ in}^6$$

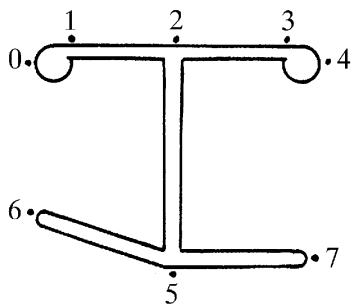
[1] Beginning node of element j

SECTION PROPERTIES

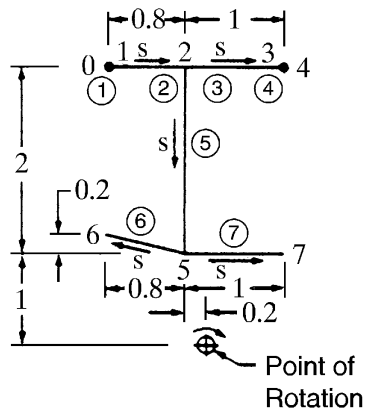
5.2 Warping Constant (Continued)

FIGURE 5–5 TWO EXAMPLE PROBLEMS USING NUMERICAL METHOD (CONTINUED)

Solving example problem 2 (I–section with bulbs):



Step 1 & 2



Step 1			Step 3		Step 5			Step 6	
1	2	3	4	5	6	7	8	9	10
j	i 1	t_j	b_j	r_{t_j}	$b_j r_{t_j}$	C_j	A_j	D_j	I_j
—	—	—	—	—	4×5	$7\ n-1 + 6\ n$	3×4	$8\left(7 - \frac{6}{2}\right)$	$8\left[7^2 - 6\left(7 - \frac{6}{3}\right)\right]$
1	0	R=0.08	R=0.08	3.0	0	0	0.02 2	0	0
2	1	0.08	0.8	3.0	2.4	2.4	0.064	0.0768	0.1229
3	2	0.10	1.0	3.0	3.0	5.4	0.10	0.3900	1.5960
4	3	R=0.08	R=0.08	3.0	0	5.4	0.02 2	0.1080	0.5832
5	2	0.15	2.0	–0.20	–0.40	2.0 3	0.30	0.6600	1.4560
6	5	0.08	0.825	–0.922	–0.760	1.24	0.066	0.1069	0.1763
7	5	0.10	1.0	1.0	1.0	3.0	0.10	0.2500	0.6333
$\sum$	—	—	—	—	—	—	0.670	1.5917	4.5677

Step 7:

$$\Gamma_1 = \sum \boxed{10} - \frac{\left(\sum \boxed{9}\right)^2}{\sum \boxed{8}} = 4.5677 - \frac{(1.5917)^2}{0.670} = 0.7863 \text{ in}^6$$

(See note in §5.2.b Step 3 about point of rotation.)

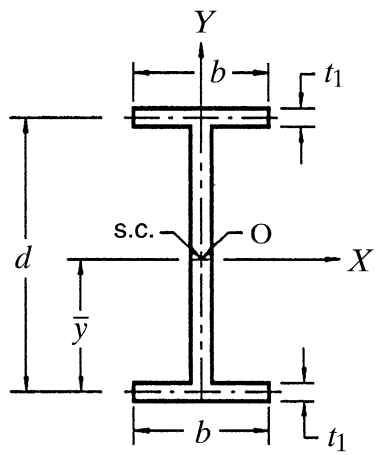
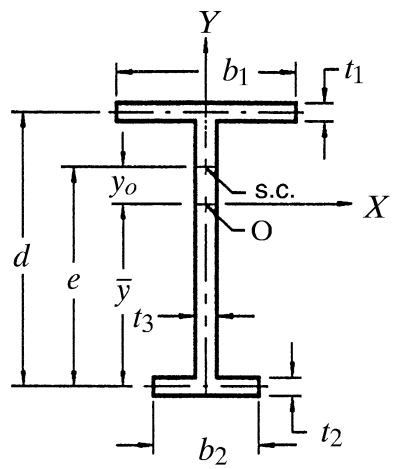
1 Beginning point of element

2 Bulb area

3 Beginning of element 5 is at point 2

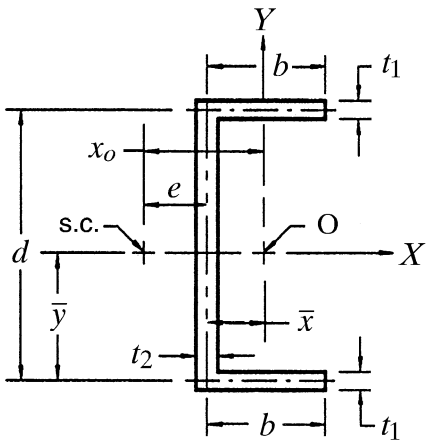
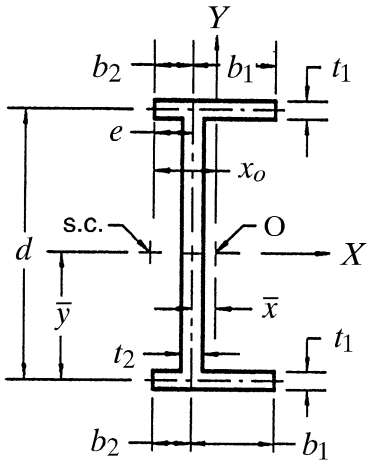
5.3 Typical Sections

FIGURE 5–6 TYPICAL SECTIONS (CONTINUED ON NEXT PAGE)

	"I" SECTIONS	
	EQUAL FLANGES	UNEQUAL FLANGES
Section	 s.c. = shear center	 s.c. = shear center
Shear Center	$x_o = 0$ $y_o = 0$	$x_o = 0$ $y_o = e - \bar{y}$ $e = \dfrac{d t_1 b_1^3}{t_1 b_1^3 + t_2 b_2^3}$
Warping Constant	$\Gamma_1 = \dfrac{t_1 b^3 d^2}{24}$	$\Gamma_1 = \dfrac{d^2}{12}\left(\dfrac{t_1 b_1^3 t_2 b_2^3}{t_1 b_1^3 + t_2 b_2^3}\right)$

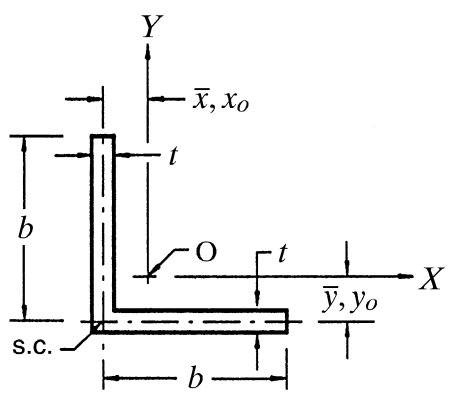
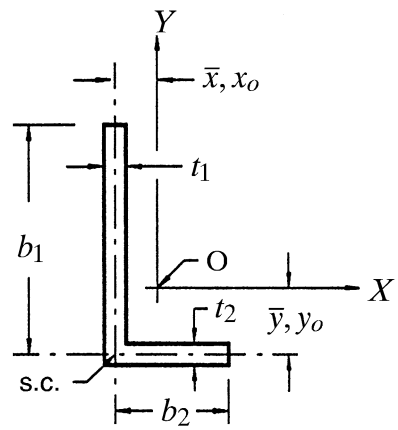
5.3 Typical Sections (Continued)

FIGURE 5–6 TYPICAL SECTIONS (CONTINUED)

	CHANNEL SECTIONS	
	EQUAL FLANGES	UNEQUAL FLANGES
Section	 s.c. = shear center	 s.c. = shear center
Shear Center	$x_o = e + \bar{x}$ $y_o = 0$ $e = \dfrac{3t_1b^2}{6bt_1 + dt_2}$	$x_o = e + \bar{x}$ $y_o = 0$ $e = \dfrac{3\left(b_1^2 - b_2^2\right)t_1}{6(b_1 + b_2)t_1 + dt_2}$
Warping Constant	$\Gamma_1 = \dfrac{t_1b^3d^2}{12}\left(\dfrac{3bt_1 + 2dt_2}{6bt_1 + dt_2}\right)$	$\Gamma_1 = \dfrac{d^2t_1}{4}\left[\dfrac{2}{3}(b_1^3 + b_2^3) - \dfrac{3(b_1^2 - b_2^2)^2t_1}{6(b_1 + b_2)t_1 + dt_2}\right]$

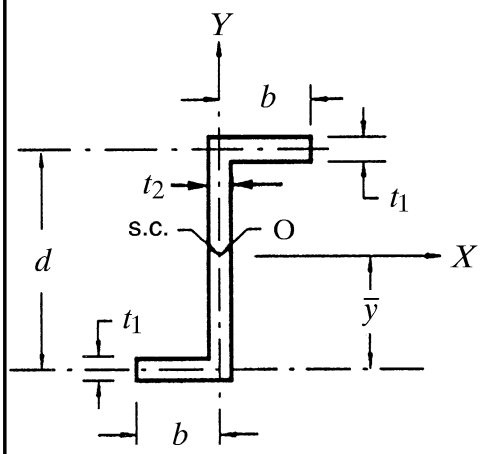
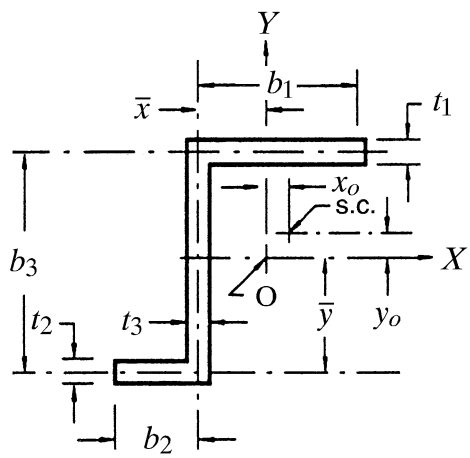
5.3 Typical Sections (Continued)

FIGURE 5–6 TYPICAL SECTIONS (CONTINUED)

	ANGLE SECTIONS	
	EQUAL LEGS	UNEQUAL LEGS
Section	 s.c. = shear center	 s.c. = shear center
Shear Center	$x_o = \bar{x}$ $y_o = \bar{y}$	$x_o = \bar{x}$ $y_o = \bar{y}$
Warping Constant	$\Gamma_1 \approx 0$ $\Gamma_2 = \frac{b^3 t^3}{18}$	$\Gamma_1 \approx 0$ $\Gamma_2 = \frac{t_1^3 b_1^3 + t_2^3 b_2^3}{36}$

5.3 Typical Sections (Continued)

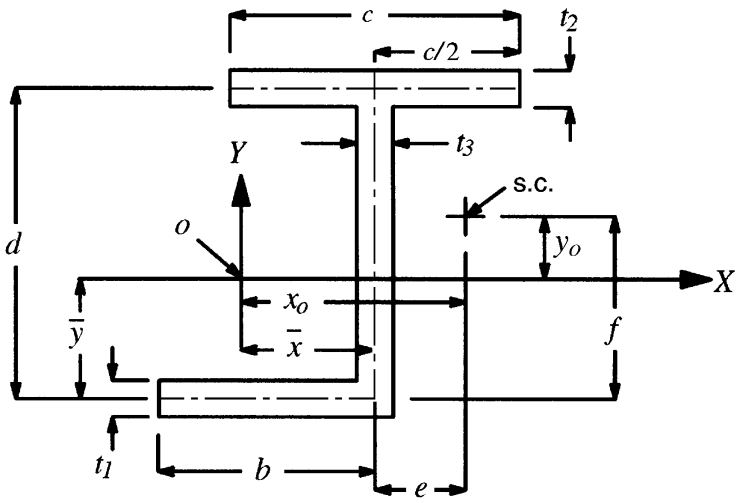
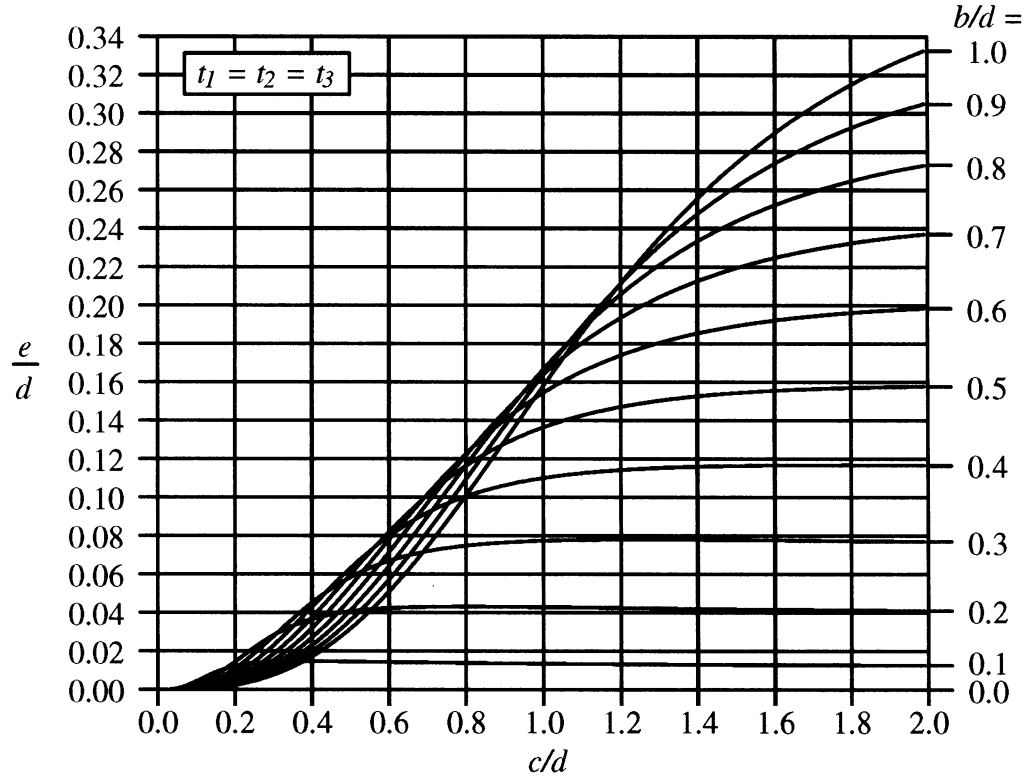
FIGURE 5–6 TYPICAL SECTIONS (CONTINUED)

	"Z" SECTIONS	
	EQUAL LEGS	UNEQUAL LEGS
Section	 s.c. = shear center	 s.c. = shear center
Shear Center	$x_o = 0$ $y_o = 0$	$x_o = t_1 b_1^2 b_3 \left[\left(\frac{b_1}{3} - \frac{\bar{x}}{2} \right) Z_{xy} - \left(\frac{b_3 - \bar{y}}{2} \right) Z_y \right] - \bar{x}$ $y_o = t_1 b_1^2 b_3 \left[\left(\frac{b_1}{3} - \frac{\bar{x}}{2} \right) Z_x + \left(\frac{\bar{y} - b_3}{2} \right) Z_{xy} \right] - \bar{y}$ (see FIGURE 5–1 for Z_i)
Warping Constant	If $t_1 = t_2 = t$, then $\Gamma_1 = \frac{t b^3 d^2}{12} \left(\frac{b + 2d}{2b + d} \right)$	$\Gamma_1 = t_2 b_2 \frac{(S_y b_2)^2}{3} + t_3 b_3 \Big[(S_y b_2)^2 - (S_x b_3 S_y b_2) + \frac{(S_x b_3)^2}{3} \Big] + t_1 b_1 \Big[(S_y b_2 - S_x b_3)^2 + (S_x b_3 - S_y b_2)(b_3 - S_y) b_1 + \frac{(b_3 - S_y)^2}{3} b_1^2 \Big] - \frac{1}{A} \left\{ \frac{-t_2 S_y b_2^2}{2} + t_3 b_3 \left(\frac{S_x b_3}{2} - S_y b_2 \right) + t_1 b_1 \left[S_x b_3 - S_y b_2 + \left(\frac{b_3 - S_y}{2} \right) b_1 \right] \right\}^2$ Where $S_x = \bar{x} + x_o,\ S_y = \bar{y} + y_o,\ A = t_1 b_1 + t_2 b_2 + t_3 b_3$

SECTION PROPERTIES

5.3 Typical Sections (Continued)

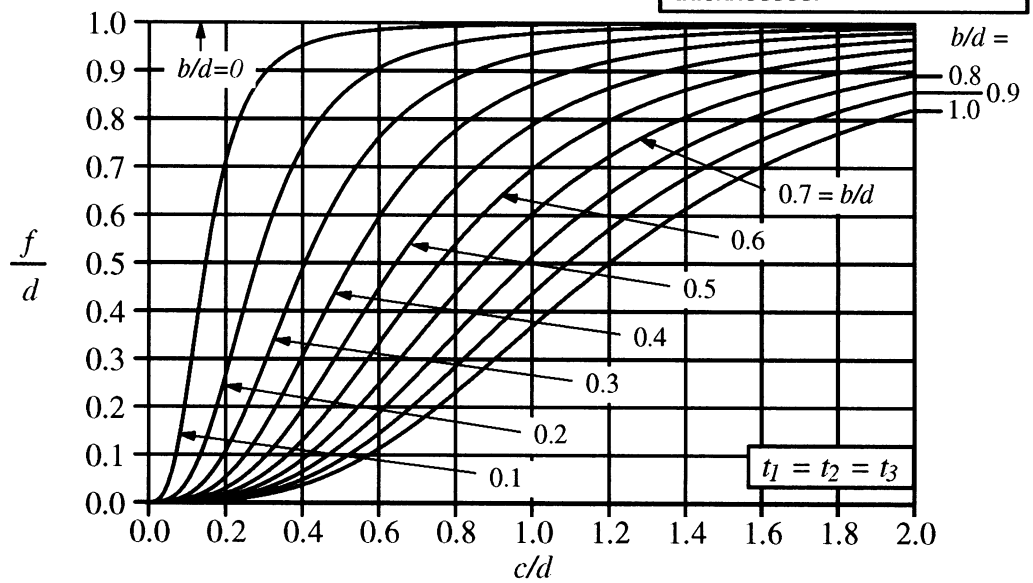
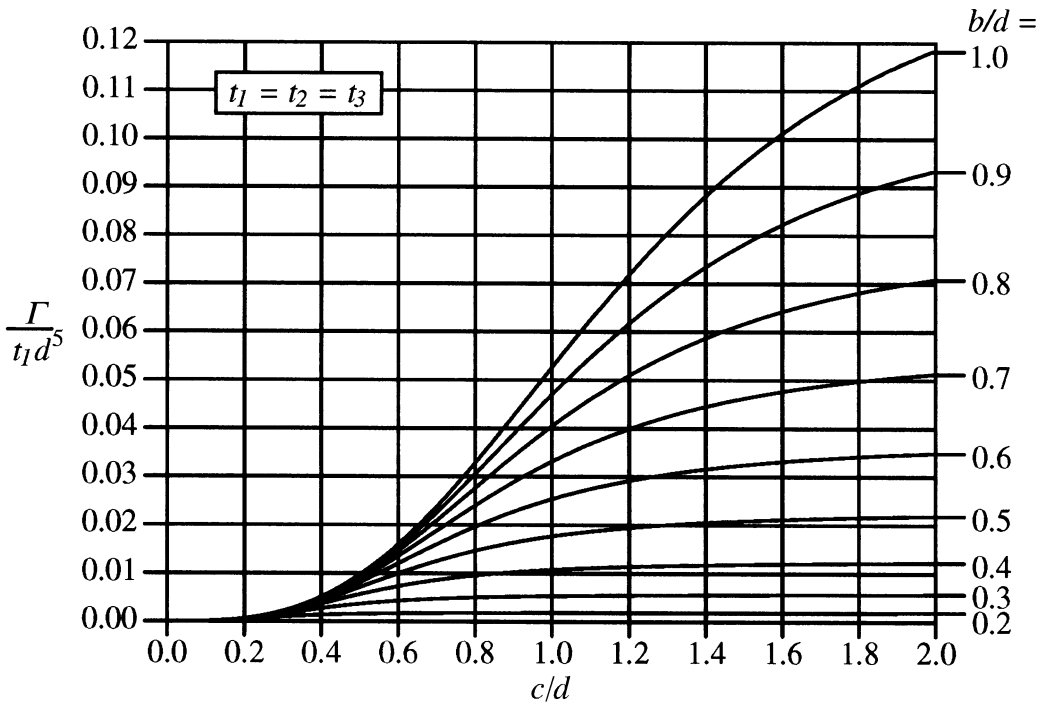
FIGURE 5–6 TYPICAL SECTIONS (CONTINUED)

	"J" SECTION
Section	
Shear Center X Location	$x_o = \bar{x} + t_1 b^2 d\left\{\left(\frac{\bar{x}}{2} - \frac{b}{3}\right) Z_{xy} + \frac{\bar{y}}{2} Z_y\right\}$ Chart assumes $t_1 = t_2 = t_3$. Equation is valid for all thicknesses. 

SECTION PROPERTIES

5.3 Typical Sections (Continued)

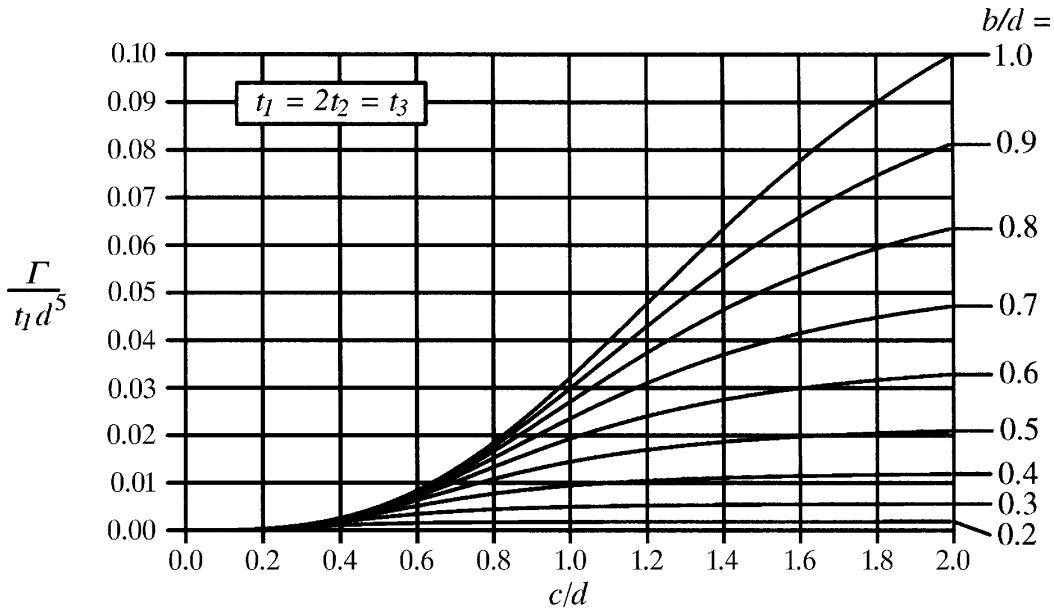
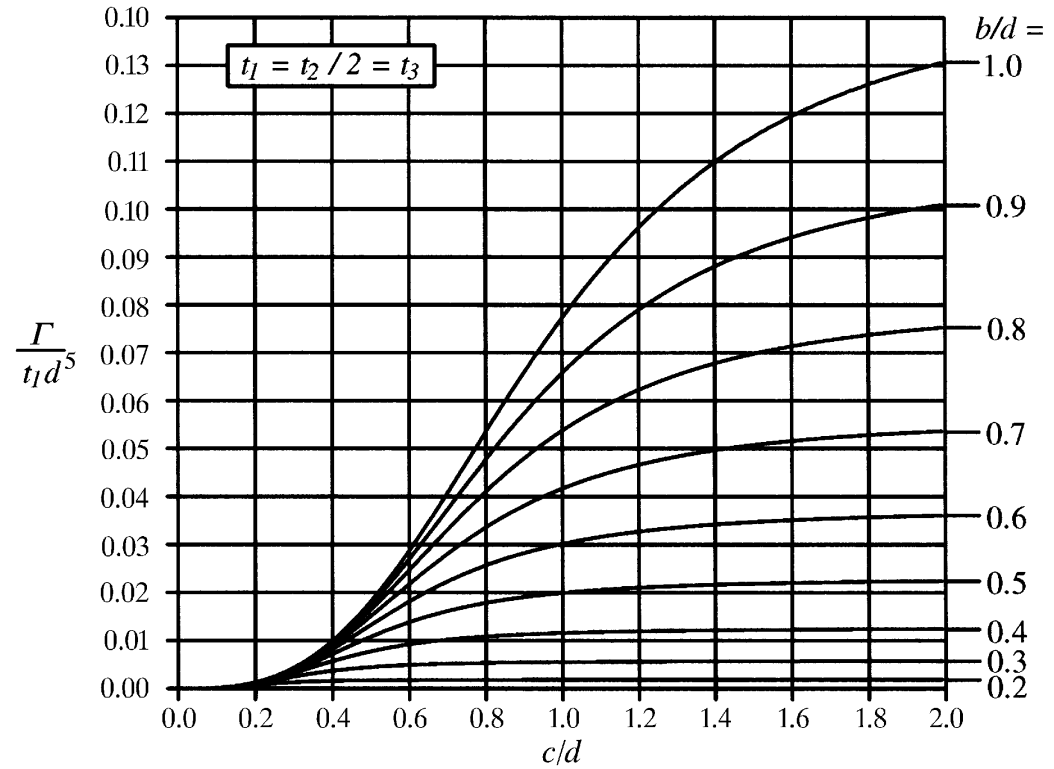
FIGURE 5–6 TYPICAL SECTIONS (CONTINUED)

	"J" SECTION (Continued)
Shear Center Y Location	$y_o = d - \bar{y} + t_1 b^2 d\left\{\left(\frac{\bar{x}}{2} - \frac{b}{3}\right) Z_x + \frac{\bar{y}}{2} Z_{xy}\right\}$ Chart assumes $t_1 = t_2 = t_3$. Equation is valid for all thicknesses. 
Warping Constant $(t_1 = t_2 = t_3)$	

SECTION PROPERTIES

5.3 Typical Sections (Continued)

FIGURE 5–6 TYPICAL SECTIONS (CONTINUED)

	"J" SECTION (Continued)
Warping Constant $(t_1 = 2t_2 = t_3)$	
Warping Constant $(t_1 = t_2/2 = t_3)$	

SECTION PROPERTIES

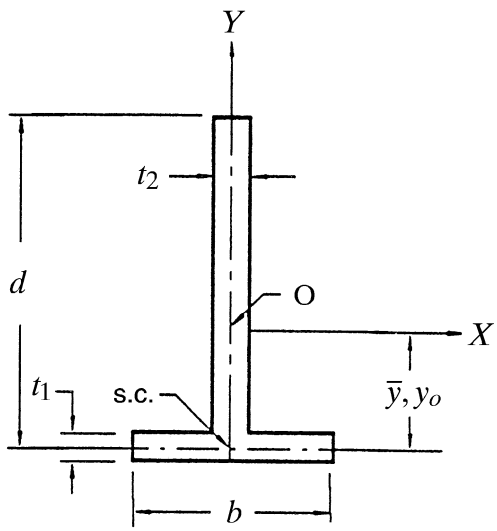
5.3 Typical Sections (Continued)

FIGURE 5–6 TYPICAL SECTIONS (CONTINUED)

Warping Constant (General case)	"J" SECTION (Continued)
	$\Gamma_1 = \frac{1}{3}c^3t_2(d-f)^2 + \frac{1}{12}dt_3\left[3c^2(d-f)^2 + 2ed(2ed+3cf-3cd)\right]$ $+ \frac{1}{12}bt_1\left[3c^2(d-f)^2 + 12edf(c-b) + 12ed^2(e-c) + 2bf\,(3cd+2bf-3cf)\right]$ $- \frac{1}{4(ct_2+dt_3+bt_1)}\left\{c^2t_2(d-f) + dt_3(dc-de-cf) + bt_1[d(c-2e)+f(b-c)]\right\}^2$ $\Gamma_2 = \frac{1}{12}\left[bt_1^3\left(\frac{b^2}{3}+be+e^2\right) + ct_2^3\left(\frac{c^2}{12}+e^2\right) + dt_3^3\left(\frac{d^2}{3}-df+f^2\right)\right]$

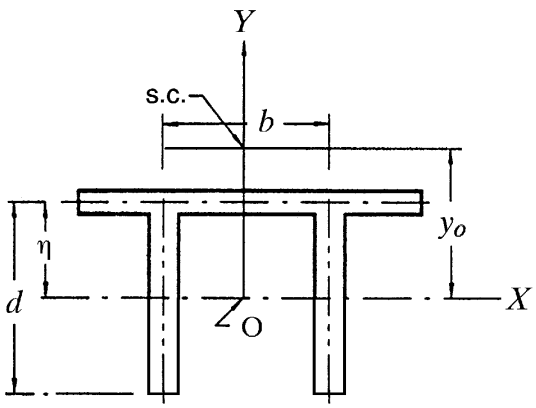
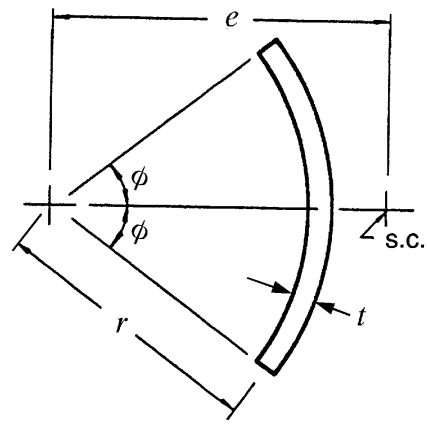
5.3 Typical Sections (Continued)

FIGURE 5–6 TYPICAL SECTIONS (CONTINUED)

	"T" SECTION
Section	 s.c. = shear center
Shear Center	$x_o = 0$ $y_o = \bar{y}$
Warping Constant	$\Gamma_1 = 0$ $\Gamma_2 = \frac{t_1^3 b^3}{144} + \frac{t_2^3 d^3}{36}$

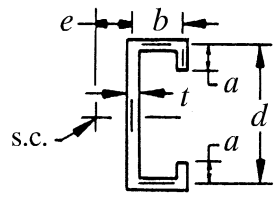
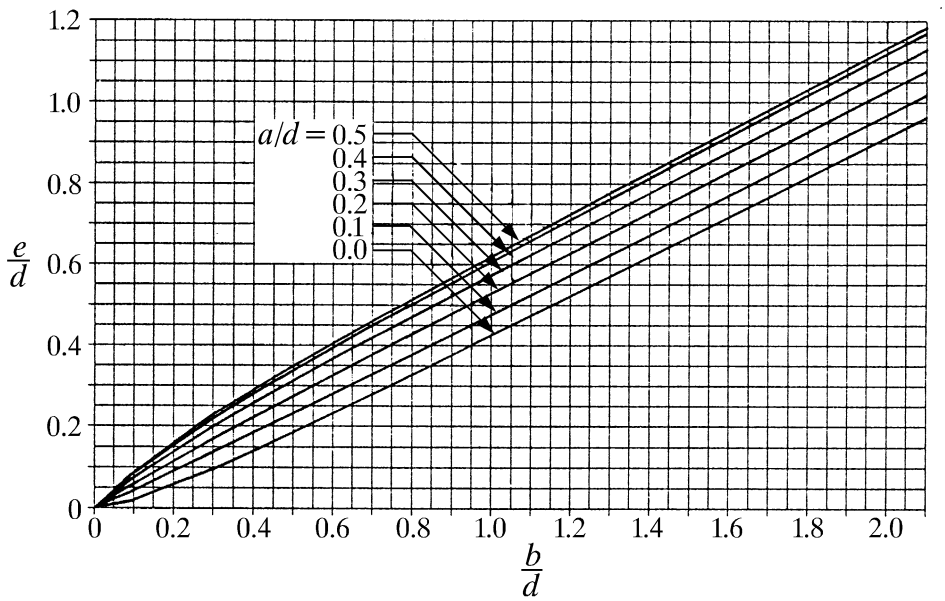
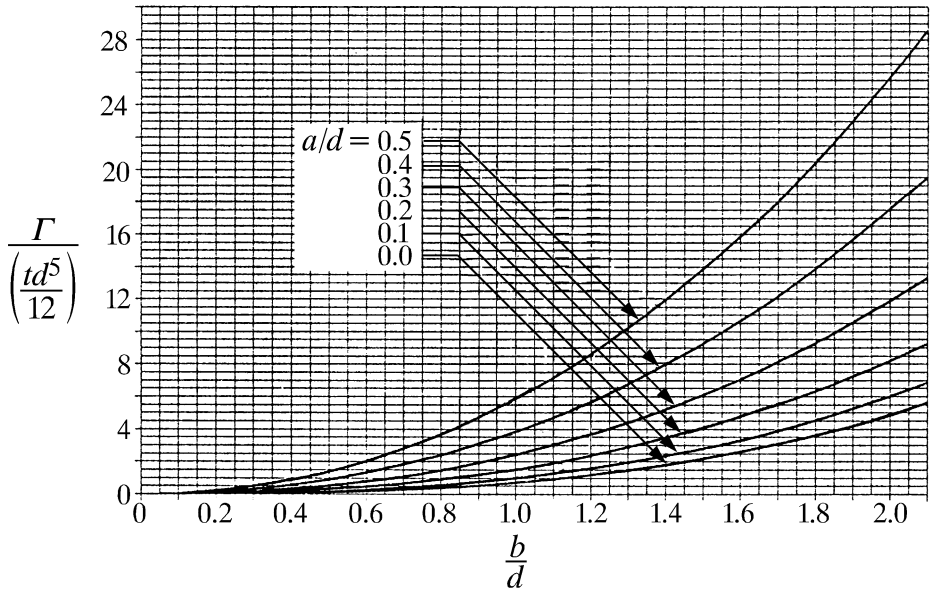
5.3 Typical Sections (Continued)

FIGURE 5–6 TYPICAL SECTIONS (CONTINUED)

	"TT" SECTION	CIRCULAR ARC
Section	 s.c. = shear center	 s.c. = shear center
Shear Center	$x_o = 0$ $y_o = \eta\left(1 + \frac{b^2A}{4I_y}\right)$	$e = 2r\left(\frac{\sin\phi - \phi\cos\phi}{\phi - \sin\phi\cos\phi}\right)$
Warping Constant	$\Gamma = \frac{b^2}{4}\left[I_x + \eta^2A\left(1 - \frac{b^2A}{4I_y}\right)\right]$	$\Gamma = \frac{2tr^5}{3}\left[\phi^3 - \frac{6(\sin\phi - \phi\cos\phi)^2}{\phi - \sin\phi\cos\phi}\right]$

5.3 Typical Sections (Continued)

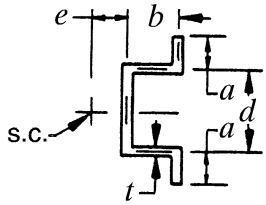
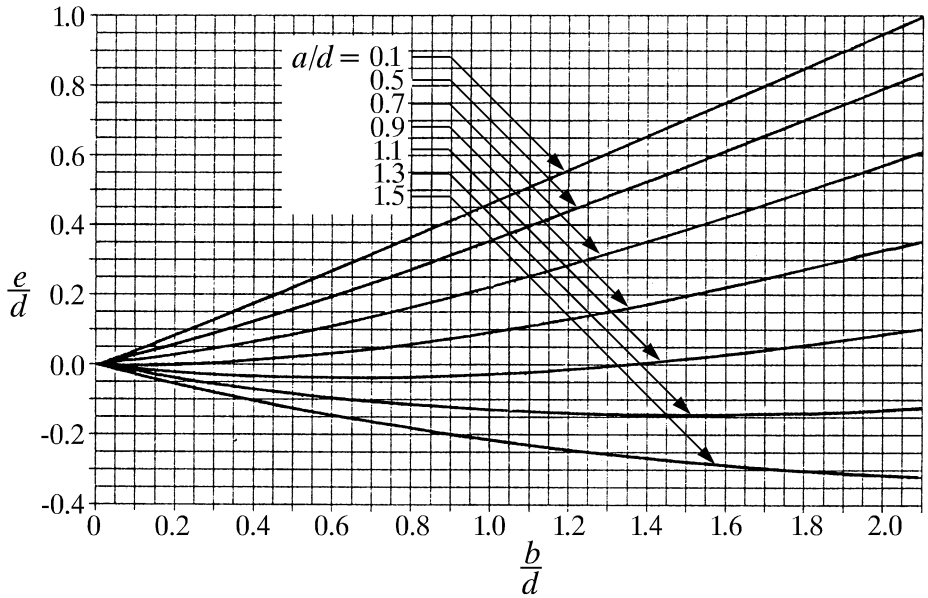
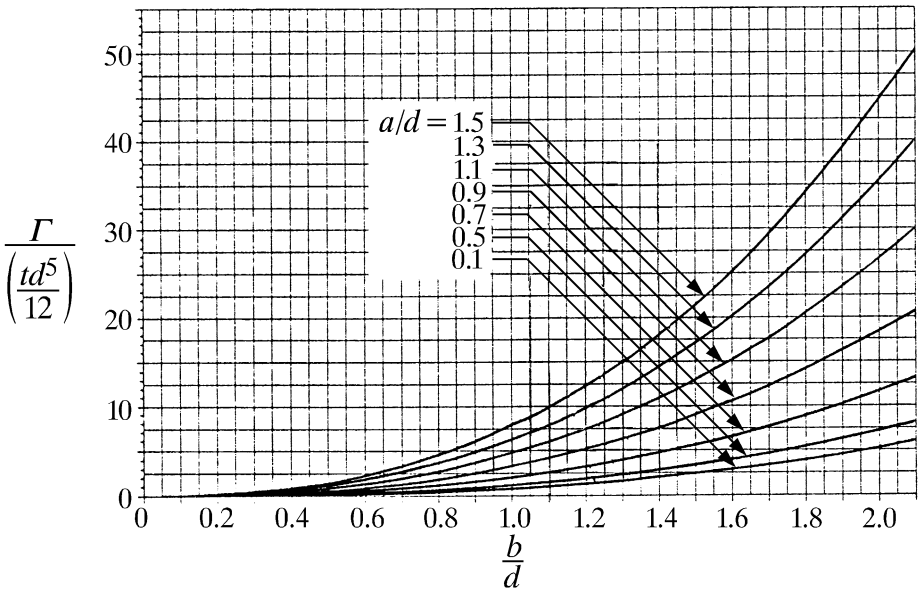
FIGURE 5–6 TYPICAL SECTIONS (CONTINUED)

	LIPPED CHANNEL SECTION
Section	 s.c. = shear center
Shear Center	
Warping Constant	

SECTION PROPERTIES

5.3 Typical Sections (Continued)

FIGURE 5–6 TYPICAL SECTIONS (CONCLUDED)

	TOP HAT SECTION
Section	 s.c. = shear center
Shear Center	
Warping Constant